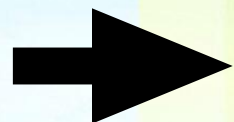


Business Process Automation with VBA and Python

Mr. Eddie Chow / 9 May 2026



Table Of Contents



Introduction to business process automation

Business Process automation with VBA

Business process automation with Python

Introduction to project management for business process automation

Development and implementation of business process automation

Final Group Presentation



Lecturer Biography

<https://www.linkedin.com/in/eddie-chow-a3860464/>



- Founder of InnoVi, Co-founder of 4 companies, covering from property, retail, education and ESG
- 20+ Years' programming experience, including 7 Years' IT Management Experience
- 7 years' teaching experience in tertiary education
- 10+ years' coaching or mentoring experience on large project development. .
- Development experience on generative ai, agentic ai, machine learning/deep learning, computer vision
- Researcher in Cutting-edge AI Research, particular in area in NLP, Computer vision
- AI Tech Speaker, AI Mentor
- Personal Website : eddiecityu.github.io

Assessment

- **Continuous Assessment**

In-class discussions and computer assignments which include the usage of different applications & methodologies

- **Final Assessment**

20-mins (3 persons) group project presentation which integrate knowledge in this module, apply different AI methodologies and software to business applications of process automation



What is Process Automation, RPA, IPA?
Overview of the technological building blocks related to process automation
Contemporary tools for process automation
Challenges and opportunities of business process automation
Business implications of process automation



Intended Learning Outcomes

1. describe the contemporary trends of process automation and explain the opportunities and challenges of business process automation;
2. outline key steps in process automation project management and illustrate the significance of each step;
3. apply computational tools to implement process automation;
4. discuss the development of process automation and practical cases for business.

Memory Test?

- NO Midterm / Final exams

Essays?

- NO research paper
- NO project report

Practical Work!

- 2 in-class coding exercises
- 1 Group Project



Taking
Exams



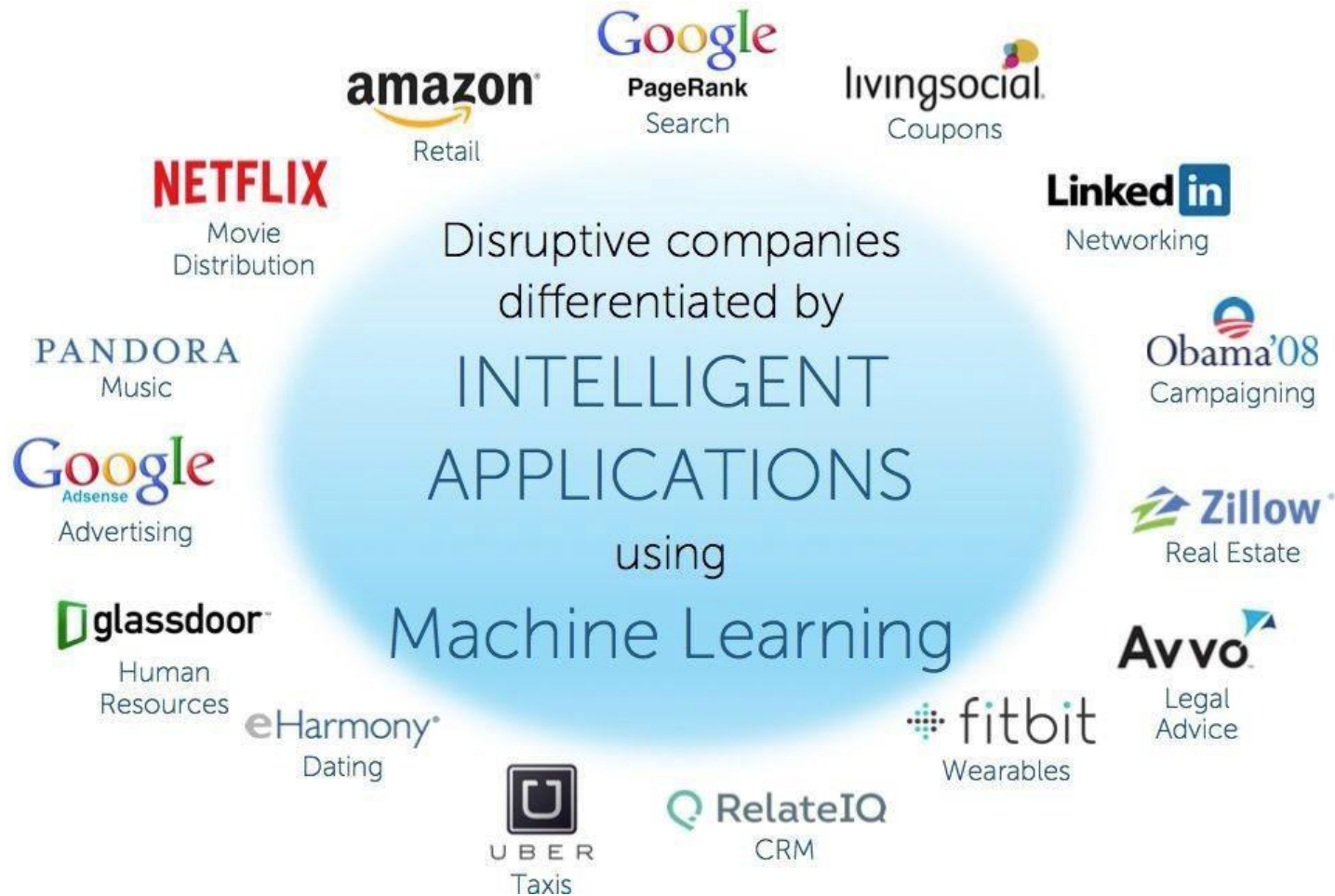
Building
Solutions

Lecture 1 - Business Process Automation with VBA and Python



Introductions: Getting To Know You
Forming Groups of 2 - 3

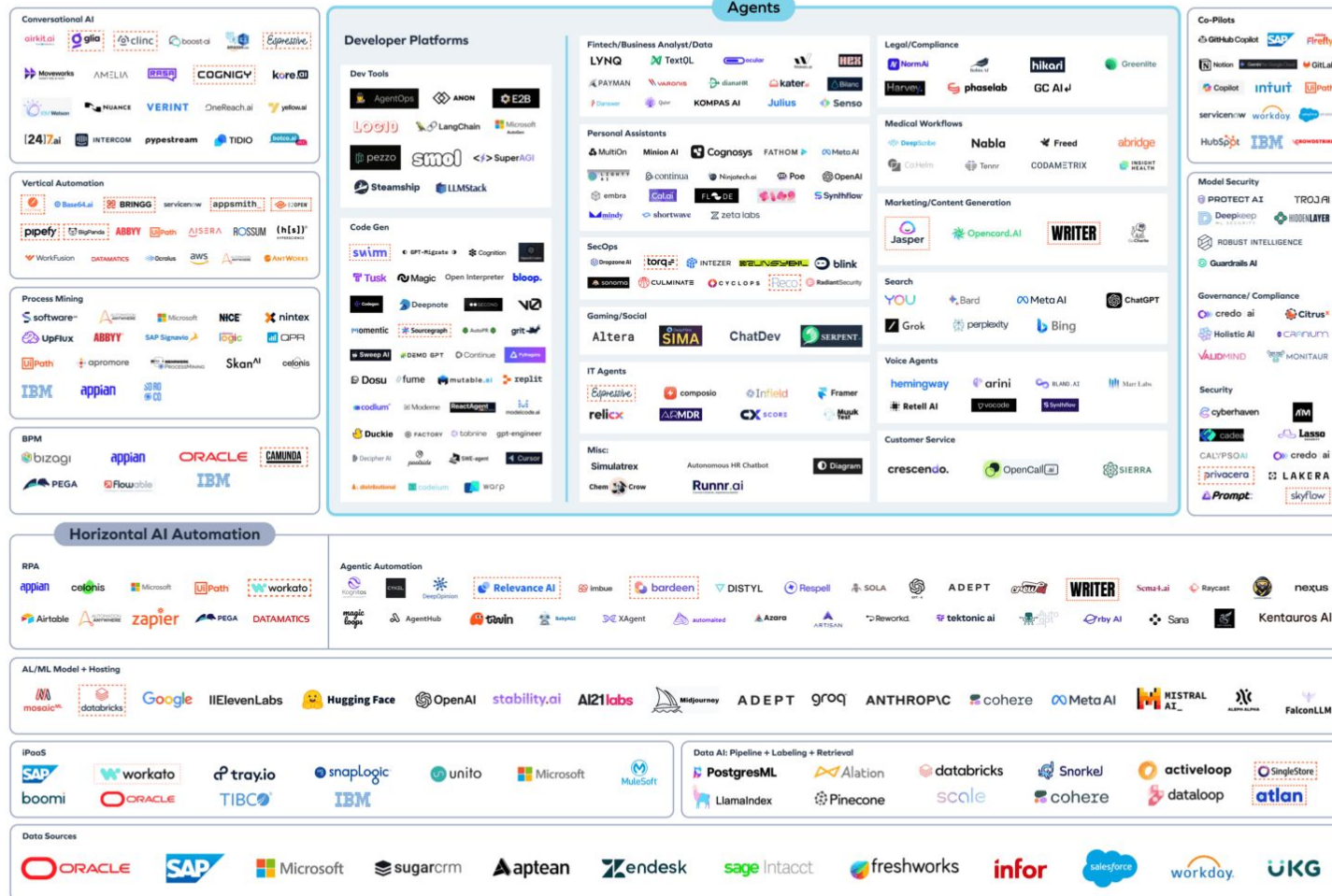
The new disruptors



The new disruptors

AI Automation Market Map

INSIGHT
PARTNERS



<https://www.insightpartners.com/ideas/ai-agents-disrupting-automation/>

Introduction to Business Process Automation

What is Business Automation?

From ChatGPT...

JA What is business automation? asking for a friend :)

 Business automation refers to the process of using technology to automate repetitive or routine tasks and processes within a business, such as data entry, invoicing, and inventory management. It can also involve the integration of various software applications and systems to streamline workflows and improve overall efficiency.

Introduction to Business Process Automation

Types of Business Automation

1. Basic Automation
2. Process Automation
3. Advanced Automation
4. Intelligent Automation



1. *The Basics of Business Automation*. (2021, May 12). The Basics of Business Automation | IBM.
<https://www.ibm.com/cloud/blog/basics-of-business-automation>

Introduction to Business Process Automation

Types of Business Automation – Basic Automation

- **Target** → Simple / Repetitive tasks
- **Why**
 - Eliminate human errors
 - Accelerate pace of transactional work
- **Example** → Robotic Process Automation (RPA)*



* To be covered in the next section

Introduction to Business Process Automation

Robotic Process Automation (RPA) - Introduction

- Rule-based software
- Automating high-volume activities
- Free up human workers (for more meaningful tasks)

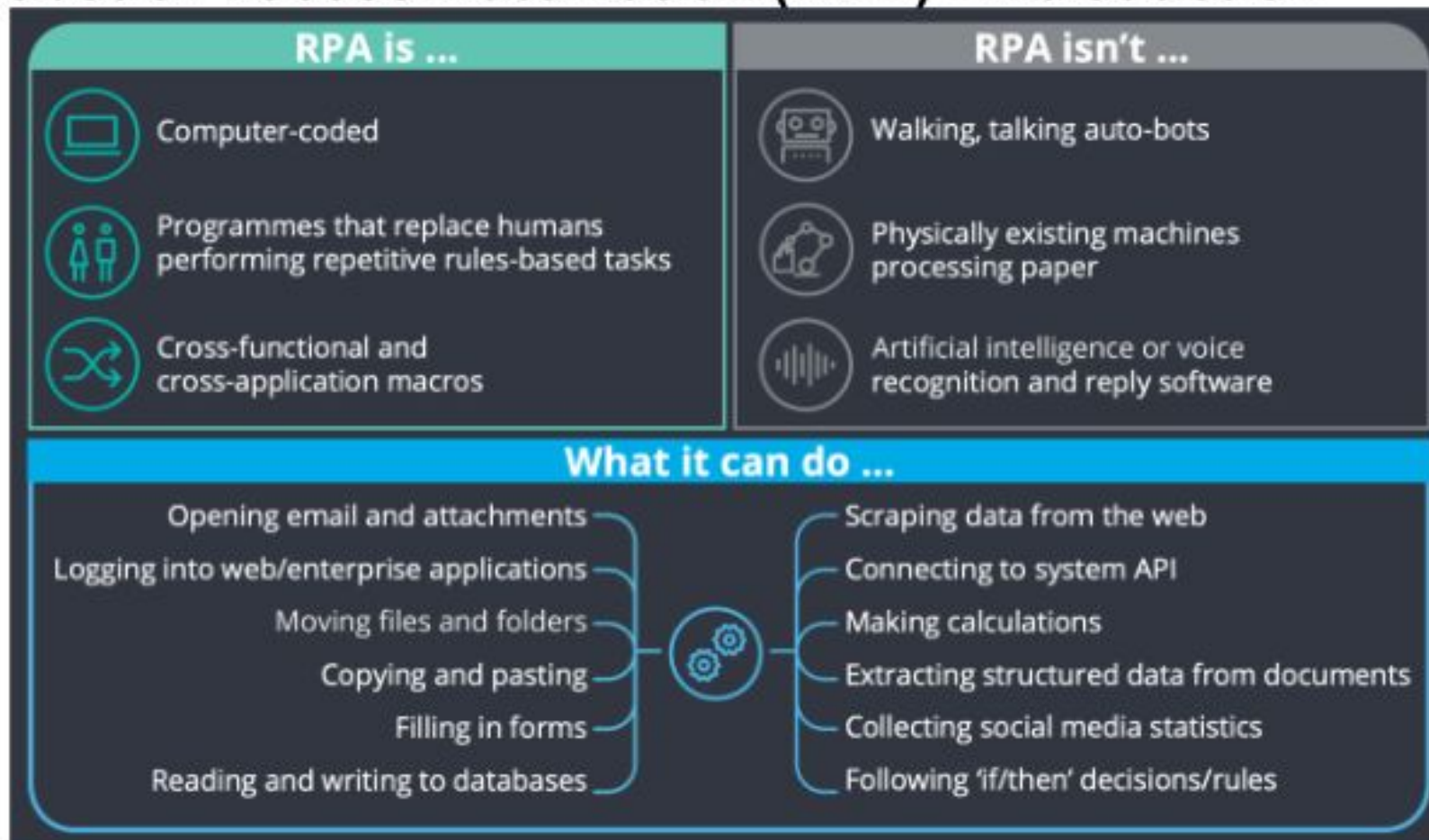


RPA bot

1. What is Robotic Process Automation (RPA)? | IBM. What Is Robotic Process Automation (RPA)? | IBM. <https://www.ibm.com/topics/rpa>

Introduction to Business Process Automation

Robotic Process Automation (RPA) - Introduction



1. Deloitte. (2018). *The robots are ready. Are you?*

<https://www2.deloitte.com/content/dam/Deloitte/tr/Documents/technology/deloitte-robots-are-ready.pdf>

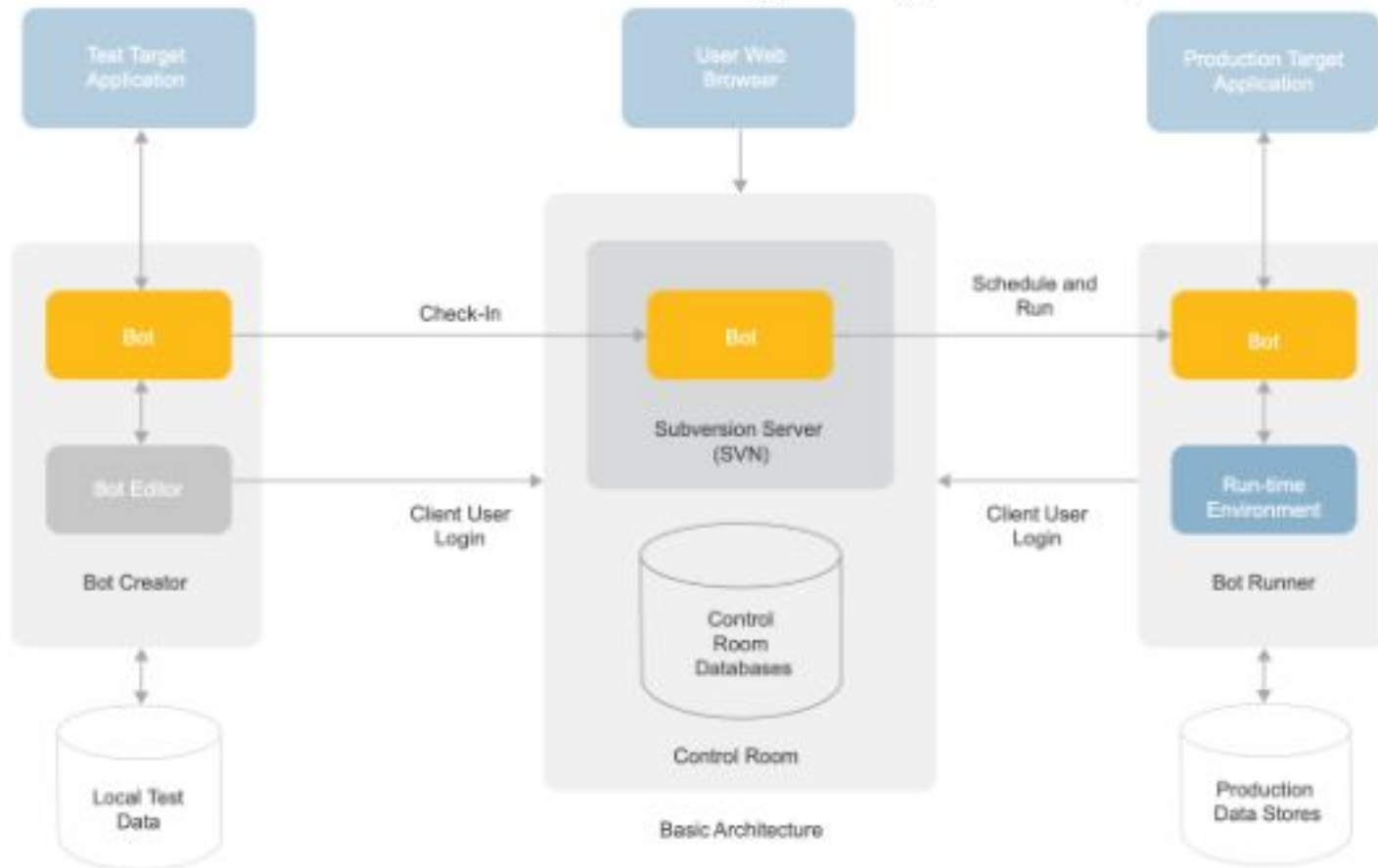
Robotic Process Automation (RPA) - Components

- **Bot modeling**
 - Bot Creator → development/ testing
 - Bot Runner → execution of tested codes in production
- **Control Room**
 - Deployment / Scheduler
 - Audit trails
 - Performance Analytics
 - Role-based access control



Introduction to Business Process Automation

Robotic Process Automation (RPA) - Components



1. Automation Anywhere. *Enterprise-class Security for Robotic Process Automation*.
https://www.automationanywhere.com/sites/default/files/internal-assets/uberflip/security-whitepaper_en.pdf

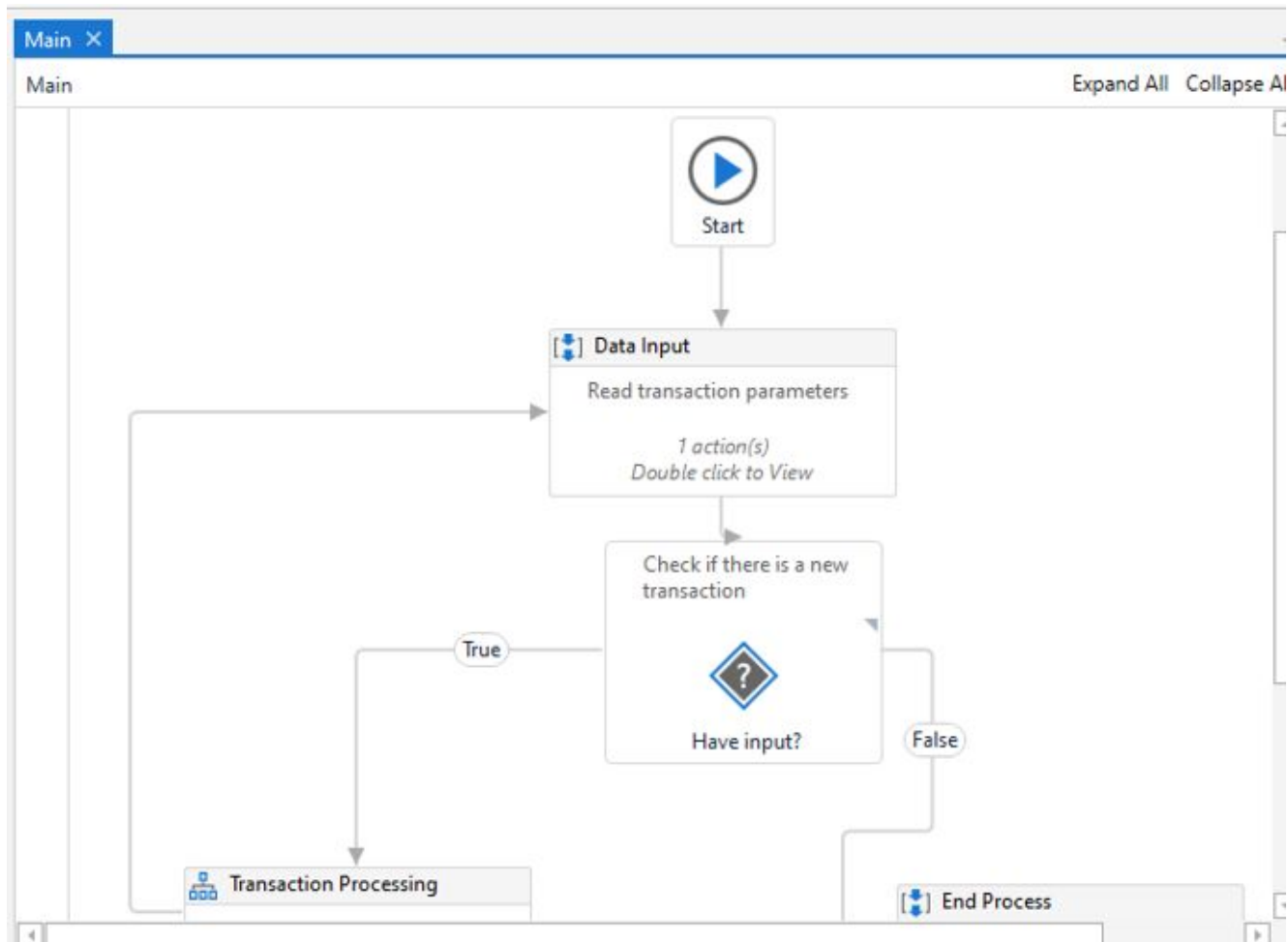
Introduction to Business Process Automation

Types of Business Automation – Process Automation

- **Target** → Process requiring dedicated solutions
- **Why**
 - Increase productivity and efficiency
- **Example** → Process Mining, workflow automation



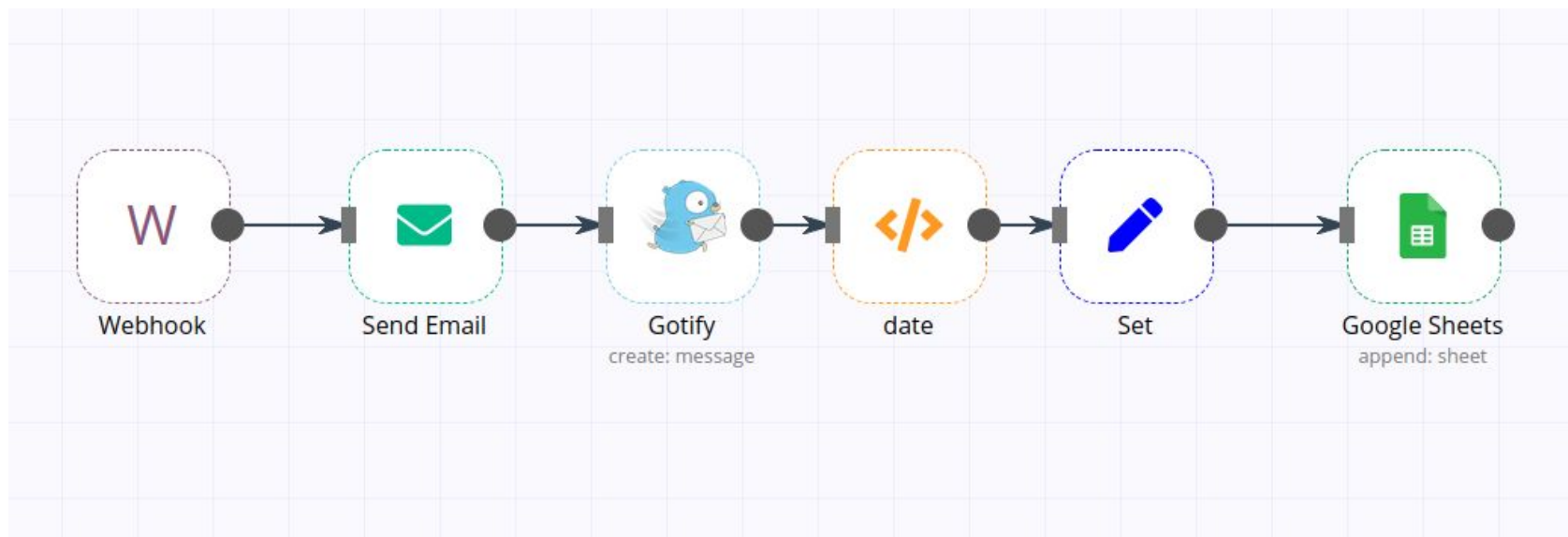
Process Automation Software Example 1 : UiPath



<https://www.uipath.com/rpa/robotic-process-automation>

Process Automation Software Example 2 : n8n

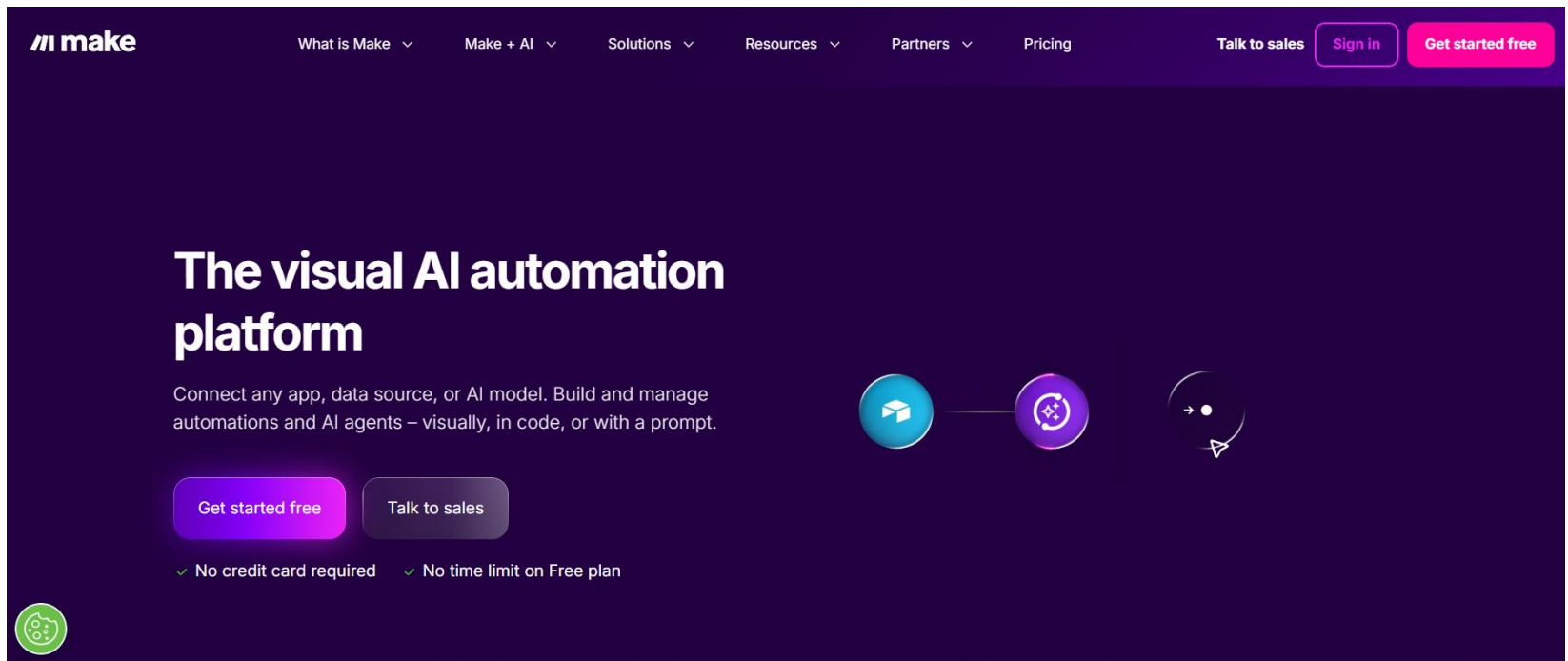
Simple enough to ship in hours, sophisticated enough to scale. n8n lets you automate business processes without limits on your logic.



<https://n8n.io/features/>

Practical Exercise - Process Automation (Make.com)

Go to <https://www.make.com/en>



Click “Get started free” or enter <https://www.make.com/en/login>

Practical Exercise - Process Automation (Make.com)

Sign in with Google

My Organization

Organization settings + Create a new scenario

ORGANIZATION TEAMS USERS SUBSCRIPTION PAYMENTS INSTALLED APPS VARIABLES SCENARIO PROPERTIES

Dashboard

Current plan - Free	
Price	\$0.00
Operations/mo	1,000
Unused operations	998
Billing interval	Monthly

Upgrade

Usage reset
2025年8月5日 11:36:41

Active scenarios 0/2

Operations ~2/1,000 (0% used)

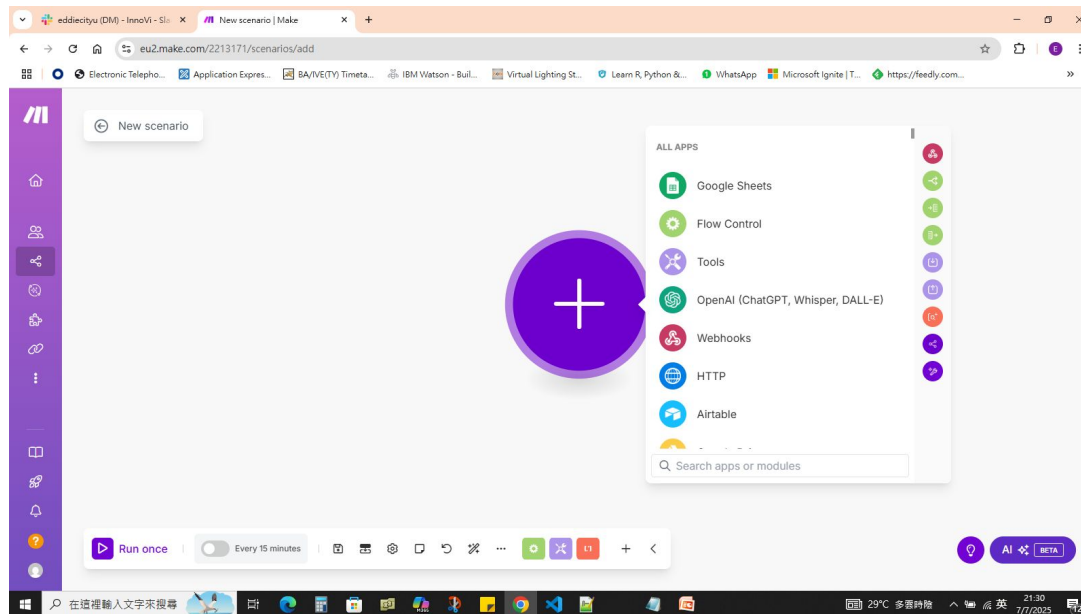
Data transfer ~592.0 B/512.0 MB (0% used)

Search for help

Click “Create a new scenario”

Practical Exercise - Process Automation (Make.com)

An interface with module is displayed

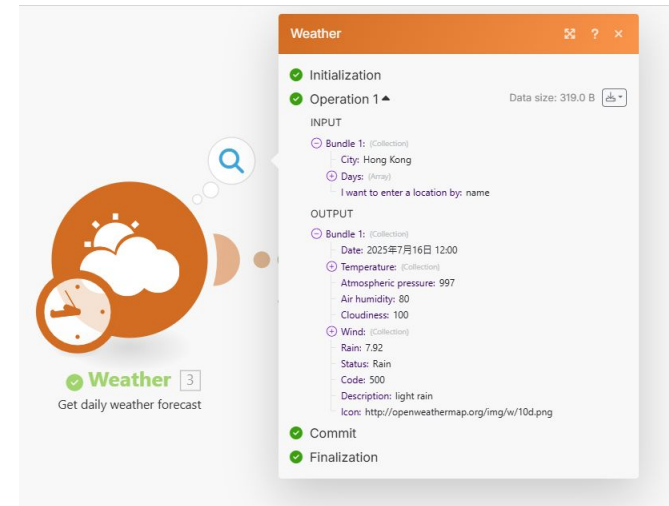
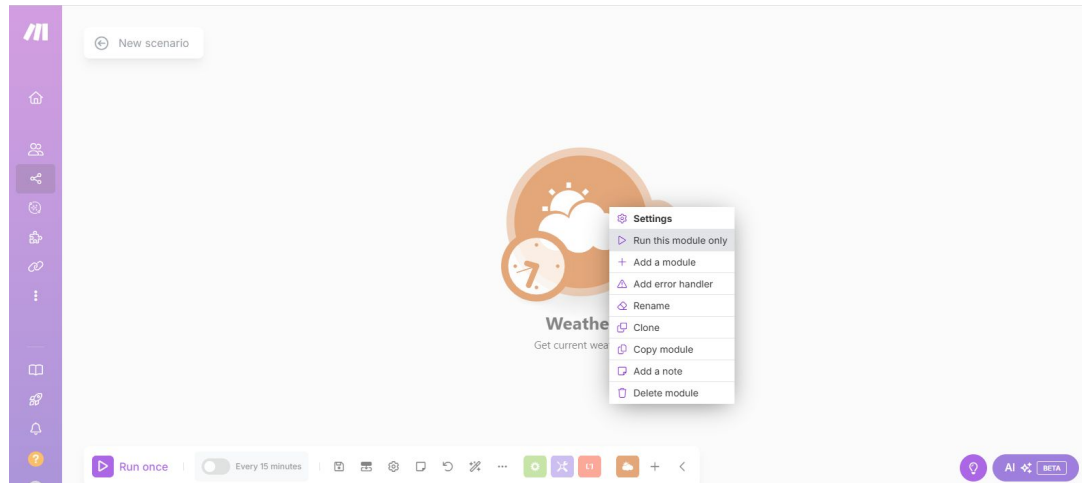


Create Module “Weather” -> Click “Get daily weather forecast”

Practical Exercise - Process Automation (Make.com)

Enter Days “9 days from today” and City “Hong Kong”
-> Click “Save”

Right-Click and select “Run module”



You can view an output file in JSON format

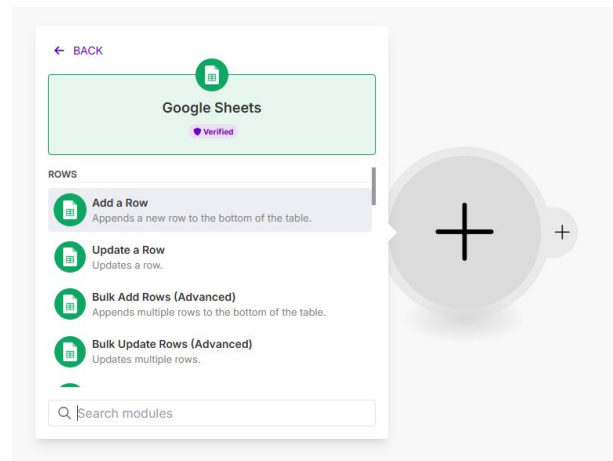
Practical Exercise - Process Automation (Make.com)

Open drive.google.com -> create an google sheet named "Weather Data"

Create an header as follow:

Date	Temperature	Air humidity	Wind	Rain
------	-------------	--------------	------	------

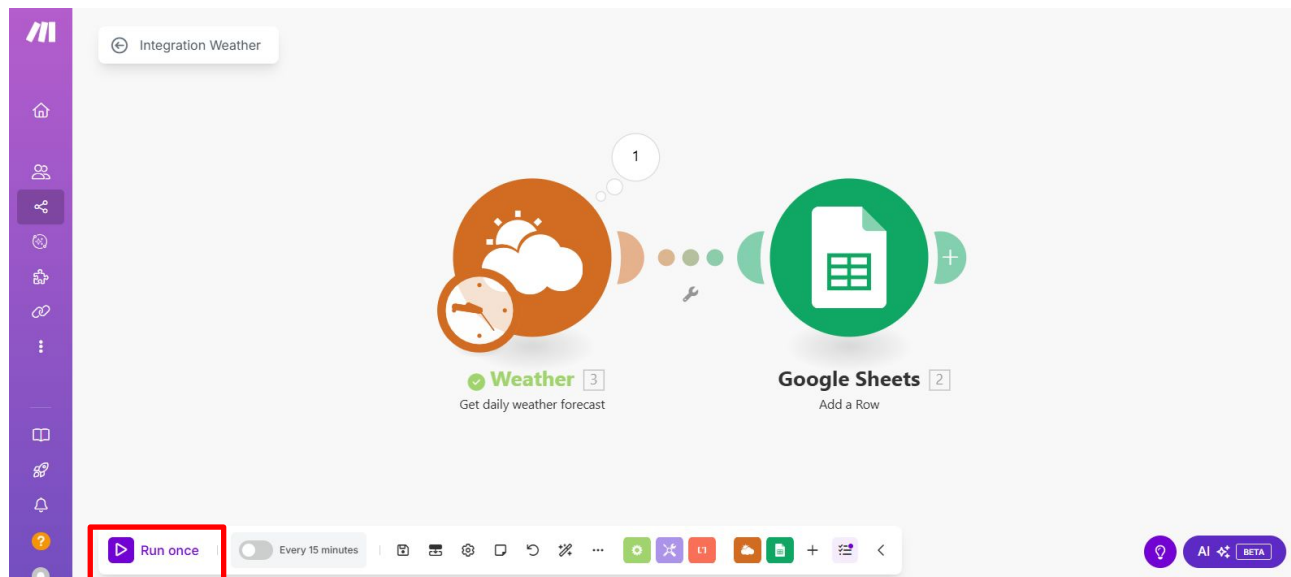
Return to make interface, in the right side of Weather icon, Click "Add another module" -> create "Google Sheets" -> Select "Add a Row"



Automation for updating weather data in google sheet

In Spreadsheet ID, click button “Click here to choose file”

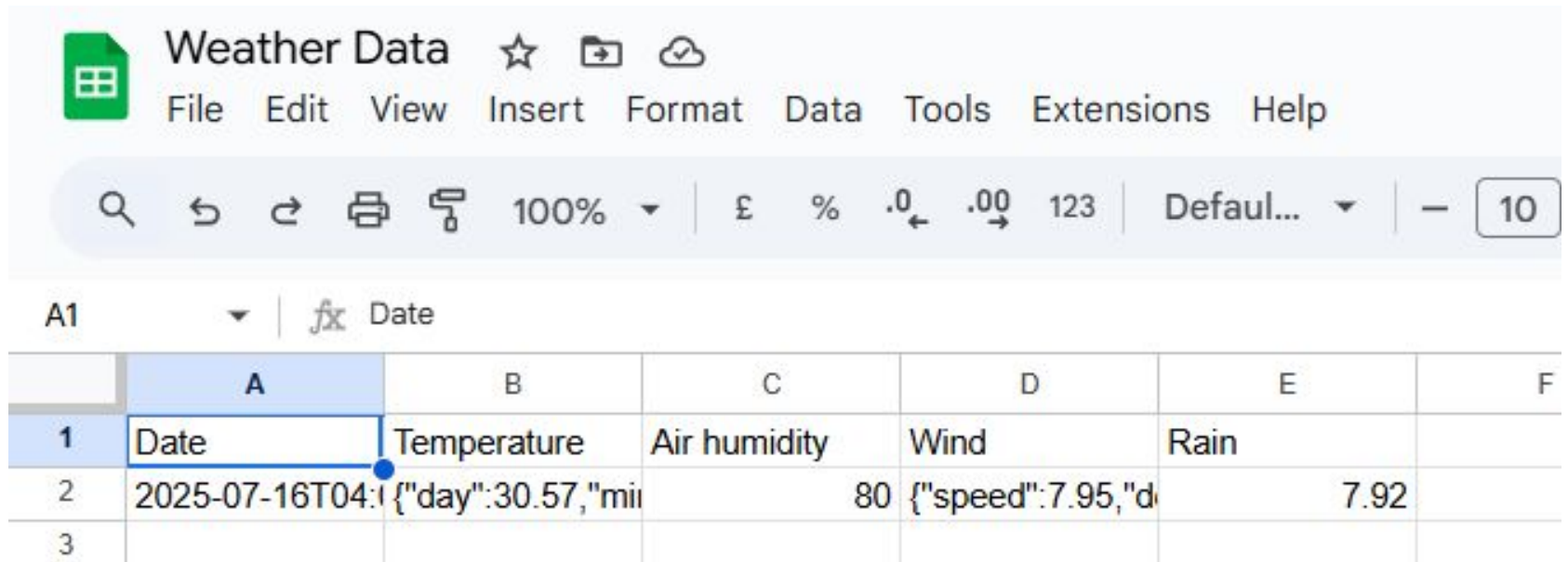
1. Select “Weather Data” -> Enter Sheet Name “Sheet1”
2. Click Date (A), select “3. Date” and other 4 columns
3. Click “Save”



In the button, click “Run once”

Automation for updating weather data in google sheet

Open the google sheet named “Weather Data”, the data is inserted in the second row as following:



The screenshot shows a Google Sheet titled "Weather Data". The interface includes a menu bar (File, Edit, View, Insert, Format, Data, Tools, Extensions, Help) and a toolbar with various icons and settings. The spreadsheet has columns labeled A through F. Row 1 contains headers: "Date", "Temperature", "Air humidity", "Wind", and "Rain". Row 2 contains data: "2025-07-16T04:10:00Z", "30.57", "80", and "7.92". The data in row 2 is truncated in the original image.

	A	B	C	D	E	F
1	Date	Temperature	Air humidity	Wind	Rain	
2	2025-07-16T04:10:00Z	30.57	80	7.92		
3						

Introduction to Business Process Automation

Types of Business Automation – Advanced Automation

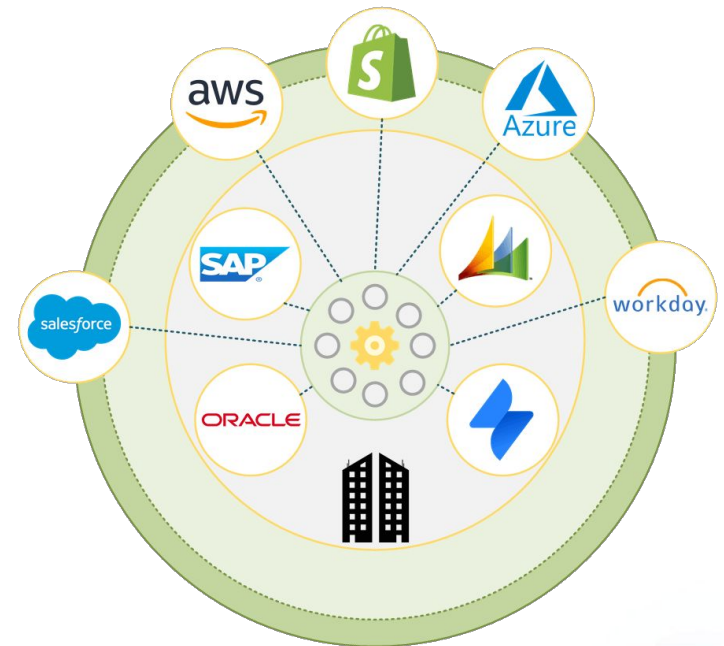
- **Target** → Complex process: Multiple systems + Human
- **Why**
 - Handles unstructured data / methods of access
- **Example** → Machine learning, natural language processing



Advanced Automation : System Integration



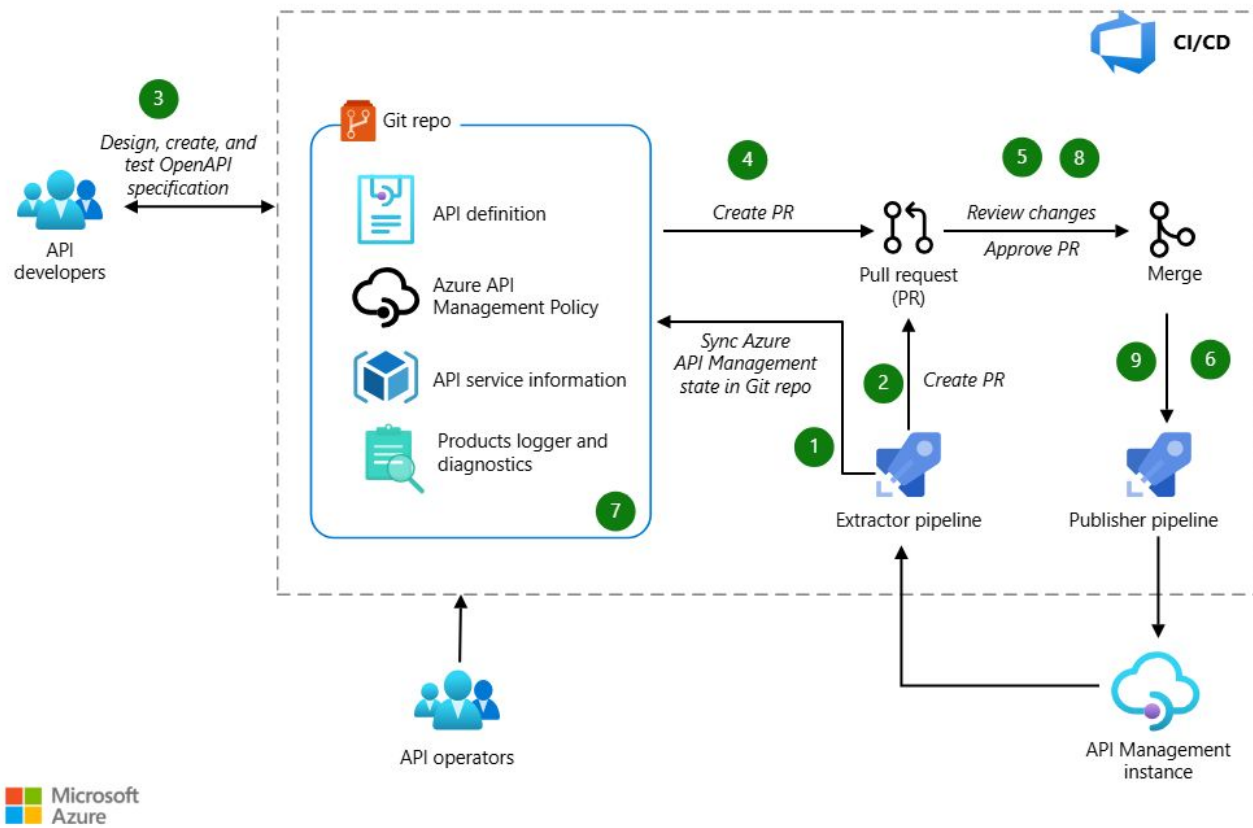
<https://fitsmallbusiness.com/crm-integration/>



<https://iwconnect.com/making-multiple-systems-work-as-one-through-integration/>

Advanced Automation : System Integration

<https://learn.microsoft.com/en-us/azure/architecture/example-scenario/devops/automated-api-deployments-apiops>



Introduction to Business Process Automation

Types of Business Automation - Intelligent Automation

- **Target** → Processes requiring AI/customized decisions
- **Why**
 - Smarter interactions, personalization
- **Example** → Voice-based interfaces



Introduction to Business Process Automation

Intelligent Process Automation (IPA)

- Using Artificial Intelligence (AI), e.g.,
 - Machine Learning (ML)
 - Natural Language Processing (NLP)
- Enhancing cognitive ability of the automation



Introduction to Business Process Automation

Intelligent Process Automation (IPA)

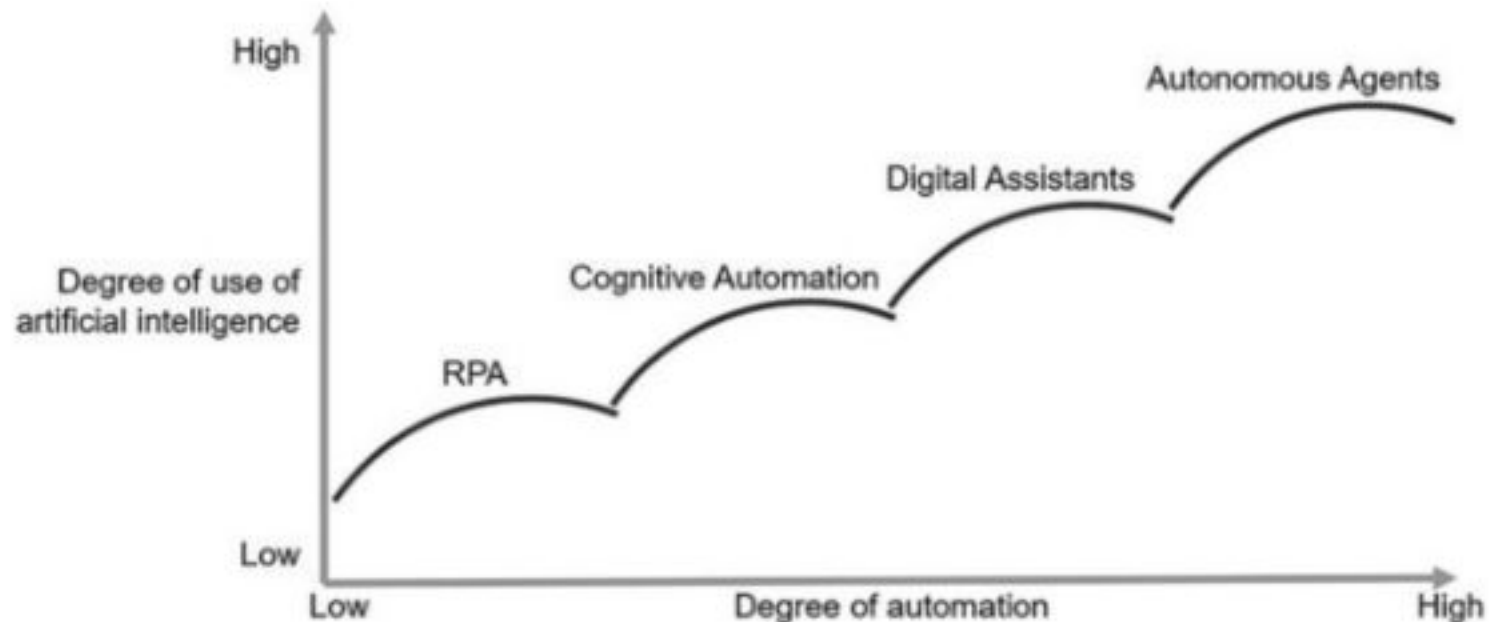
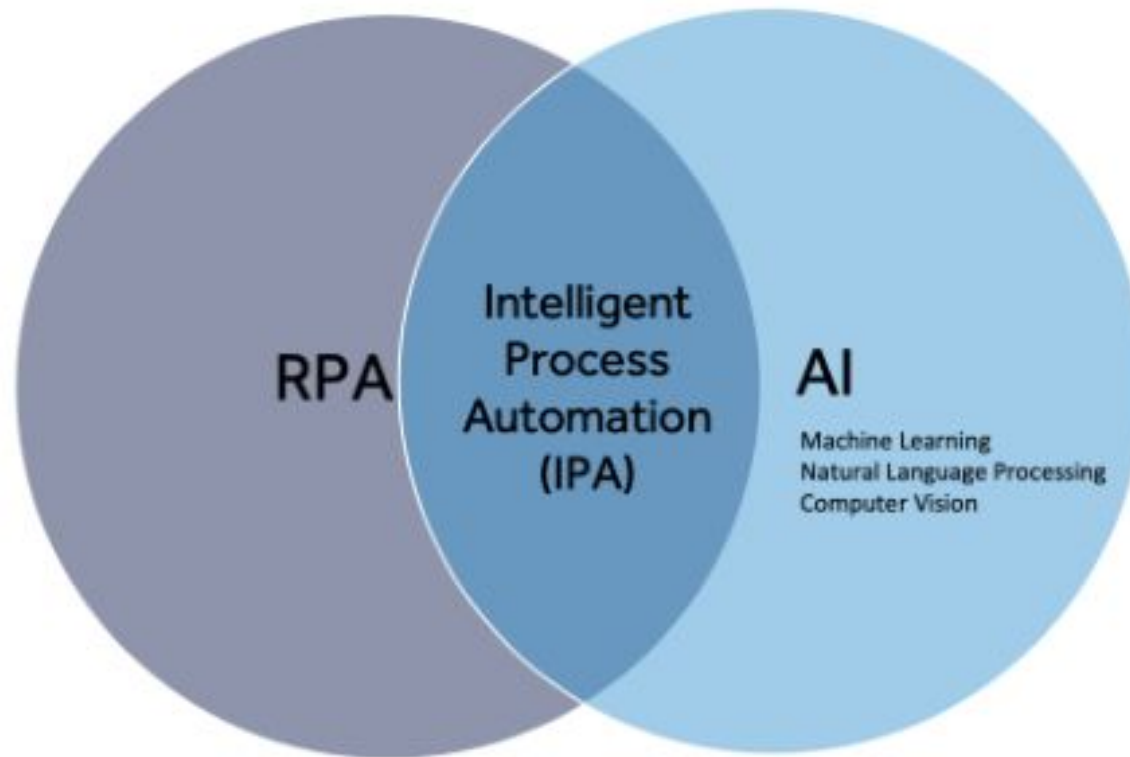


Fig. 2.2 Classification of RPA according to the degree of automation and the use of artificial intelligence (based on Ostrowicz 2018, p. 4)

1. Smeets, M., Erhard, R., & Kaußler, T. (2021, July 30). *Robotic Process Automation (RPA) in the Financial Sector: Technology - Implementation - Success for Decision Makers and Users*. Springer. <https://doi.org/10.1007/978-3-658-32974-7>

Introduction to Business Process Automation

Intelligent Process Automation (IPA)



1. Kosmopoulos, C. (2021, May 11). *What is Intelligent Process Automation (IPA)?* | Blueprint. What Is Intelligent Process Automation (IPA)? | Blueprint. <https://www.blueprintsys.com/blog/rpa/what-is-intelligent-process-automation-ipa>

Introduction to Business Process Automation

Types of Business Automation - Checkpoint

1. Daily email template to enter Market commentary
2. HR time off workflow
3. Online supermarket order processing
4. Voice-automated credit card hotline



Intelligence Automation Software Example : Microsoft Power Automate

Link - microsoft.com/en/power-platform/products/power-automate

← Back Every week, post a message in a chat or channel. Save Test Copilot New designer

»

Recurrence

Post message in a chat or channel

1 of 3 Next

Every week, post a message in a chat or channel.

Connected to **Microsoft Teams**

Ask a question or describe how you want to change this flow

0/2000

Make sure AI-generated content is accurate and appropriate before using.

<https://learn.microsoft.com/en-us/power-automate/create-cloud-flow-using-copilot#create-a-cloud-flow-using-copilot>



What is Process Automation, RPA, IPA?
Overview of the technological building blocks related to process automation
Contemporary tools for process automation
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Introduction to Business Process Automation

Tools for Automation - Overview

Which tool is being used ?

As some companies use more than one tool, the percentages in this graphic total to more than 100%

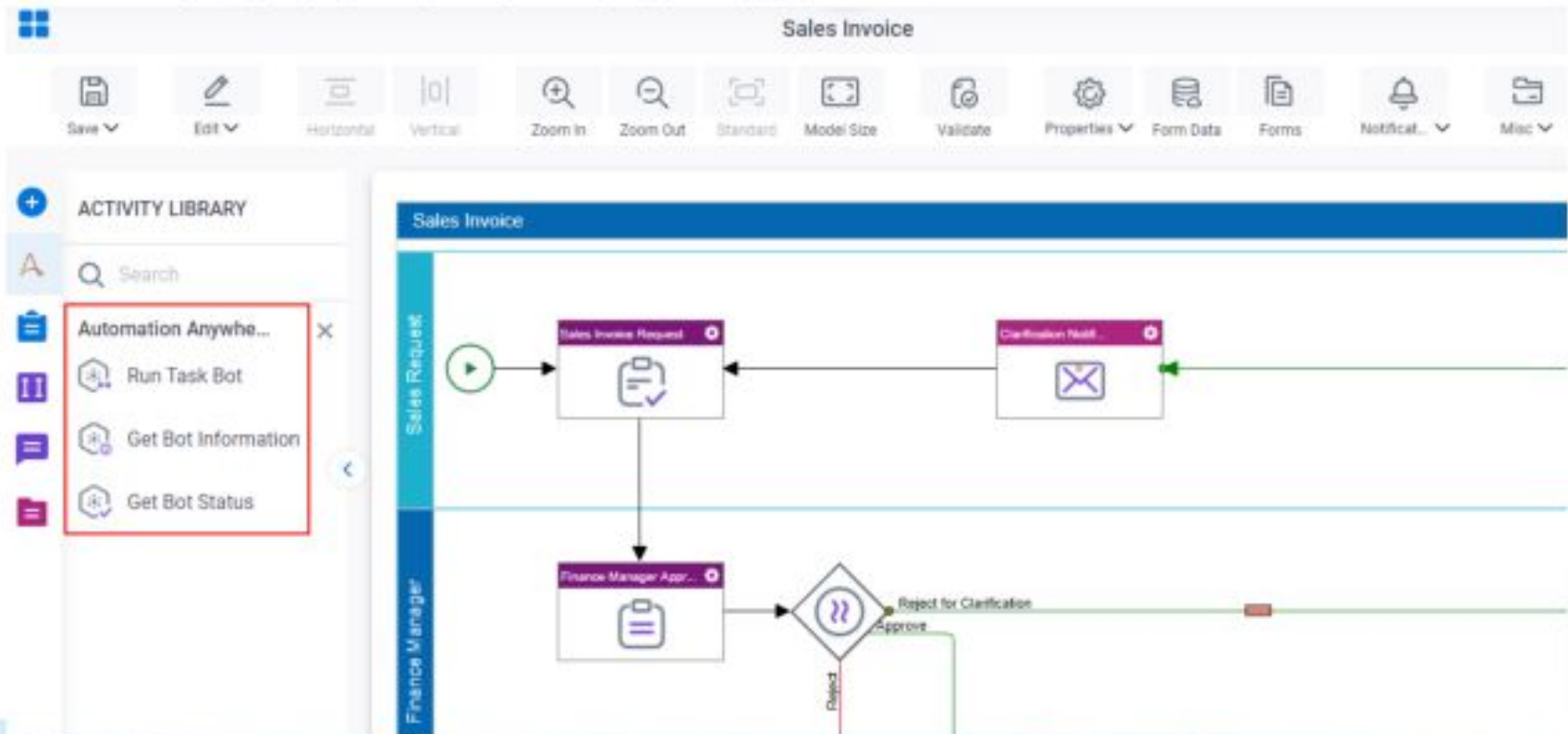


1. 2019 actuarial robotic process automation (RPA) survey report. PwC.
<https://www.pwc.com/gx/en/industries/financial-services/publications/pwc-2019-actuarial-robotic-process-automation-survey-report.html>

Introduction to Business Process Automation

Tools for Automation - Types

- **Low-code platforms**
 - Automation Anywhere, XCEPTOR, Microsoft Power Automate



Introduction to Business Process Automation

Tools for Automation - Types

- **In-house Solutions**
 - Excel VBA, Java, Python, Web

Microsoft®
Excel VBA



Most in-demand functional areas

Compliance
Operational Risk
Fund Accounting
Corporate Accounting
Actuarial
Finance Operations
IT Audit



In-demand technical skills and experience

IFRS 17 Reporting and Accounting Policy
RBC and Capital Reporting
IT skills (Python, C+ BASIC, C, C++, COBOL, Java, R, Tableau, SQL, VBA, and Macro)
Operational Risk, Information and IT Risk, Risk Transformation, Risk Analytics, Internal Control Framework



1. Robert Half. 2024 Salary Guide HK.
<https://content.roberthalfonline.com/SG24/SG24-PDF/2024-Salary-Guide-HK.pdf>

Introduction to Business Process Automation

Tools for Automation – Getting a taste

- Visualping (<https://visualping.io/>)
 - Websites monitoring



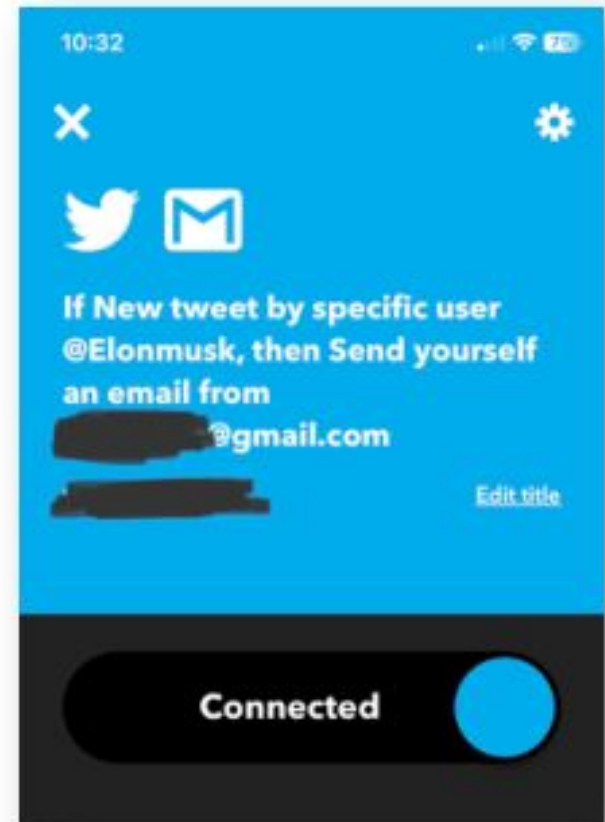
<https://hkuspace.hku.hk> » prog » cert-for-
Certificate for Module (Busin...
(2) Mr Jackie Liu. Mr. Jackie Liu is veter...
top global investment banks. With over 1...
an quantitative strategist currently working in one of the
0 years of ...

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<https://hkuspace.hku.hk> » prog » cert-for-module-busi...
Certificate for Module (Business Process Automation with VBA ...
(2) Mr Jackie Liu. Mr. Jackie Liu is veteran quantitative strategist currently working in one of the
top global investment banks. With over 10 years of ...

Introduction to Business Process Automation

Tools for Automation – Getting a taste

- IFTTT (<https://ifttt.com/>)
 - Web + App interfaces
 - Service-based triggers and actions



Tools for Automation – Considerations

- **Cost**

- **One-time cost**

- Initial effort of adaptation
 - Implementation and rollout

- **Running cost**

- License, e.g., Power Automate: \$Millions/year¹
 - Infrastructure, e.g., Cloud storage
 - Training, e.g., Automation Anywhere, 10+ hours²
 - Support, e.g., bot re-configuration

Cost		
Production	Gold	Maint.
60 🍌	240 🍌	1 🍌

1. RPA Pricing: Comparison of Leading RPA Vendors' Fees in 2023. <https://research.aimultiple.com/rpa-pricing/>

2. RPA Course Learning Trails | Automation Anywhere University. <https://university.automationanywhere.com/training/rpa-learning-trails/>

Introduction to Business Process Automation

Tools for Automation – Considerations

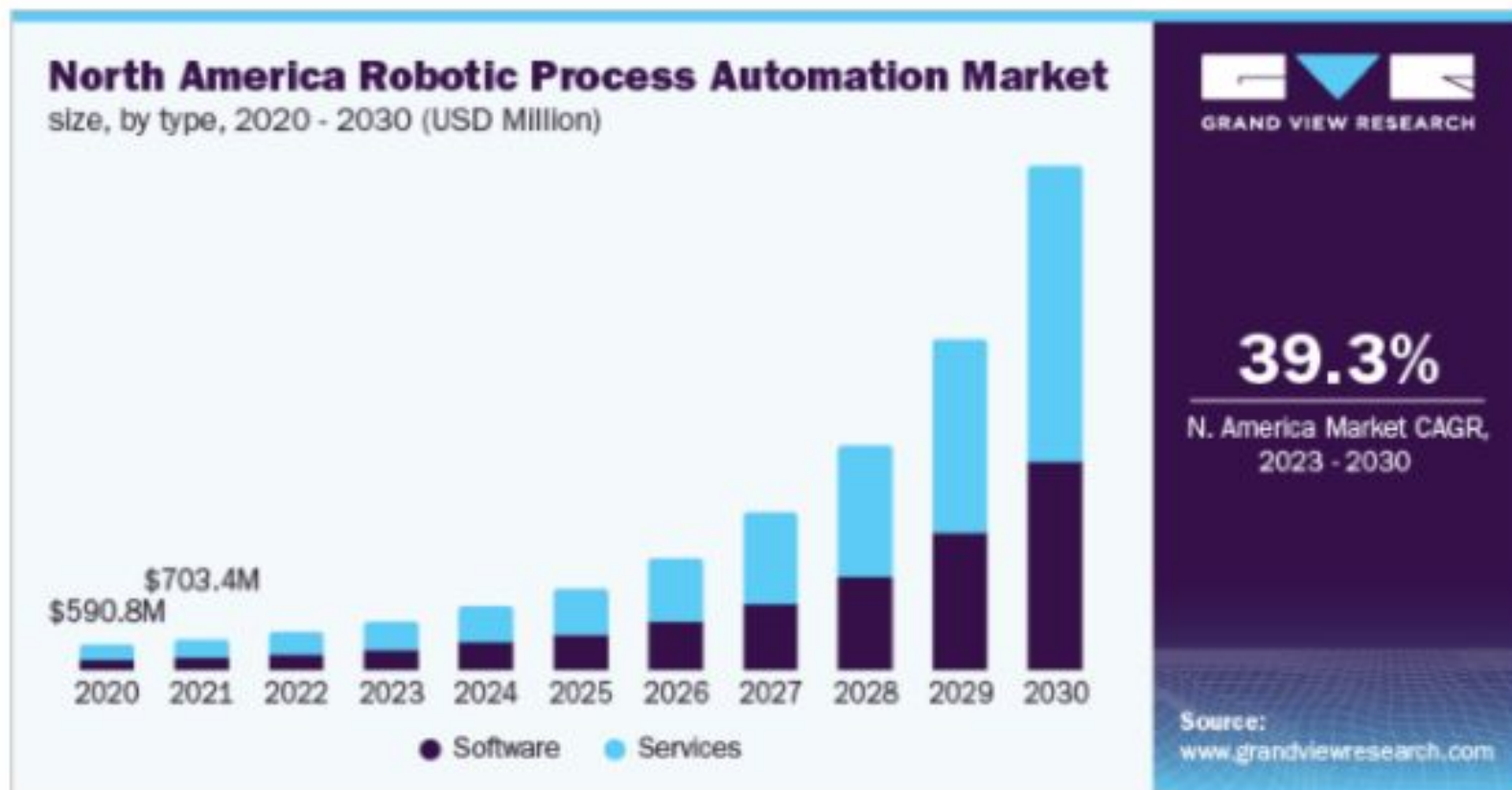
- **Flexibility**
 - Specialized vs Generalized, e.g., XCEPTOR → Tax
- **Scalability**
 - Desktop vs Enterprise, e.g., Excel VBA
 - Version Control, e.g., Use of repository
- **Time to delivery**
 - “Perfect” solution does not exist



1. Smeets, M., Erhard, R., & Kaußler, T. (2021, July 30). *Robotic Process Automation (RPA) in the Financial Sector: Technology - Implementation - Success for Decision Makers and Users*. Springer. <https://doi.org/10.1007/978-3-658-32974-7>

Introduction to Business Process Automation

Opportunities of BPA – The Money



1. Robotic Process Automation Market Size & Share Report 2030. Robotic Process Automation Market Size & Share Report 2030.
<https://www.grandviewresearch.com/industry-analysis/robotic-process-automation-rpa-market>

Introduction to Business Process Automation

Opportunities of BPA – Sectors

Industries	% of RPA Solutions
Banking, Financial Services & Insurance	51 %
Business Process Outsourcing (BPO)	14 %
Manufacturing – Consumer Packaged Goods (CPG)	7 %
Professional, Legal & Accountancy Services	7 %
Retail Trade	7 %
Technology (IT, Internet, SAAS)	7 %
Utilities	7 %

1. Top 15 RPA Use Cases & Examples in Banking in 2023. <https://research.aimultiple.com/banking-rpa/>

Opportunities of BPA – Sectors

- **Healthcare**
 - Monitoring, drug dispensing
- **Retail**
 - Auto-checkout, smart shelf, inventory management
- **Marketing**
 - Lead generation
 - social media management
- **Finance**
 - Fraud detection, Credit management, Regtech



Introduction to Business Process Automation

Opportunities of BPA – Regtech

- Follow the money

CNN

Wells Fargo ordered to pay \$3.7 billion for 'illegal activity' including unjust foreclosures and vehicle repossessions

The CFPB said the more than \$2 billion in customer refunds Wells Fargo has been ordered to pay includes more than \$1.3 billion to consumers hurt...

20 Dec 2022



Spiceworks

SEC Penalizes Major Wall Street Firms \$1.97B For Using Unauthorized Messaging Apps

If there are allegations of wrongdoing or misconduct, we must be ... Morgan Stanley recently agreed to a \$35 million fine by the SEC for...

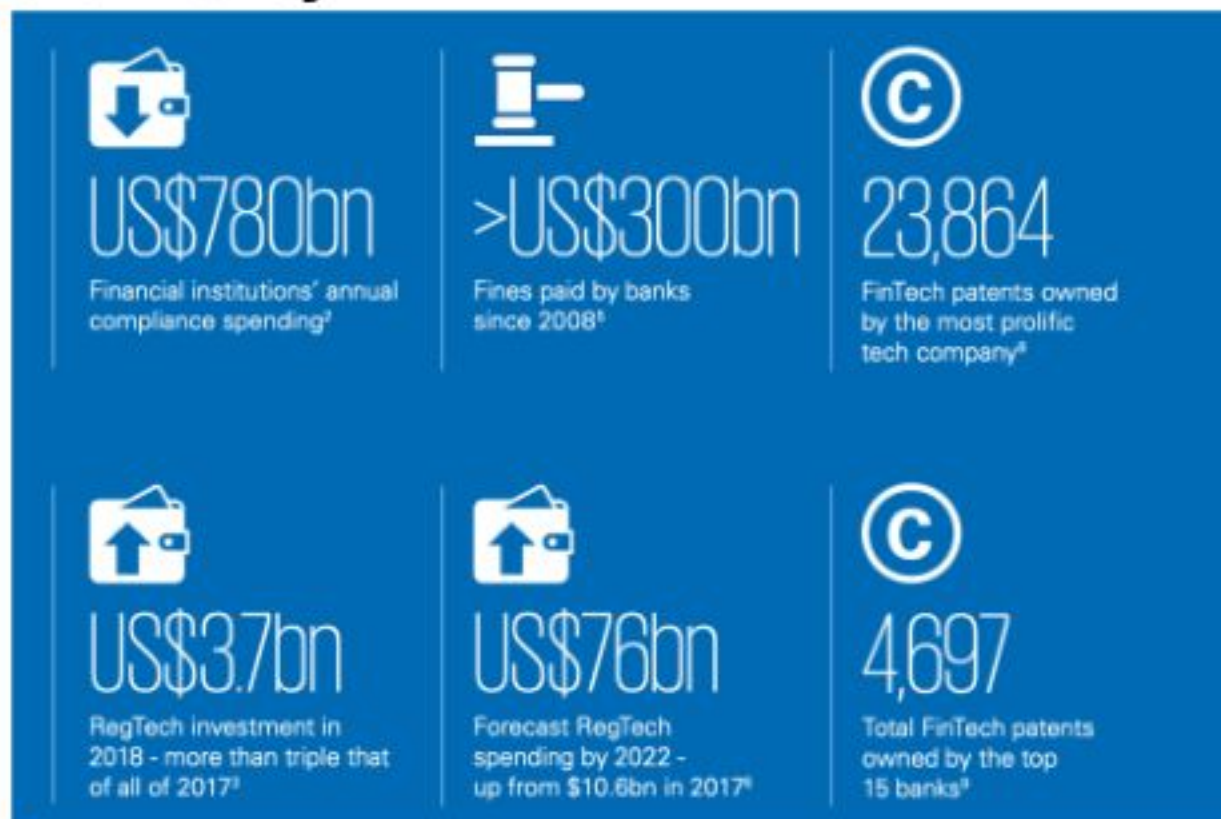
28 Sept 2022



Introduction to Business Process Automation

Opportunities of BPA – Regtech

- Follow the money

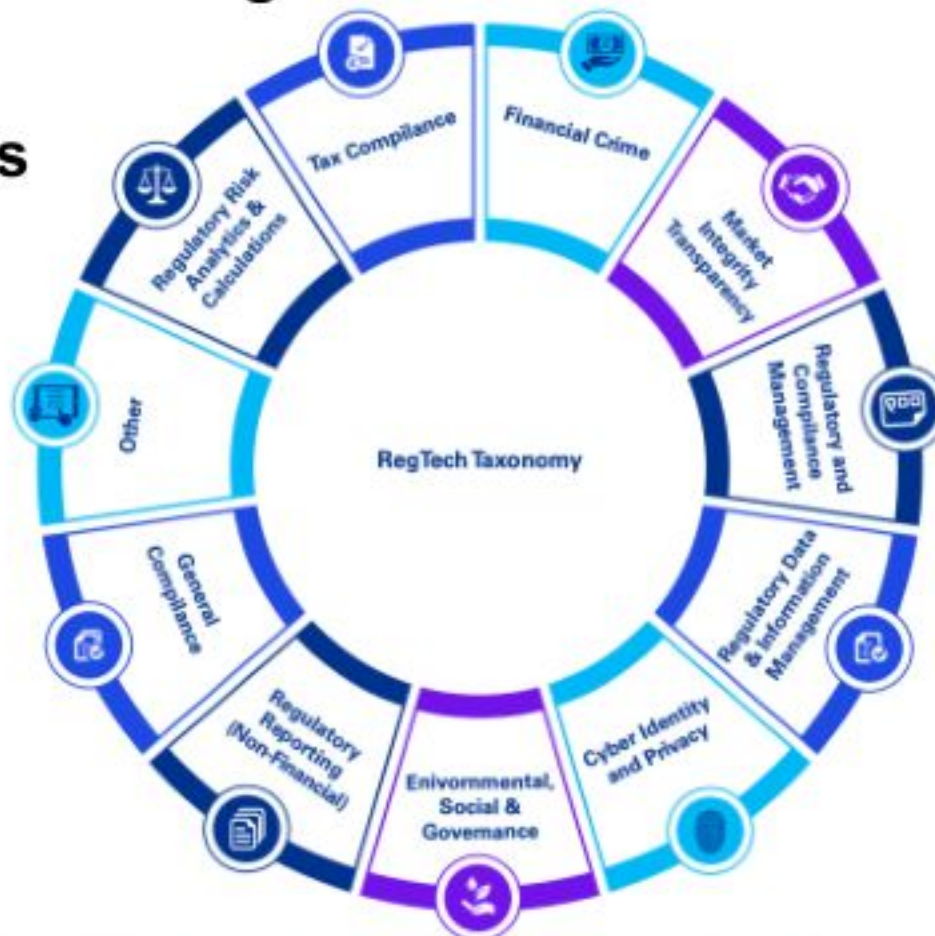


1. KPMG. (2019, June). *There's a revolution coming.*
<https://assets.kpmg.com/content/dam/kpmg/cn/pdf/en/2019/06/embracing-the-challenge-of-the-new-regtech-era.pdf>

Introduction to Business Process Automation

Opportunities of BPA – Regtech

- Common themes



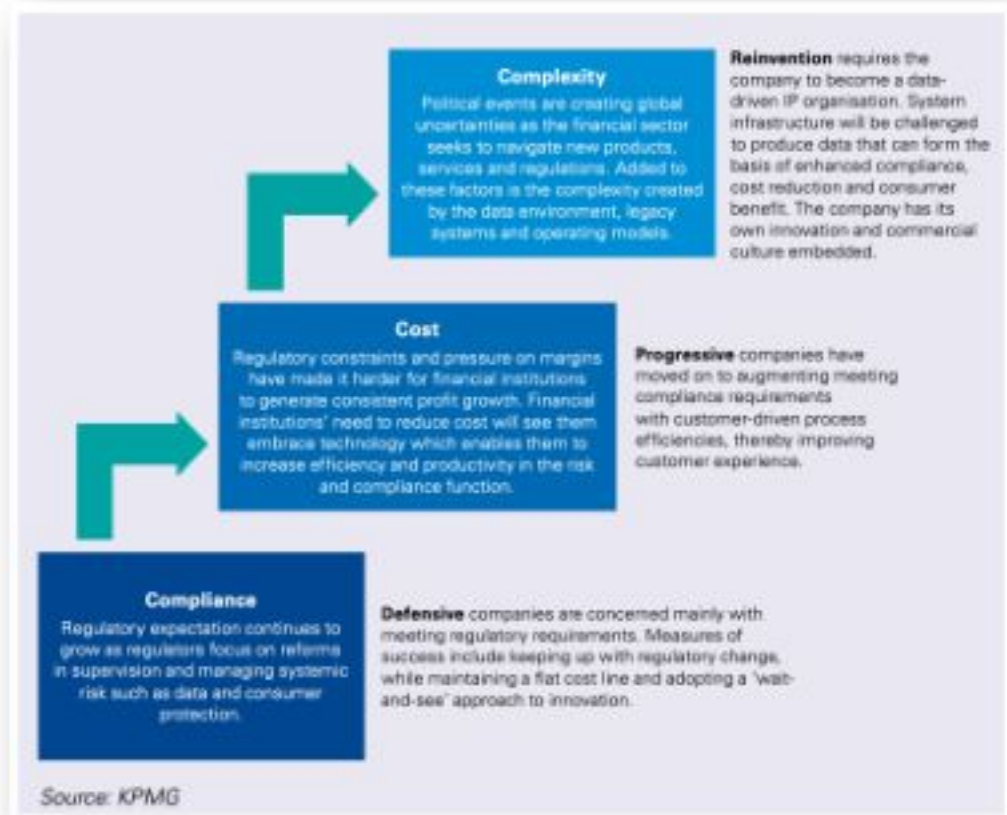
1. KPMG. (2022, November). <https://assets.kpmg.com/content/dam/kpmg/uk/pdf/2022/11/innovate-finance-regtech-industry-and-adoption.pdf>.
<https://assets.kpmg.com/content/dam/kpmg/uk/pdf/2022/11/innovate-finance-regtech-industry-and-adoption.pdf>

Introduction to Business Process Automation

Opportunities of BPA – Regtech

Adoption phases:

1. Defensive
2. Progressive
3. Reinvention



1. KPMG. (2019, June). *There's a revolution coming*.
<https://assets.kpmg.com/content/dam/kpmg/cn/pdf/en/2019/06/embracing-the-challenge-of-the-new-regtech-era.pdf>

Introduction to Business Process Automation

Opportunities of BPA – Regtech

- **HKMA 3-Year Roadmap**
 - **RPA:** automating rule-based processes
 - **Network Analysis:** Company holdings and exposures



1. HKMA. (2021, June). *Regtech Watch Issue No. 7*.
<https://www.hkma.gov.hk/media/eng/doc/key-information/guidelines-and-circular/2021/20210617e1a1.pdf>

Opportunities of BPA – Regtech

- **HKMA 3-Year Roadmap**
 - **Speech-to-text (STT):** Audio to time-stamped text
 - **Sentiment Analysis:** Realtime news monitoring



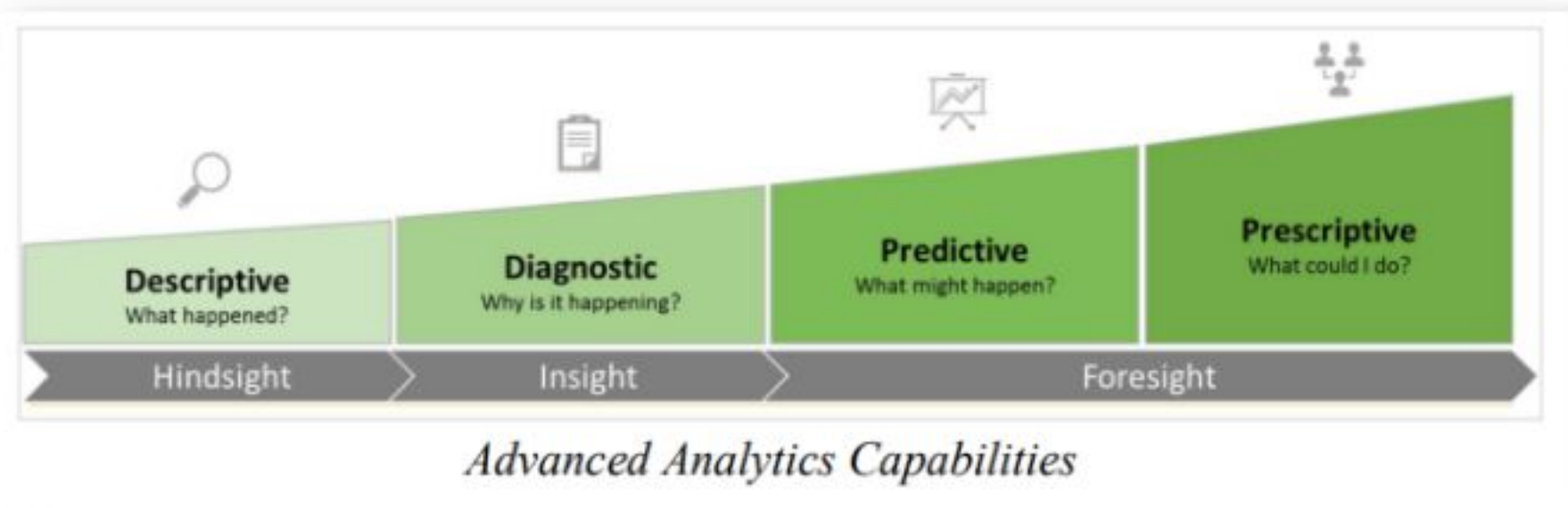
Illustrative Sentiment Analysis Heat Map

1. HKMA. (2021, June). *Regtech Watch Issue No. 7*.
<https://www.hkma.gov.hk/media/eng/doc/key-information/guidelines-and-circular/2021/20210617e1a1.pdf>

Introduction to Business Process Automation

Opportunities of BPA – Regtech

- **Analytics enabled by BPA**



1. HKMA. (2021, June). *Regtech Watch Issue No. 7*.
<https://www.hkma.gov.hk/media/eng/doc/key-information/guidelines-and-circular/2021/20210617e1a1.pdf>

Introduction to Business Process Automation

Challenges of BPA – Insight from the banks

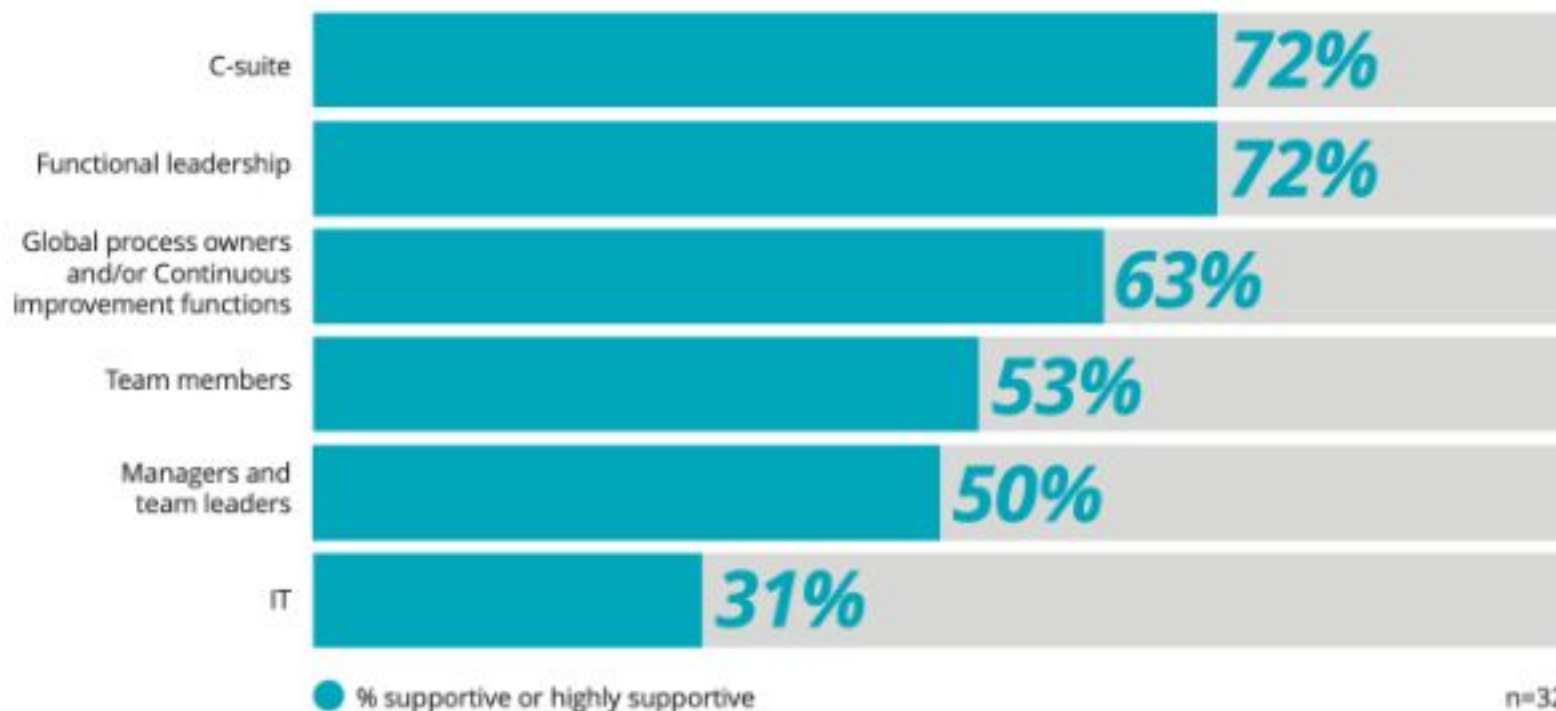


1. KPMG & HKMA. (2020). *Transforming Risk Management and Compliance: Harnessing the Power of Regtech*. <https://www.hkma.gov.hk/media/chi/doc/key-information/press-release/2020/20201102c3a1.pdf>

Introduction to Business Process Automation

Challenges of BPA – Worries from the workers?

Figure 5: How supportive of the RPA implementation were your stakeholder groups?

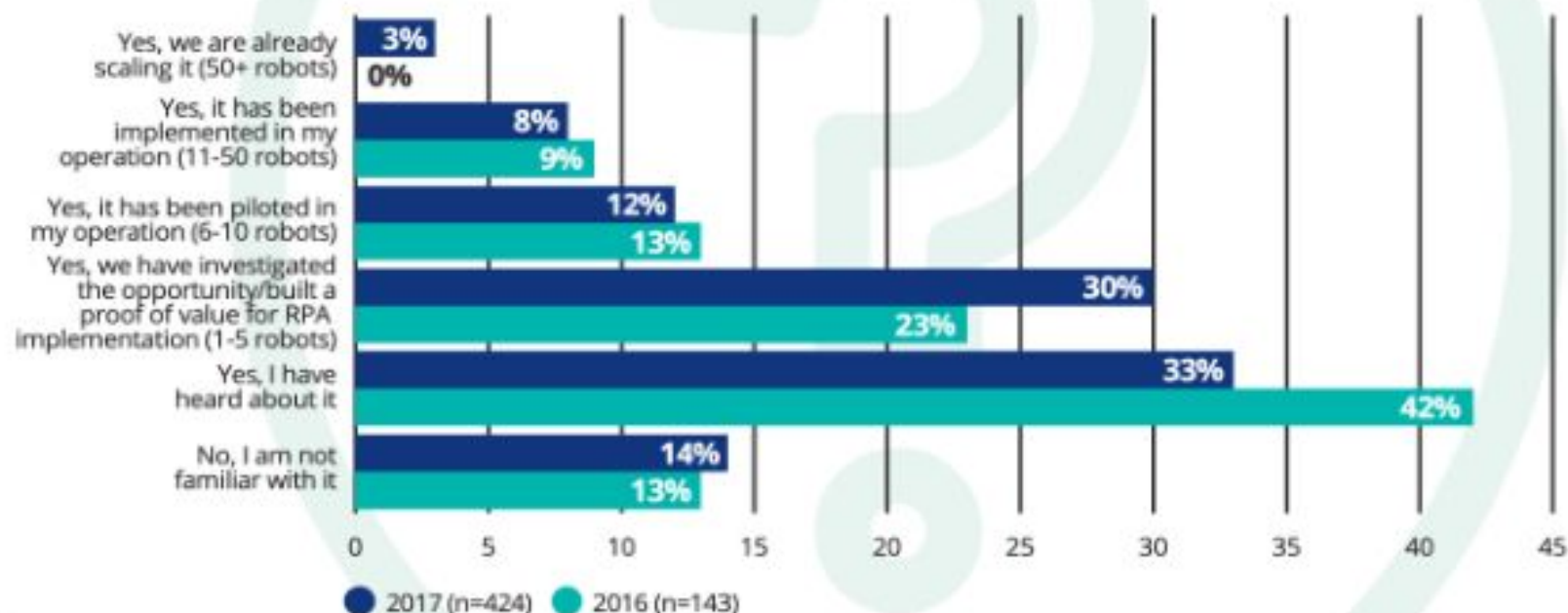


1. Deloitte. (2018). *The robots are ready. Are you?*
<https://www2.deloitte.com/content/dam/Deloitte/tr/Documents/technology/deloitte-robots-are-ready.pdf>

Introduction to Business Process Automation

Challenges of BPA – Hesitant to scale up

Figure 2: Are you familiar with RPA?



1. Deloitte. (2018). *The robots are ready. Are you?*
<https://www2.deloitte.com/content/dam/Deloitte/tr/Documents/technology/deloitte-robots-are-ready.pdf>

Introduction to Business Process Automation

Challenges of BPA

- **Shortage**
 - Budget / Resource/ Talent
- **Concerns**
 - Risk of rushing / failing
 - Moral / Structural unemployment

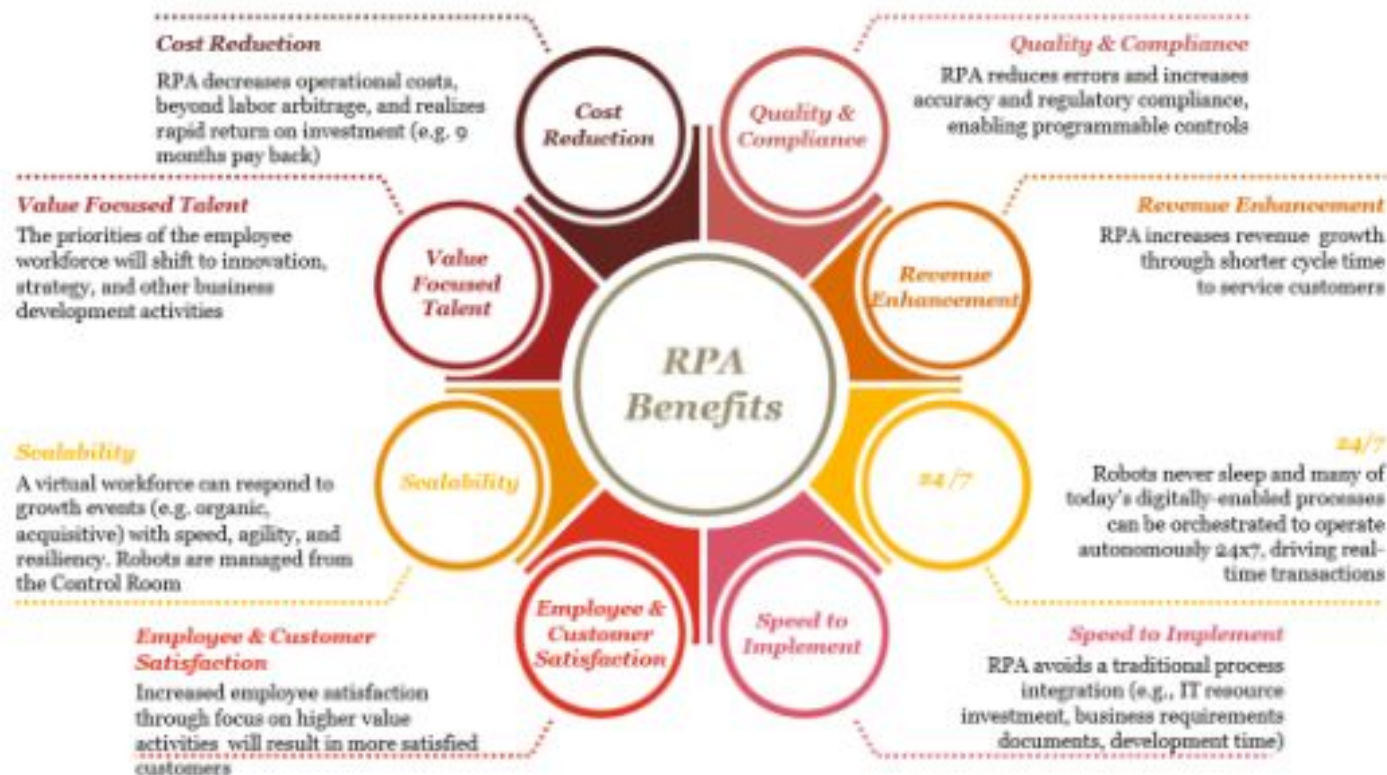


* Success depends on planning
(More on project management later)

Introduction to Business Process Automation

Implications of BPA – The Good

- Cost, Efficiency, Customer Experience



1. PwC Robotics Process Automation Solutions. PwC.
<https://www.pwchk.com/en/services/entrepreneurial-and-private-business/new-technology-digitalisation-and-transformation/process-automation-solutions.html>

Introduction to Business Process Automation

Implications of BPA – The Bad

- Automation complacency / dependency
- Risk of Technical Issues / Downtime

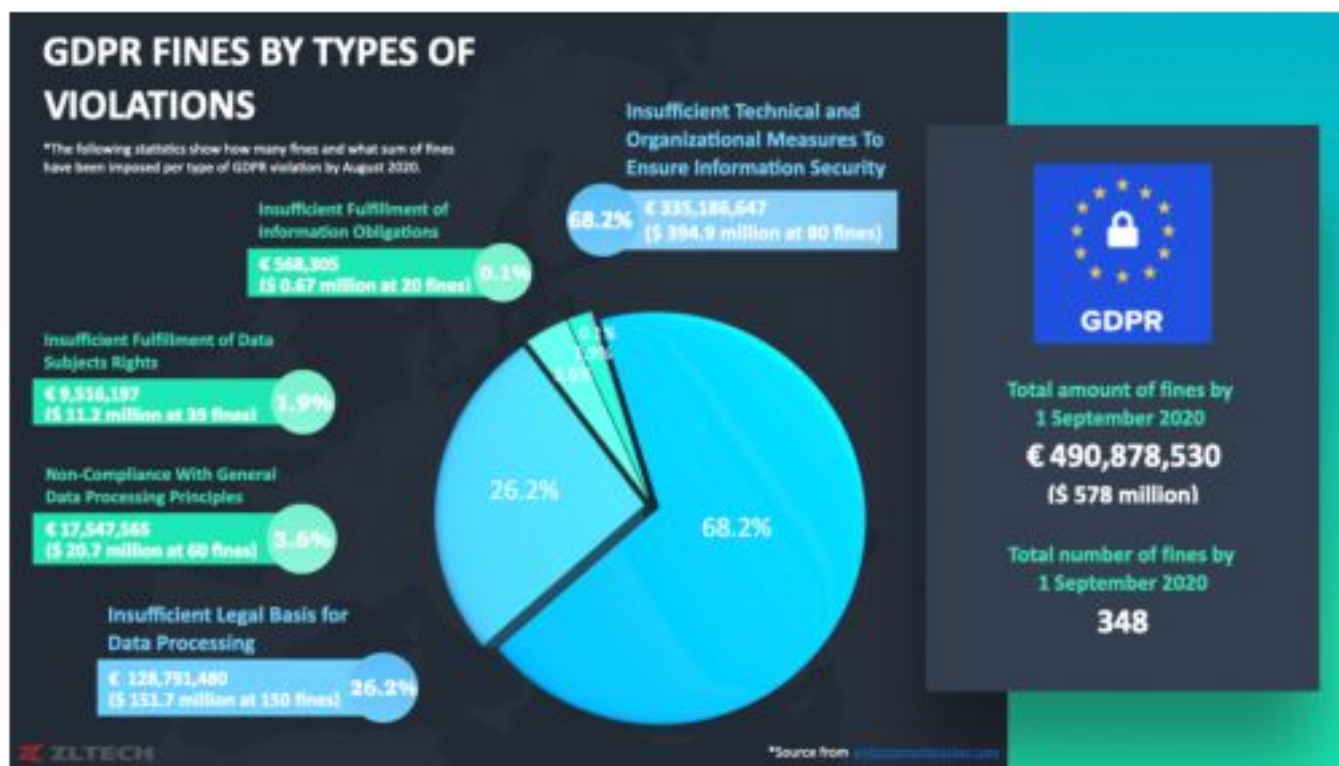


Image: Civil Aviation Safety Authority

Introduction to Business Process Automation

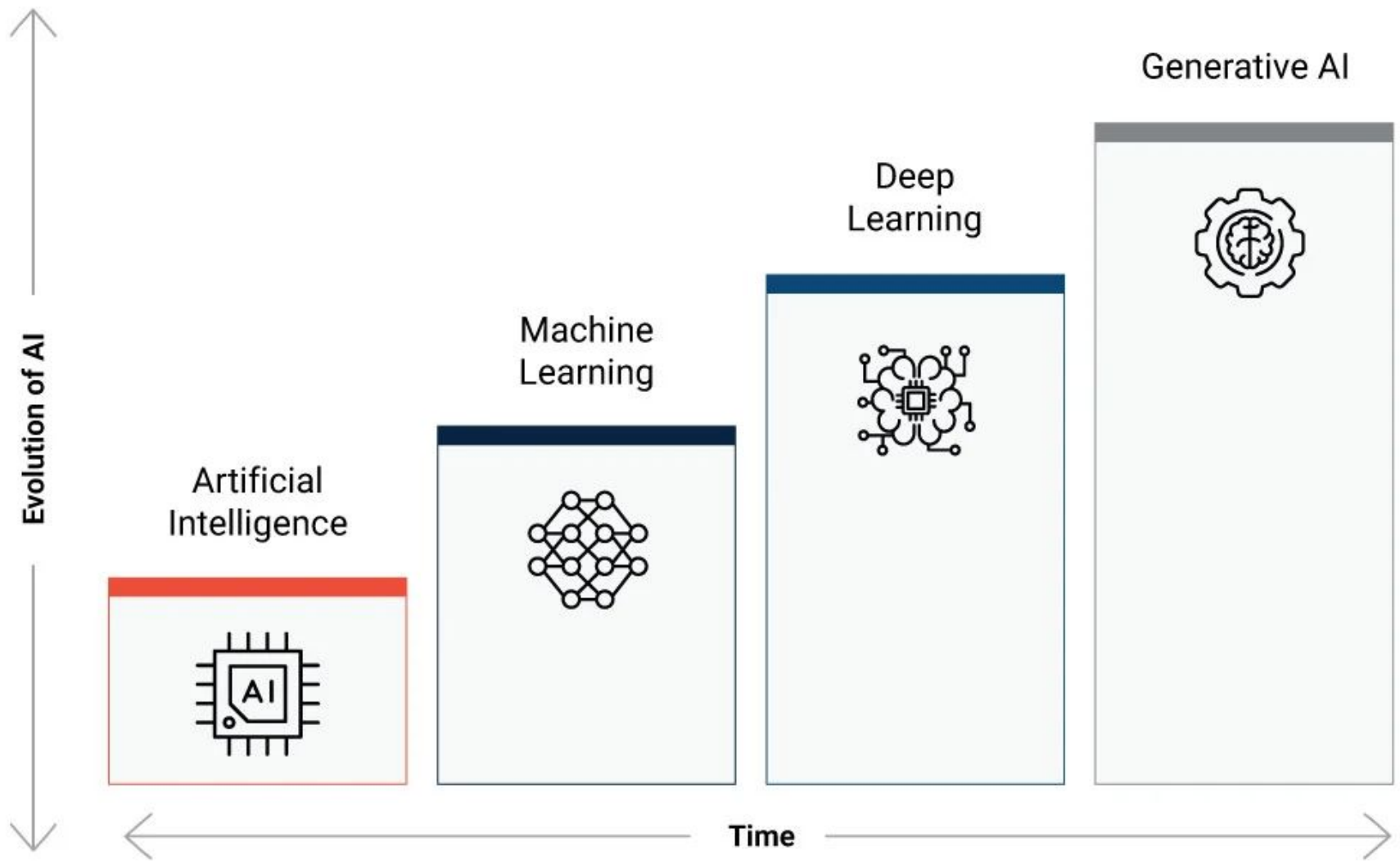
Implications of BPA – The Ugly

- Data Collection and Analysis ¹
- Vulnerability to Cyberattacks



1. Chen, B. (2020, September 9). *GDPR Recap: 28 Nations, 348 Fines, Half a Billion Euros*. ZL Tech.
<https://www.zltech.com/blog/gdpr-recap-28-nations-343-fines-half-a-billion-euros/>

Overall Artificial Intelligence Journey



Machine Learning Business Use Case

Hotel Recommendation Engine



♥

Boston Harbor Hotel

★★★★★

Downtown Boston
Come Experience our 5 Star Luxury Hotel.

Iconic Waterfront Hotel with a Convenient Location to Everything Boston Has to Offer. Every Room Boasts City or Harbor Views.

1-866-286-0843 • Expedia Rate

👁 Viewed

4.8/5 Exceptional!
(1,132 reviews)

~~\$693~~ **\$581**

nightly price
Sponsored

 **Get member price**

People who looked at the Boston Harbor Hotel also viewed these:



Four Seasons Hotel Boston

★★★★★

\$645



The Ritz-Carlton, Boston

★★★★★

\$595

Machine Learning, Deep Learning Features in Retail

- Chatbots, Virtual Assistants
- Facial Recognition
- Demographic Segmentation
- Facial Personality Analytics
- AI-enabled Customer Analytics
- Real-time insights
- Personalization
- Inventory Management
- Visual Recognition
- Customer Purchase Prediction
- Trend Prediction
- Purchasing Recommendation
- Ideal Price Point Recommendation
- Predict the trends in Fashion, e-commerce, etc
- Predictive Analysis
- Predictive Personalization

Deep Learning Learning Business Use Case

Face Recognition Algorithm



Cases : Facial recognition for customer analytics



Face Detection

Reliably detect human faces in a photo, along with their coordinates and sizes.



Emotions, Gender, Age

Recognize people's facial expressions and detects primary and secondary emotions. Also detects age and gender

<https://findface.pro/en/solutions/face-recognition-for-customer-analytics.html>

Shifting mindsets of customers: From Products To Relationships



**Products
1970s**



**Customer
Centric**



**Relationship
Centric Today**

These companies applied analytics to stay 1 step ahead of the competition



Business Case Studies: Deep Learning

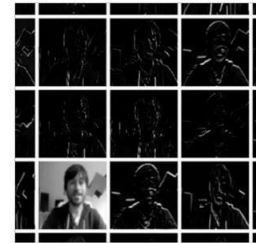
Artificial Intelligence

Common Use Cases:

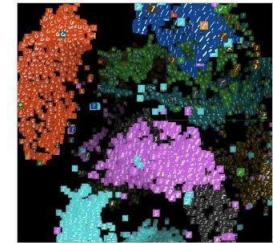
Object Detection
Speech Recognition
Natural Language Processing
Translation between languages
Creativity - Style Transfer
Art Restoration



TEACHABLE MACHINE



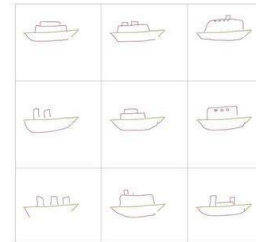
WHAT NEURAL NETWORKS SEE



VISUALIZING HIGH-DIMENSIONAL SPACE



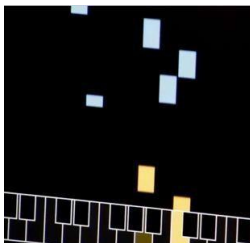
QUICK, DRAW!



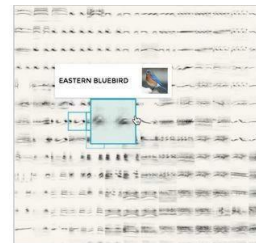
SKETCH-RNN DEMOS



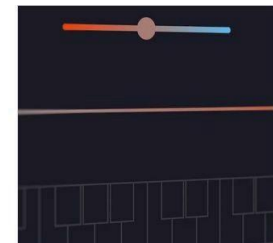
HANDWRITING WITH A NEURAL NET



AI DUET



BIRD SOUNDS



NSYNTH: SOUND MAKER

What is NLP?

Natural Language Processing (NLP) is a multidisciplinary field combining computer science, linguistics, and AI.

It enables computers to perform tasks like:

- Analyzing text for deeper meaning
- Transcribing spoken communication
- Automating complex language interactions
- Extracting insights from unstructured data



Deep Learning : NLP in real life

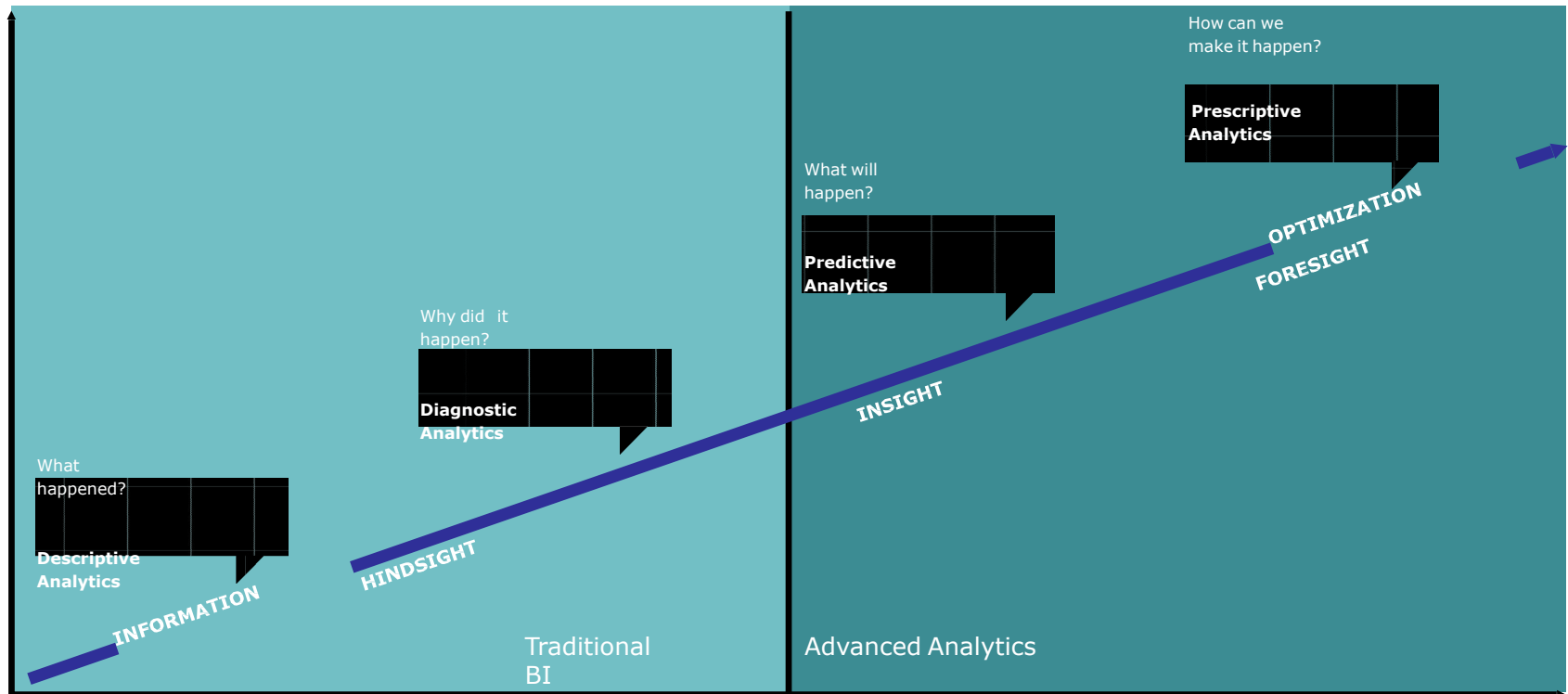
- Machine Translation
 - Google Translate translates language from one language to another.
- Text Simplification
 - Rewordify simplifies the meaning of sentences.
- Sentiment Analysis
 - Hater News gives us the sentiment of the user.
- Text Summarization
 - Smmry or Reddit's autotldr gives a summary of sentences.
- Auto-Predict
 - Google Search predicts user search results.
- Auto-Correct
 - Google Keyboard and Grammarly correct words otherwise spelled wrong.
- Speech Recognition
 - Google WebSpeech or Vocalware.
- Question Answering
 - IBM Watson's answers to a query.
- Natural Language Generation
 - Generation of text from image or video data

What Is Data Science?

From Wikipedia:

Data science, also known as data-driven science, is an interdisciplinary field about scientific processes and systems to extract knowledge or insights from data in various forms, either structured or unstructured, which is a continuation of some of the data analysis fields such as statistics, machine learning, data mining, and predictive analytics similar to Knowledge Discovery in Databases (KDD).

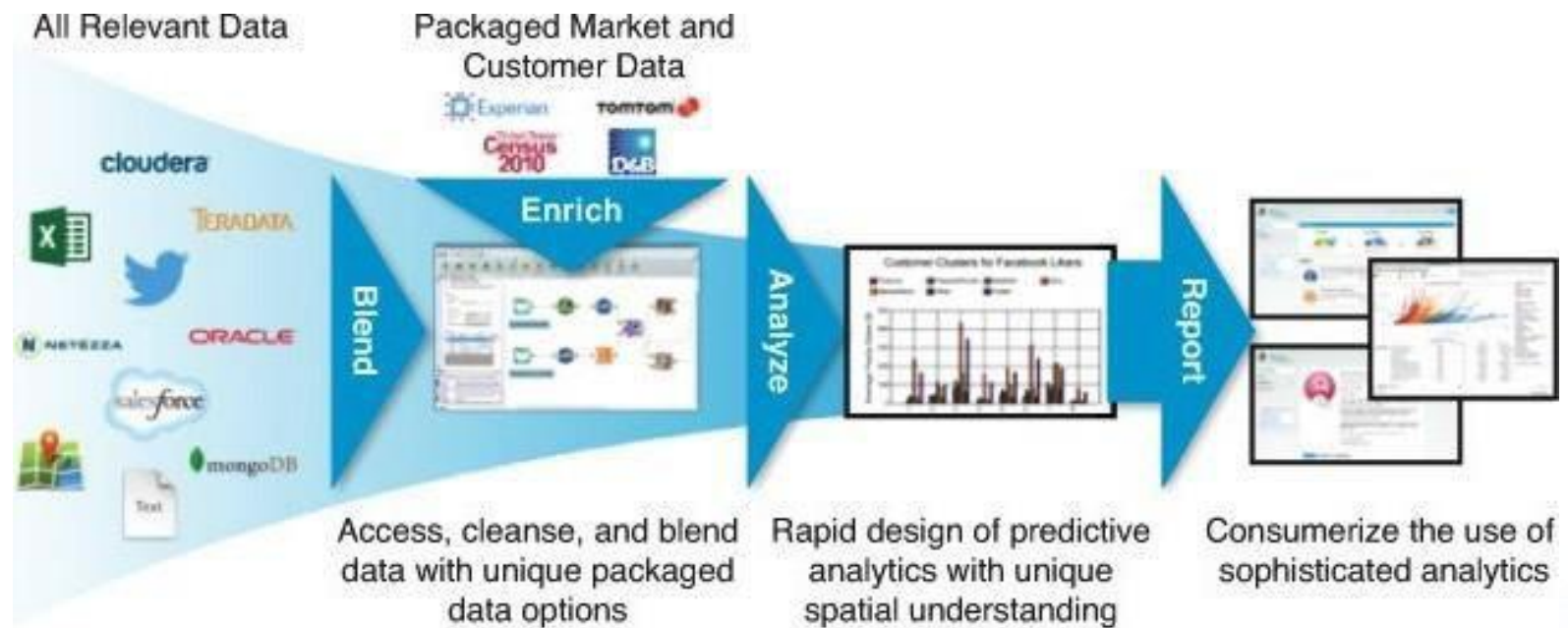
How Is Business Intelligence Different From Data Science?



Modern BI Process - Blend, Enrich, Analyze, and Report

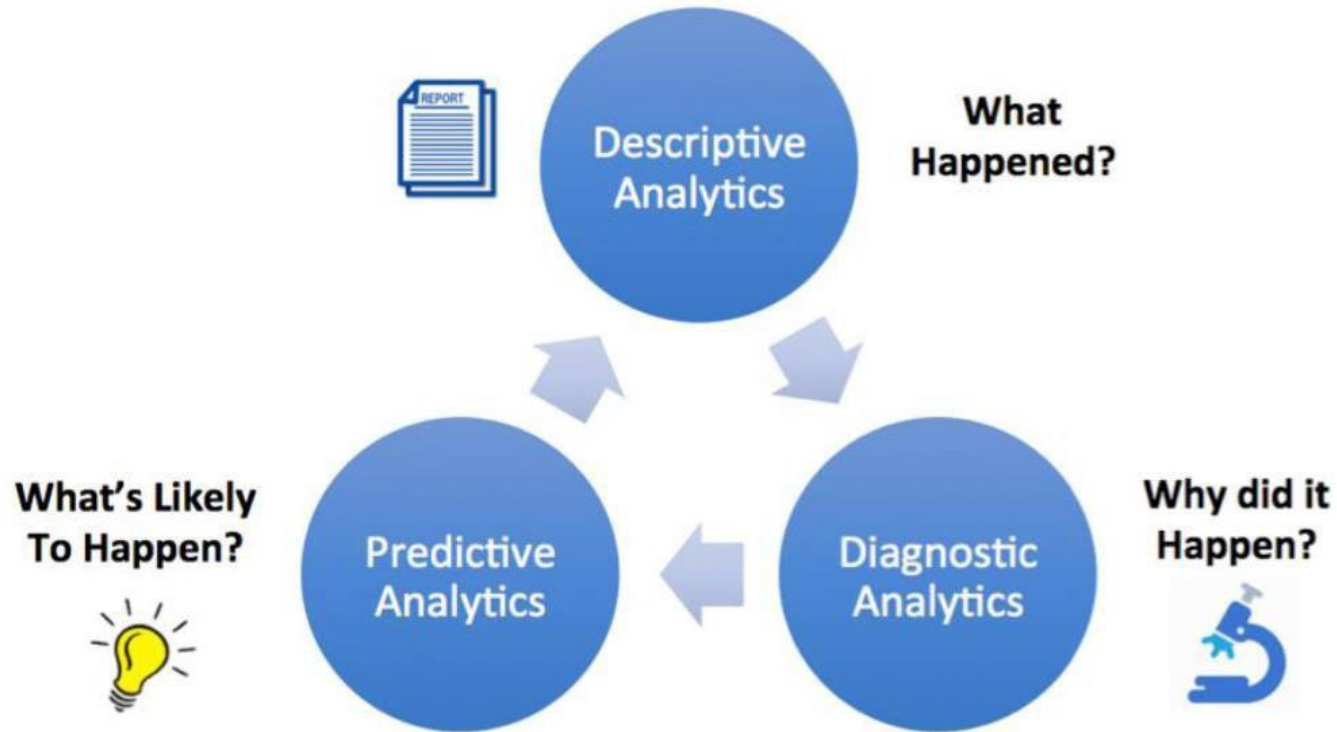
Predictive Analytics: By applying machine learning algorithms to historical project data, organizations can forecast outcomes more accurately and mitigate risks in project delivery

Business Intelligence: Automating the preparation of data for analytics tools allows organizations to gain actionable insights more rapidly, enhancing their decision-making capabilities.



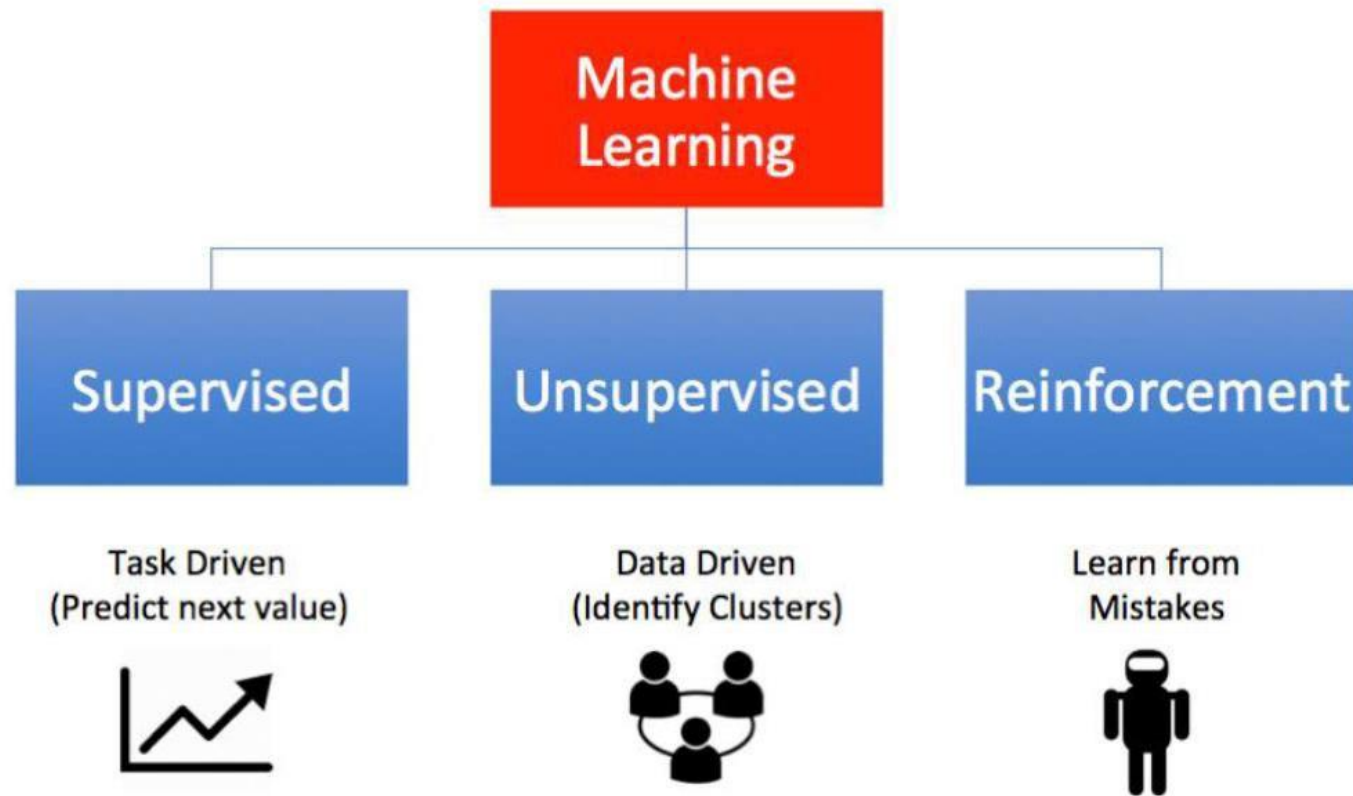
What is Business Analytics & Machine Learning?

Types of Analytics

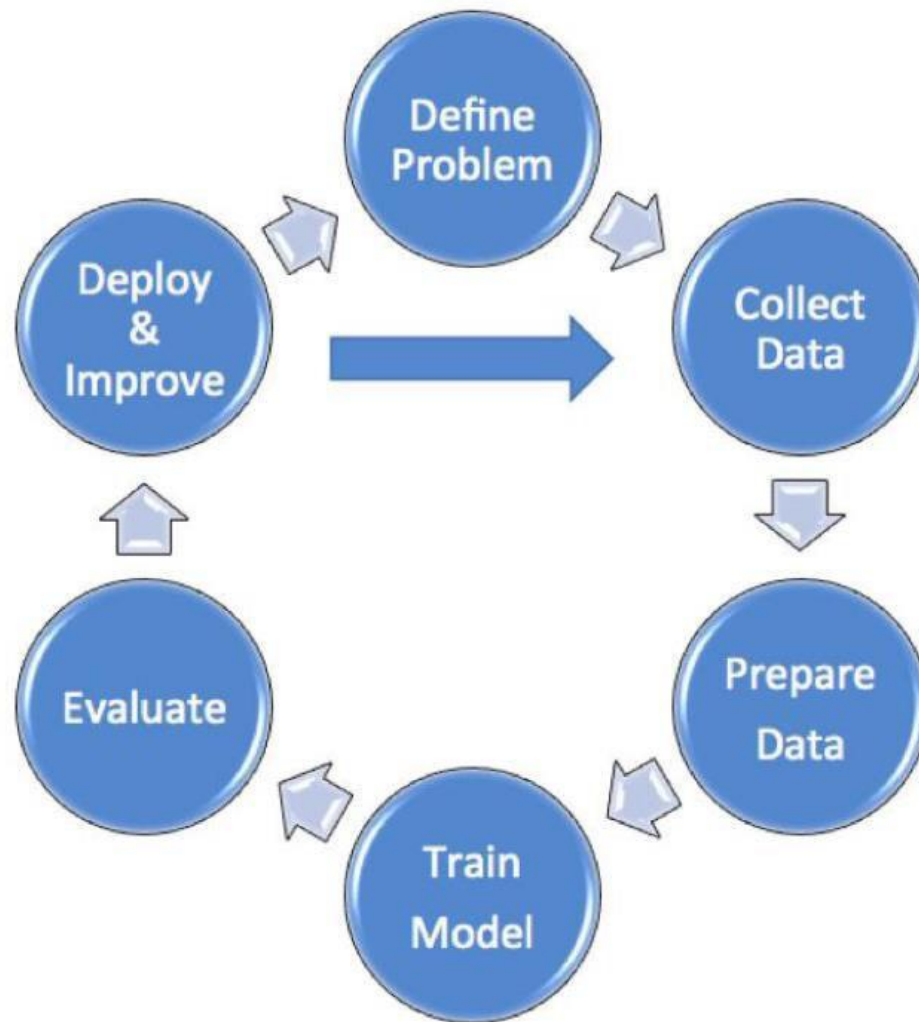


What is Business Analytics & Machine Learning?

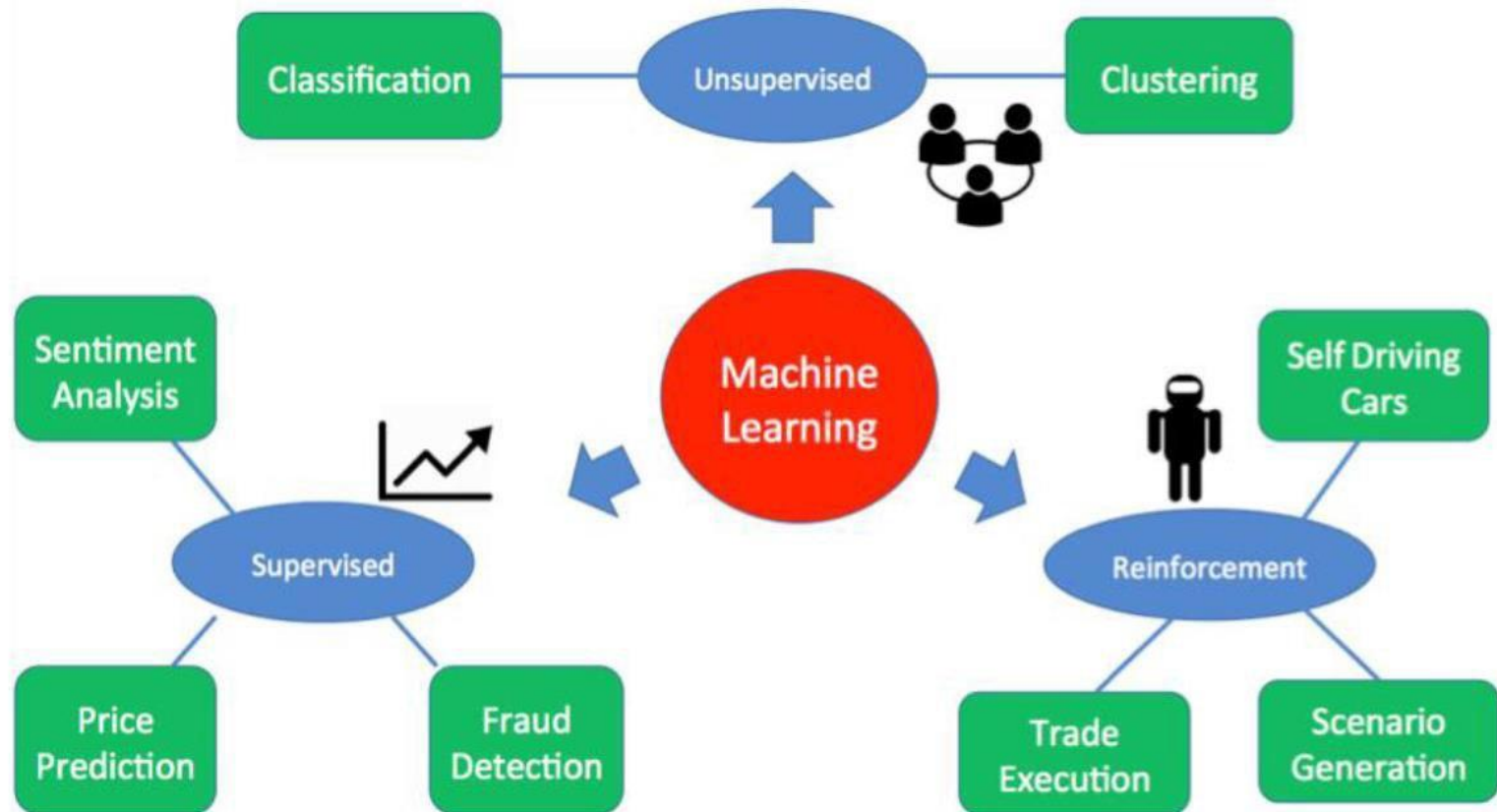
Types of Machine Learning



What is Business Analytics & Machine Learning?



What is Business Analytics & Machine Learning?



How To Do Data Mining: Machine Learning Techniques

Essential Concepts of Machine Learning

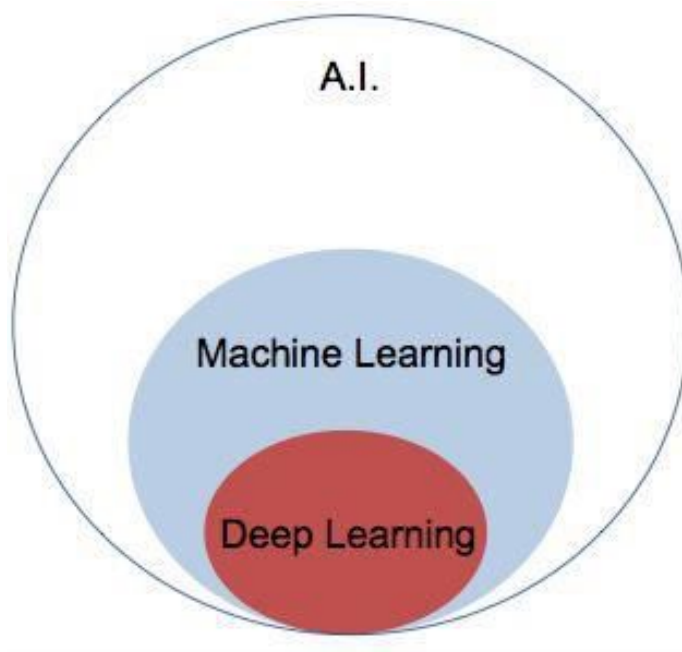
Artificial Intelligence

Machine Learning

Deep Learning

Source: <https://www.becomingadatascientist.com/2017/07/17/introductory-machine-learning-terminology-with-food/>

Clearing The Confusion About Data Science And Big Data



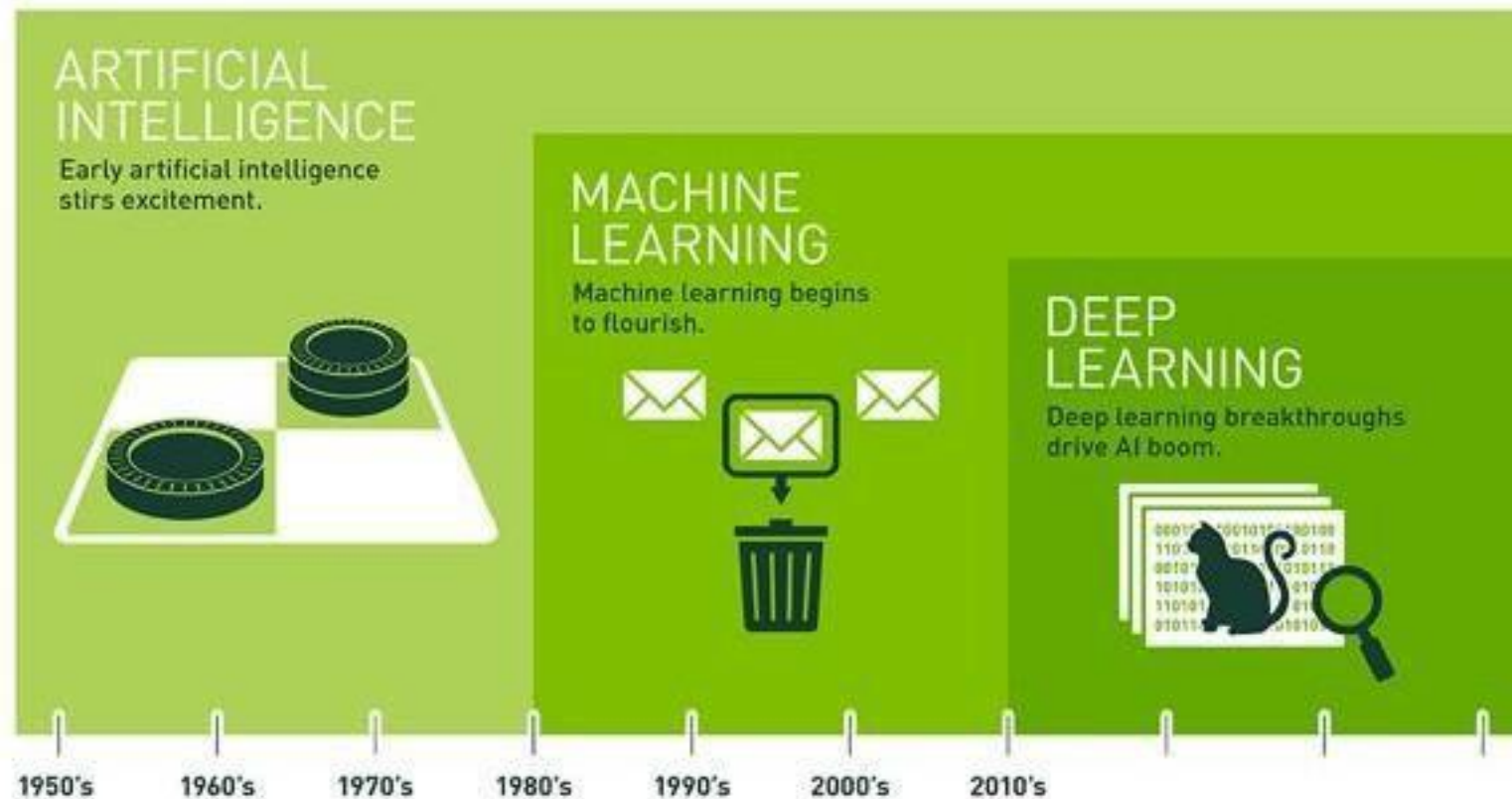
Definition of AI: a branch of computer science trying to get computers to exhibit intelligence.

- Machine Learning is a range of techniques for computers to perform cognitive functions
- Deep Learning (a type of Machine Learning that uses layers of neural networks) has been the most accurate and productive technique in AI research

Frank Chen of a 16z

Big Data Tools and Algorithm Exploration

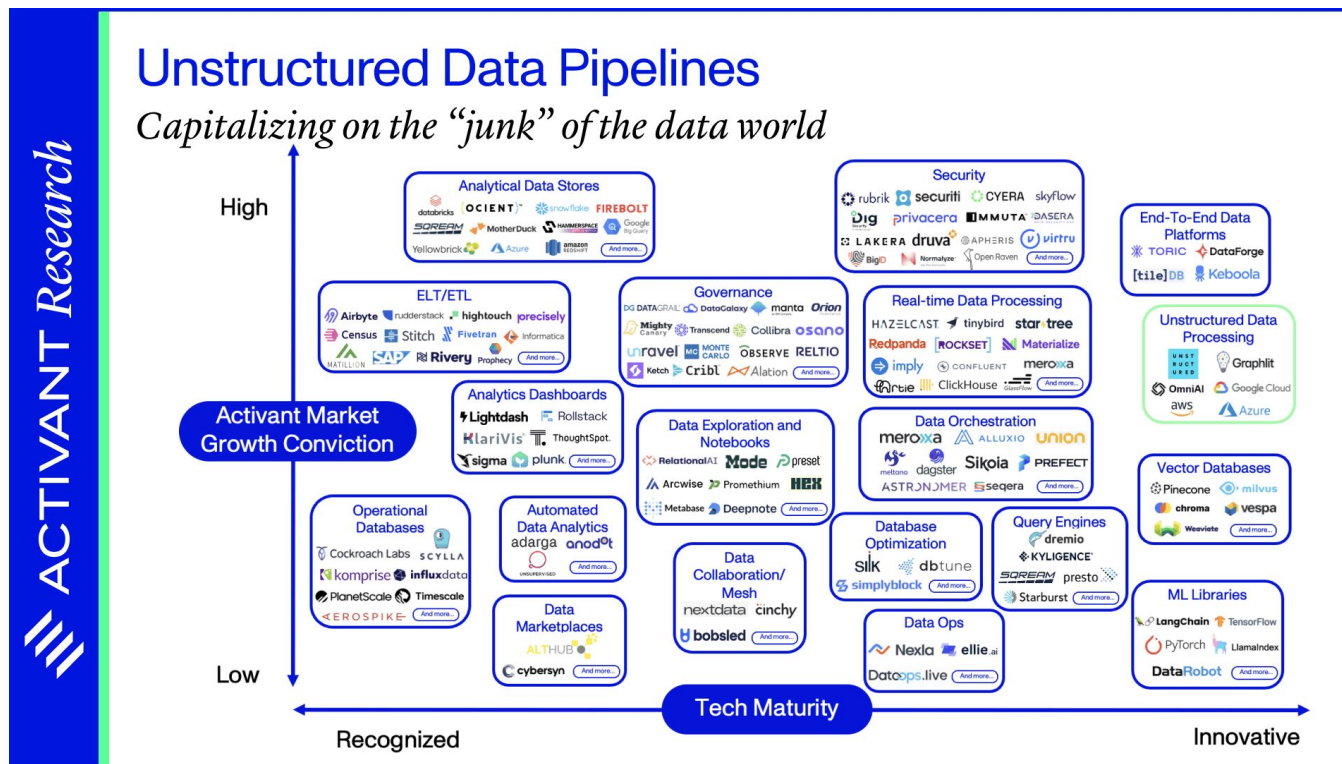
Difference Between AI, Machine Learning & Deep Learning



Since an early flush of optimism in the 1950s, smaller subsets of artificial intelligence – first machine learning, then deep learning, a subset of machine learning – have created ever larger disruptions.

IPA – Data Science

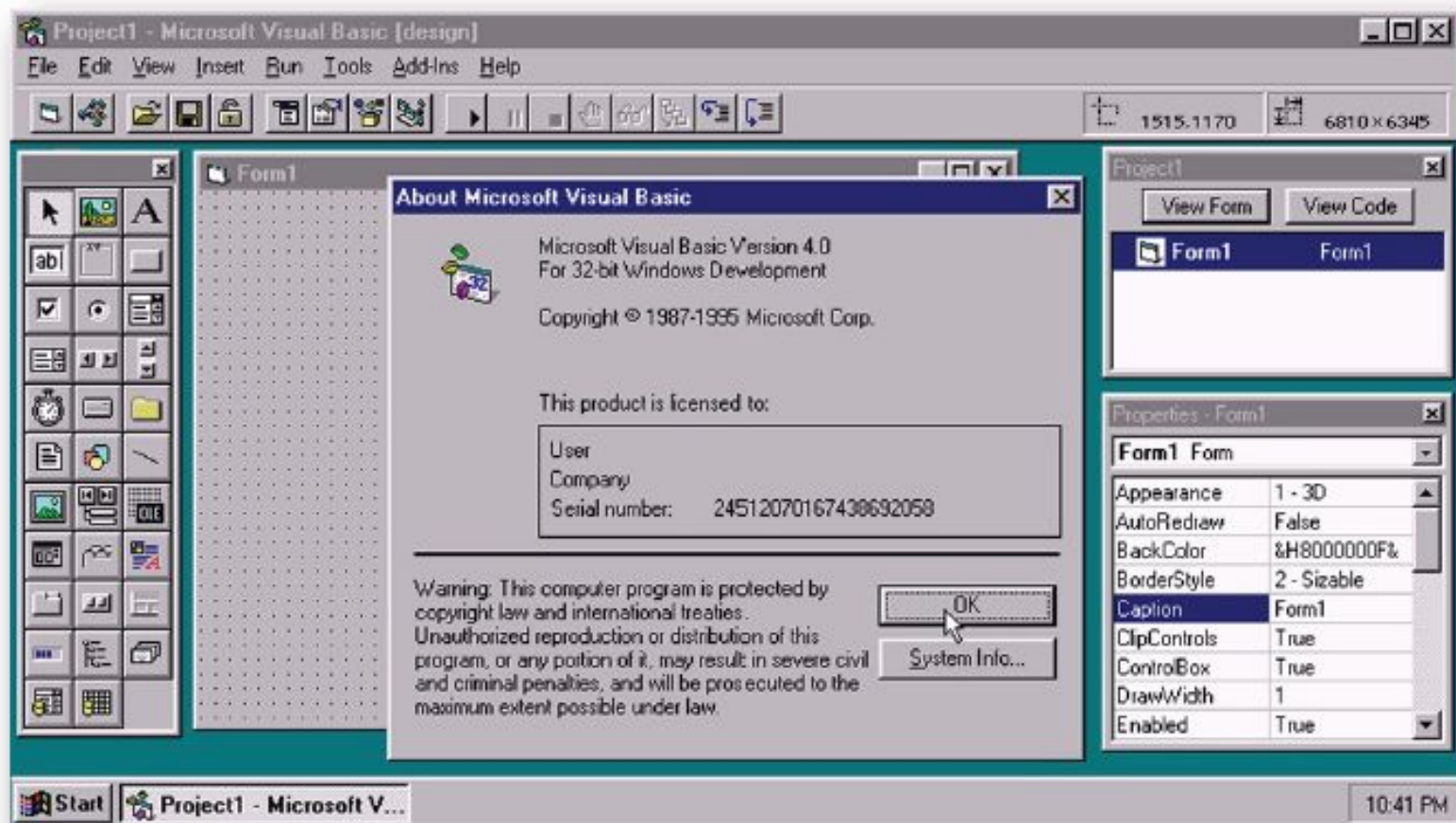
IPA's integration with data science is particularly impactful in several areas:
 Unstructured Data Processing: IPA can convert unstructured data into structured formats suitable for analysis, enabling businesses to leverage insights from previously untapped data source.



Business Process Automation with VBA

VBA - Visual Basic

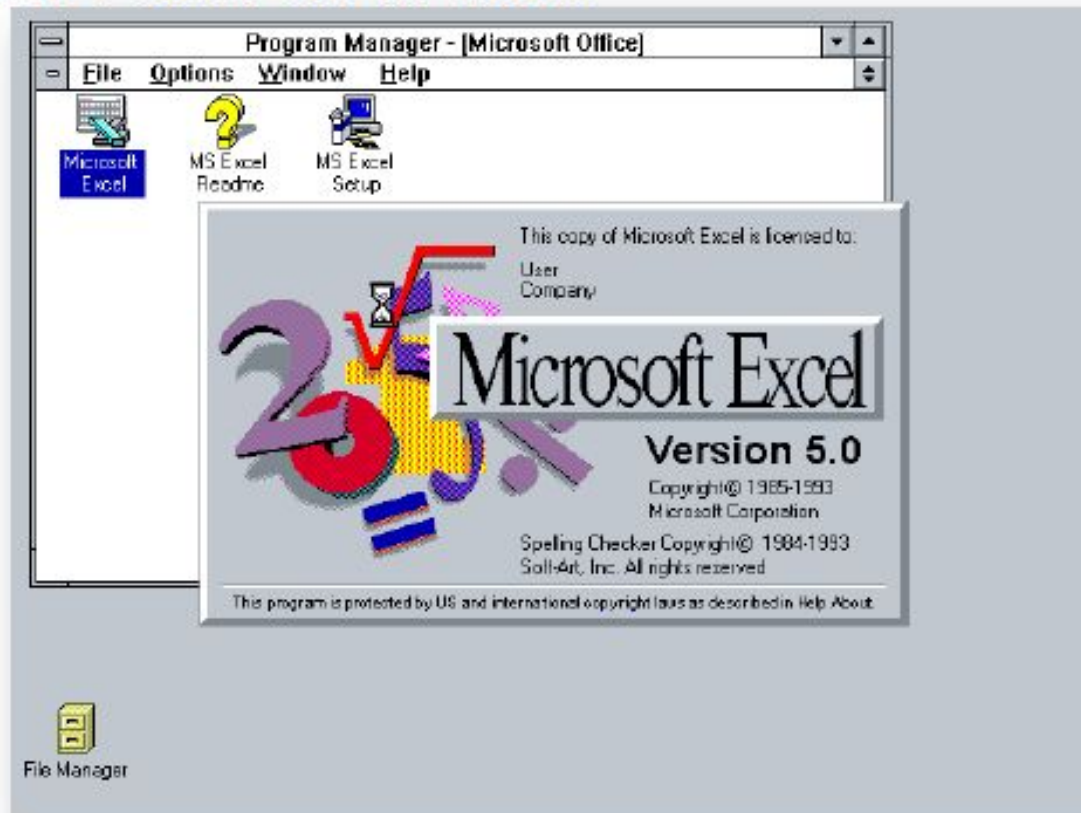
- VB was “The” Microsoft way to write programs



Business Process Automation with VBA

VBA - Visual Basic for Applications

- Introduced to Excel 5.0 in 1993

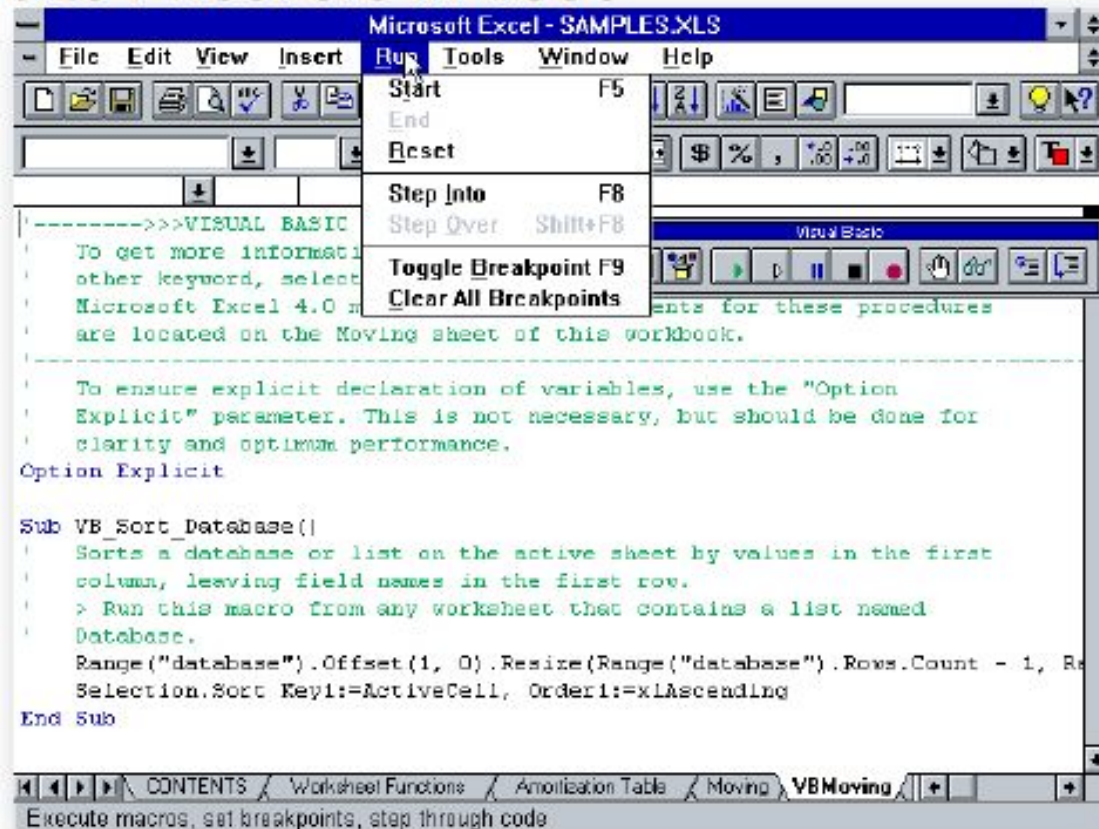


1. Microsoft Excel 5.x. WinWorld. <https://winworldpc.com/product/microsoft-excel/5x>

Business Process Automation with VBA

VBA - Visual Basic for Applications

- Introduced to Excel 5.0 in 1993

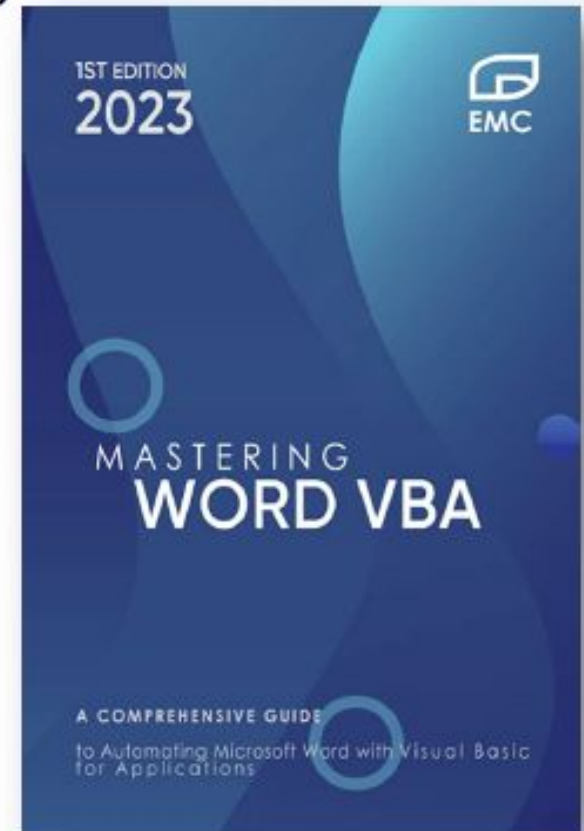


1. Microsoft Excel 5.x. WinWorld. <https://winworldpc.com/product/microsoft-excel/5x>

Business Process Automation with VBA

VBA - Visual Basic for Applications

- Gradually extended to Microsoft Office
- VB classic → ended in 2008
- VB.NET → evolution ended in 2020¹
- VBA → still supported (e.g., M365)



1. Team, N. (2020, March 11). *Visual Basic support planned for .NET 5.0*. Visual Basic Blog. <https://devblogs.microsoft.com/vbteam/visual-basic-support-planned-for-net-5-0/>

Business Process Automation with VBA

VBA – Why?

- Automation
- User interaction
- Widely used in the industry



1. O. (2022, June 8). *Getting started with VBA in Office*. Getting Started With VBA in Office | Microsoft Learn. <https://learn.microsoft.com/en-us/office/vba/library-reference/concepts/getting-started-with-vba-in-office>

Business Process Automation with VBA

VBA - Automation

- Macros → Microsoft's response to automation
- Naming:
 - Micro-instruction vs Macro-instruction
 - 1 key to perform series of commands / actions

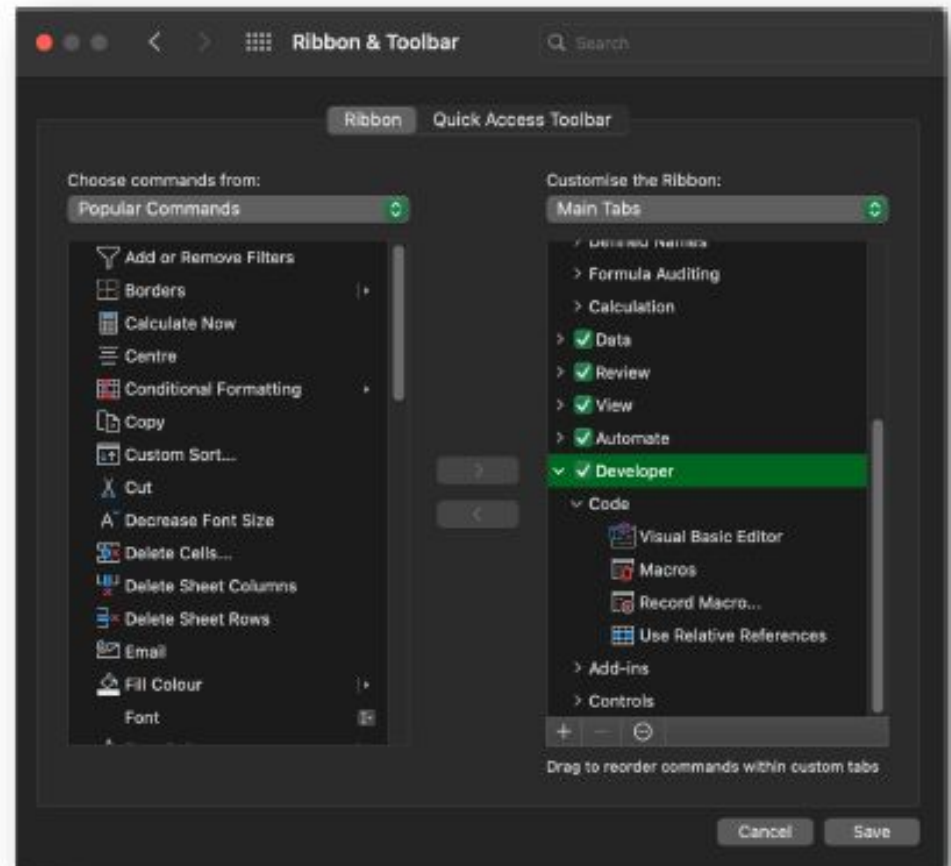


1. Why is Excel VBA called "Macros"? Why Is Excel VBA Called "Macros"? <https://www.excelforum.com/the-water-cooler/791161-why-is-excel-vba-called-macros.html>

Business Process Automation with VBA

Macros – Where?

- “Developer” tab
 - Hidden by default
 - File
 - ➔ Options
 - ➔ Customize Ribbon
- Turn on “Developer”

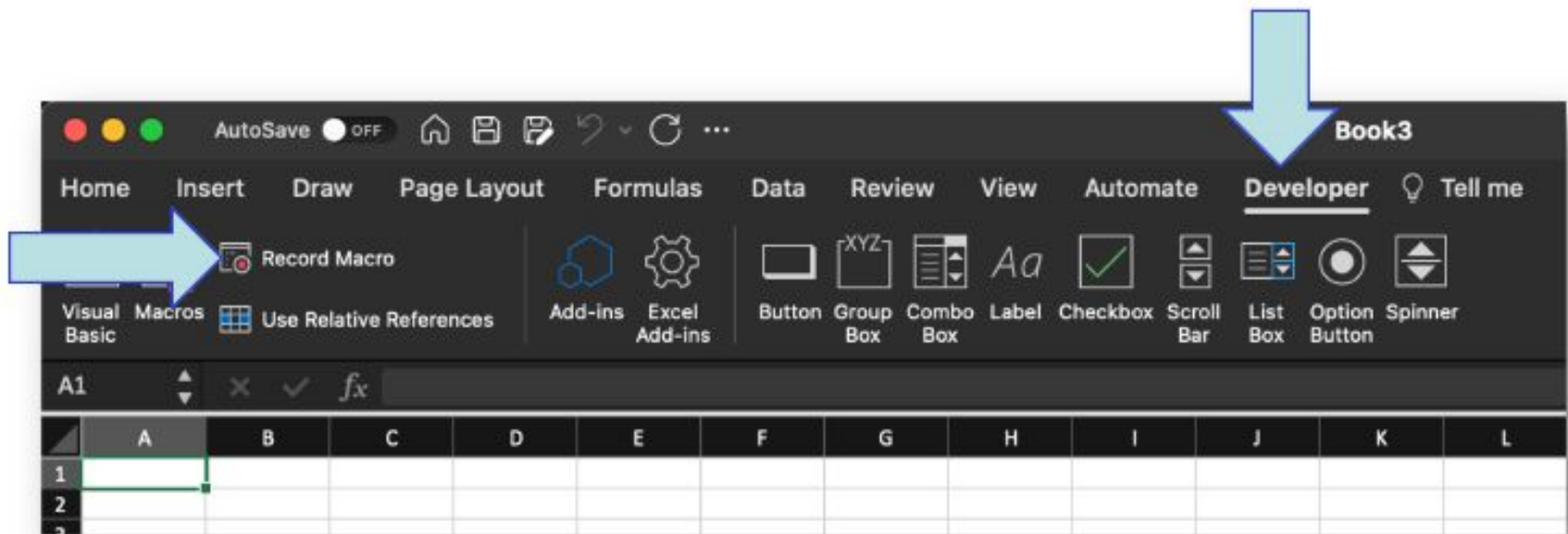


1. *Show the Developer tab - Microsoft Support.* Show the Developer Tab - Microsoft Support.
<https://support.microsoft.com/en-us/office/show-the-developer-tab-e1192344-5e56-4d45-931b-e5fd9bea2d45>

Business Process Automation with VBA

Practice 1 - Your first macro

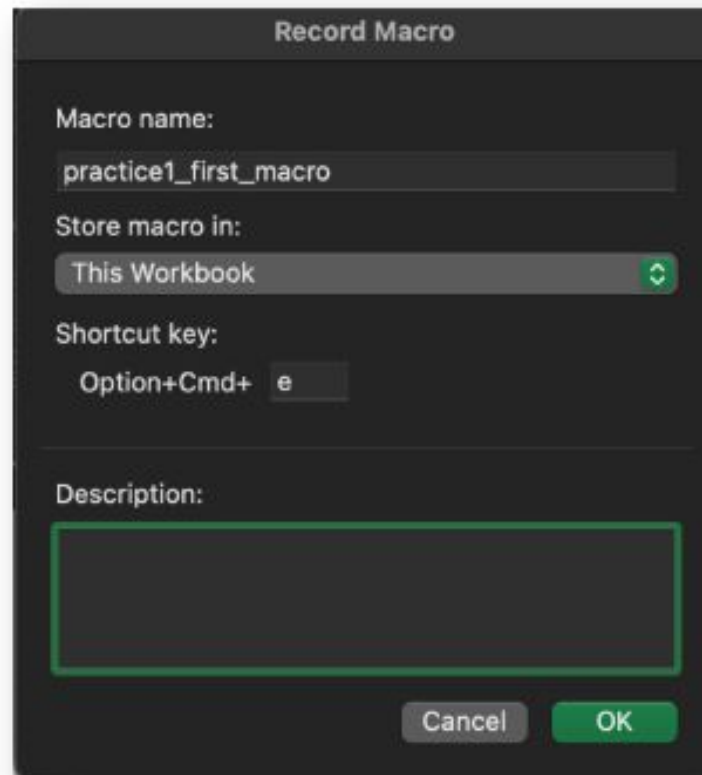
- Click “Record Macro”



Business Process Automation with VBA

Practice 1 - Your first macro

- Type in a name (use “_” instead of spaces)
- Set a shortcut key
- Click OK



Business Process Automation with VBA

Practice 1 - Your first macro

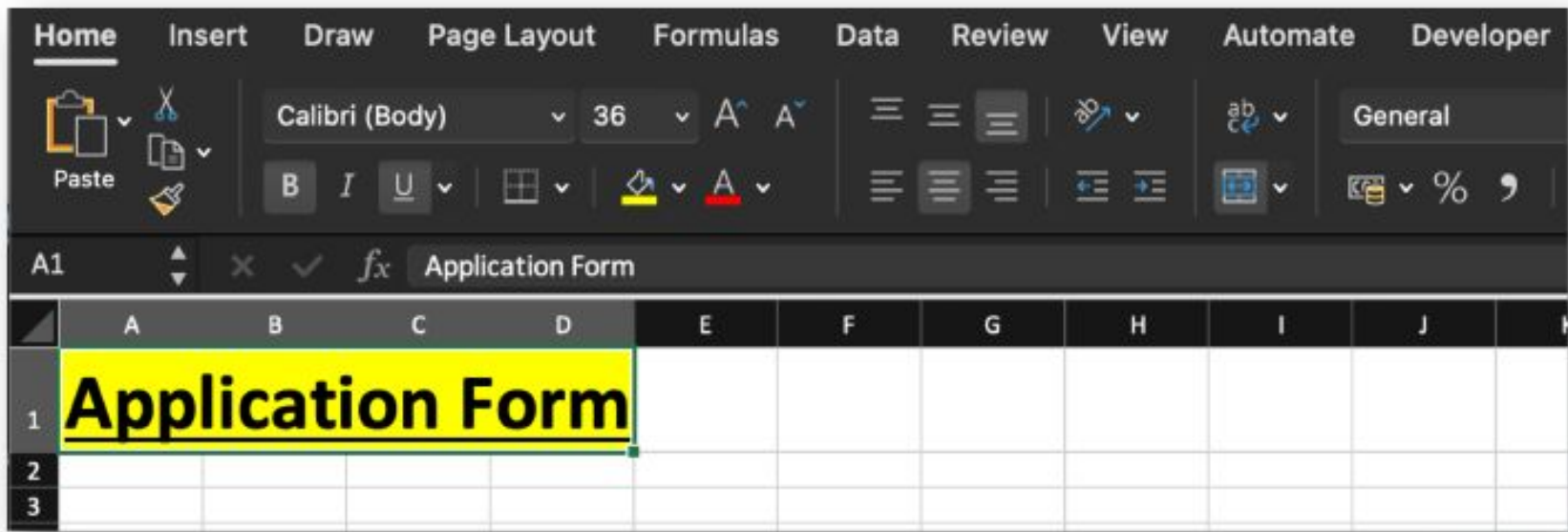
- The recording has started
- Add a new sheet with the “+” at the bottom-left



Business Process Automation with VBA

Practice 1 - Your first macro

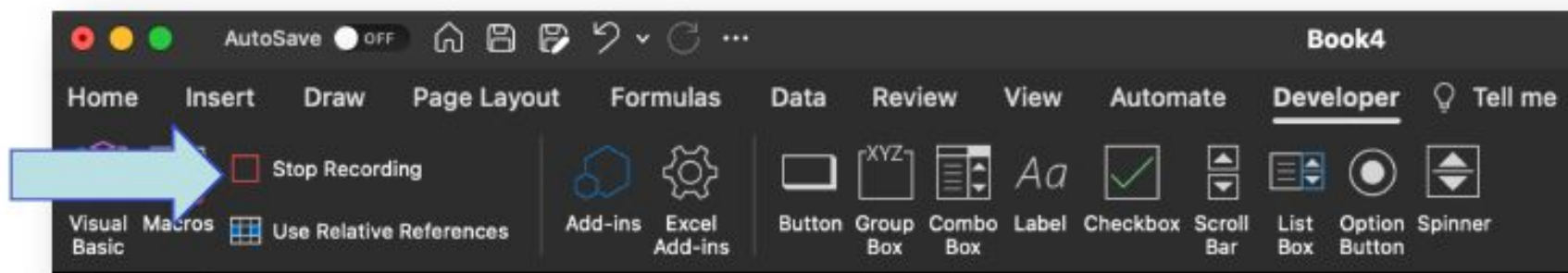
- Go back to “Home” tab
- Select Cells A1 to D1 → Merge
- Enter “Application Form” at the top-left
- Style it: e.g., Font size + Bold + Underline



Business Process Automation with VBA

Practice 1 - Your first macro

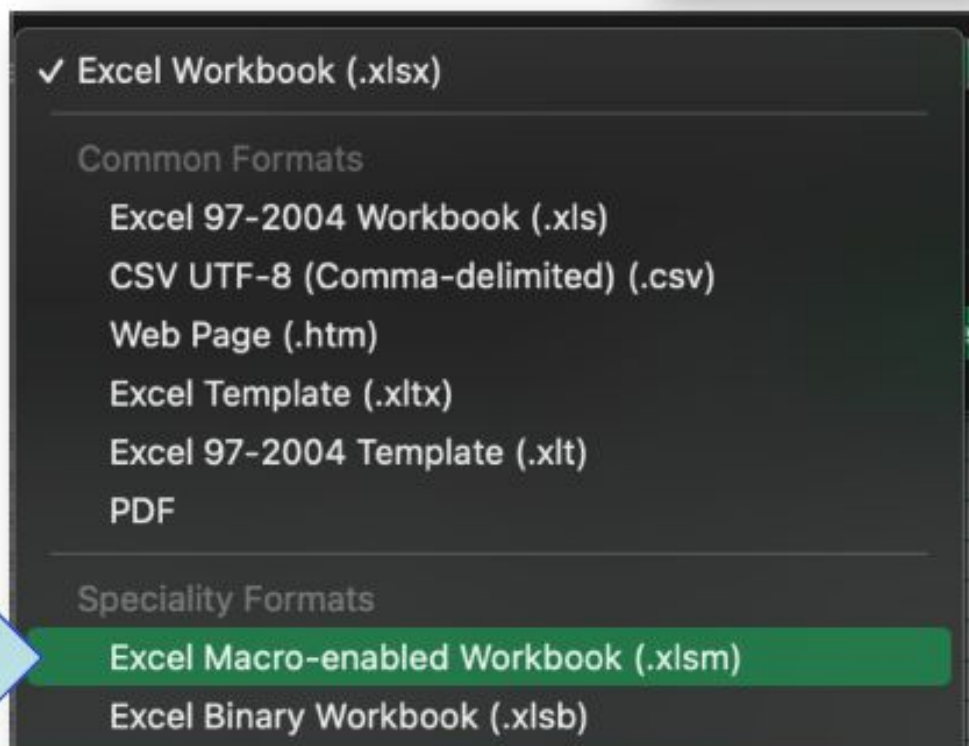
- Go back to “Developer” tab
- Hit “Stop Recording”



Business Process Automation with VBA

Practice 1 - Your first macro

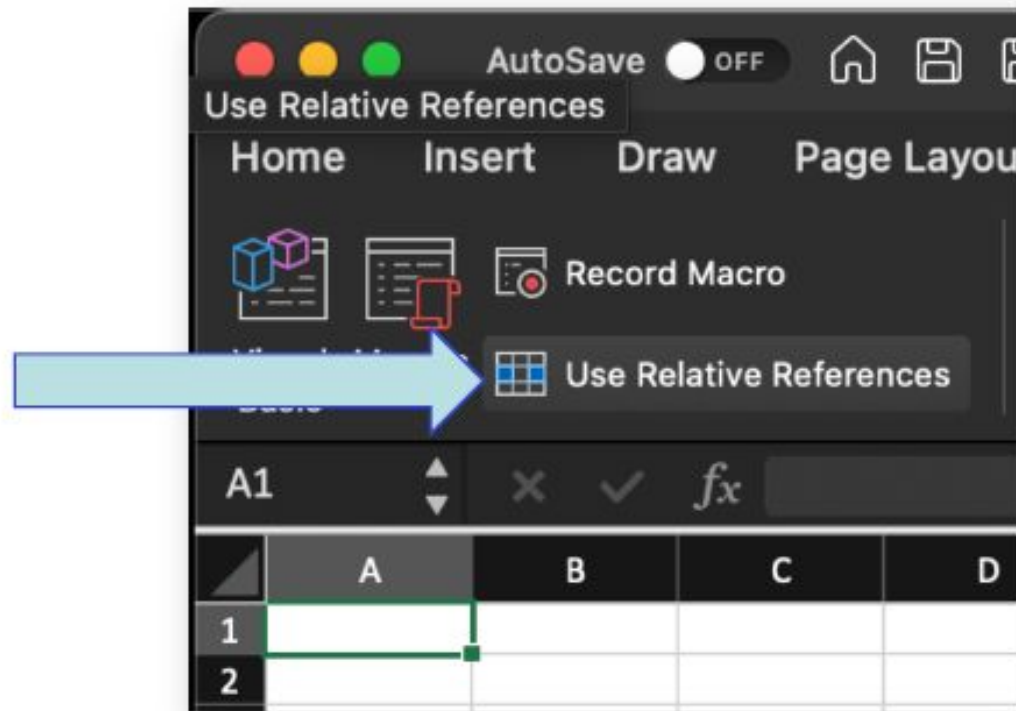
- Save as .xlsm to preserve your macros!



Business Process Automation with VBA

Relative Reference

- Record and execute based on “current selection”
- Click BEFORE “Record Macro”



Business Process Automation with VBA

Practice 2 – Relative References

- Create a new macro to generate below

Record Macro

Macro name:
practice2_relative_references

Store macro in:
This Workbook

Shortcut key:
Option+Cmd+ r

Description:

Cancel OK

Visual Basic Macros Record Macro Use Relative References Add-ins Excel Add-ins Button

F17

	A	B	C	D	E	F
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						

Name: _____

Phone: _____

Address: _____

Business Process Automation with VBA

VBA - Why not just use macros?

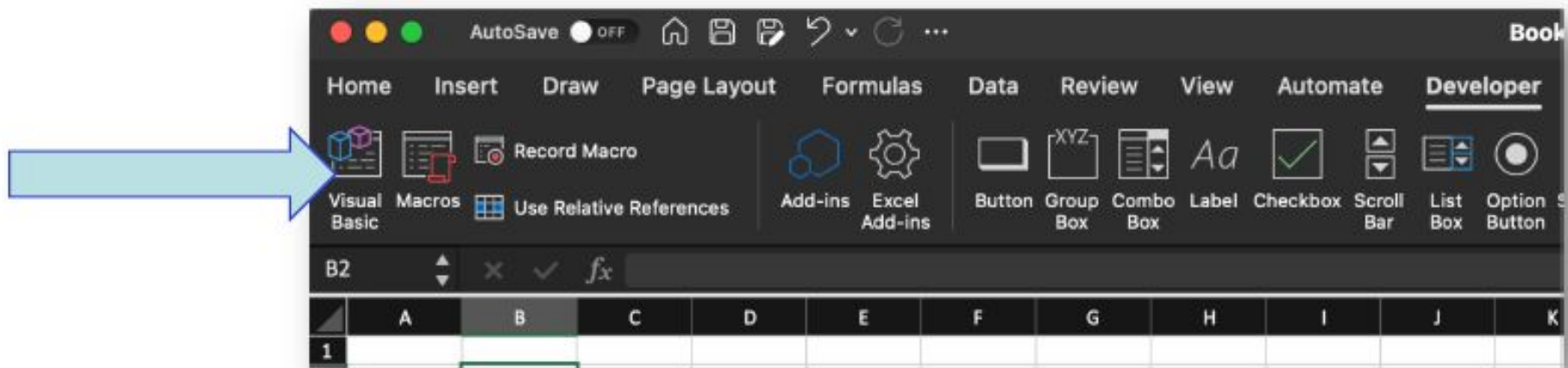
- Macros → great for repetition
- Need → More flexibility and interactivity
- Solution → Dig into the codes



Business Process Automation with VBA

Entering Visual Basic Mode

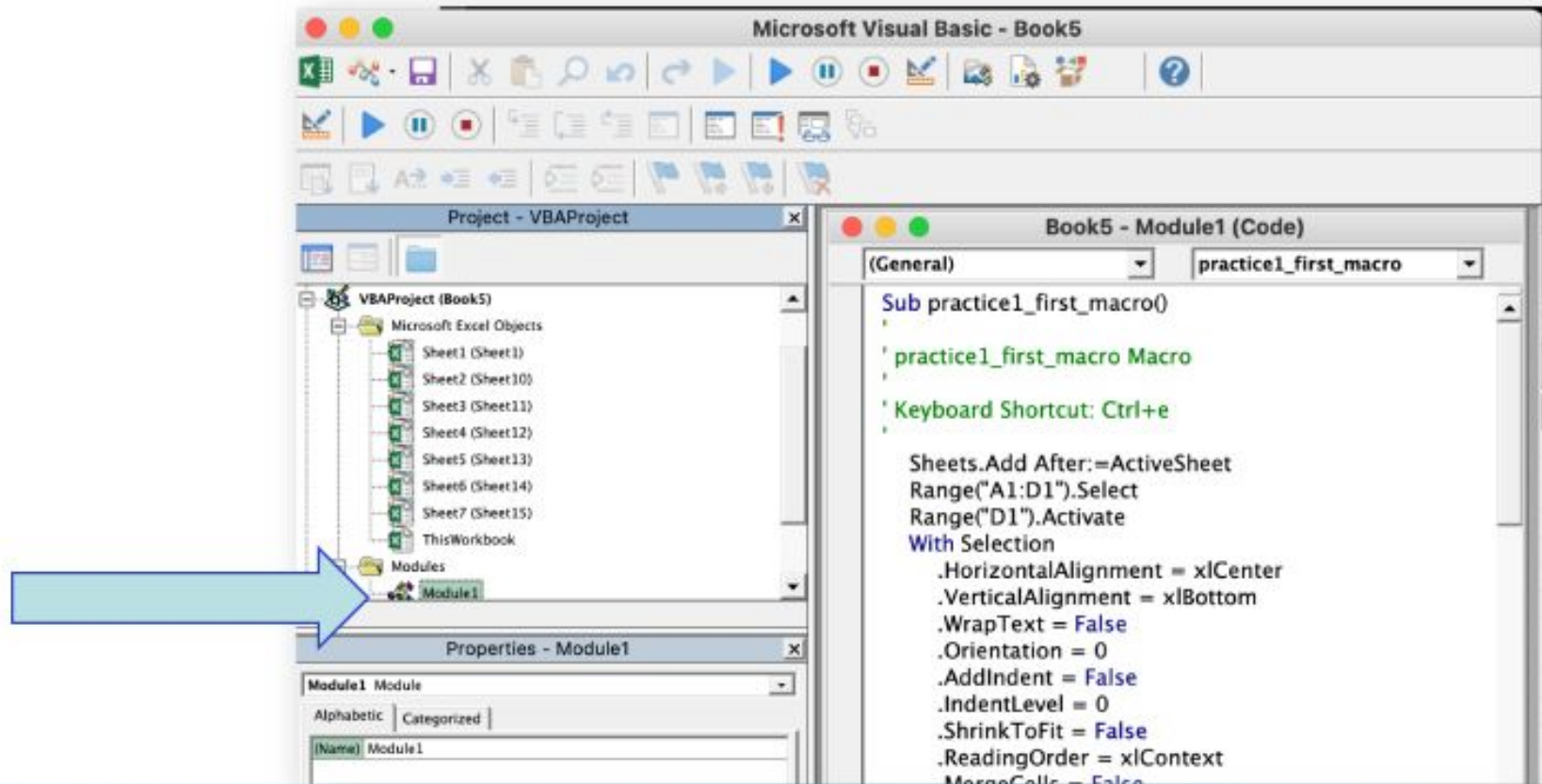
- Developer tab → Visual Basic



Business Process Automation with VBA

Entering Visual Basic Mode

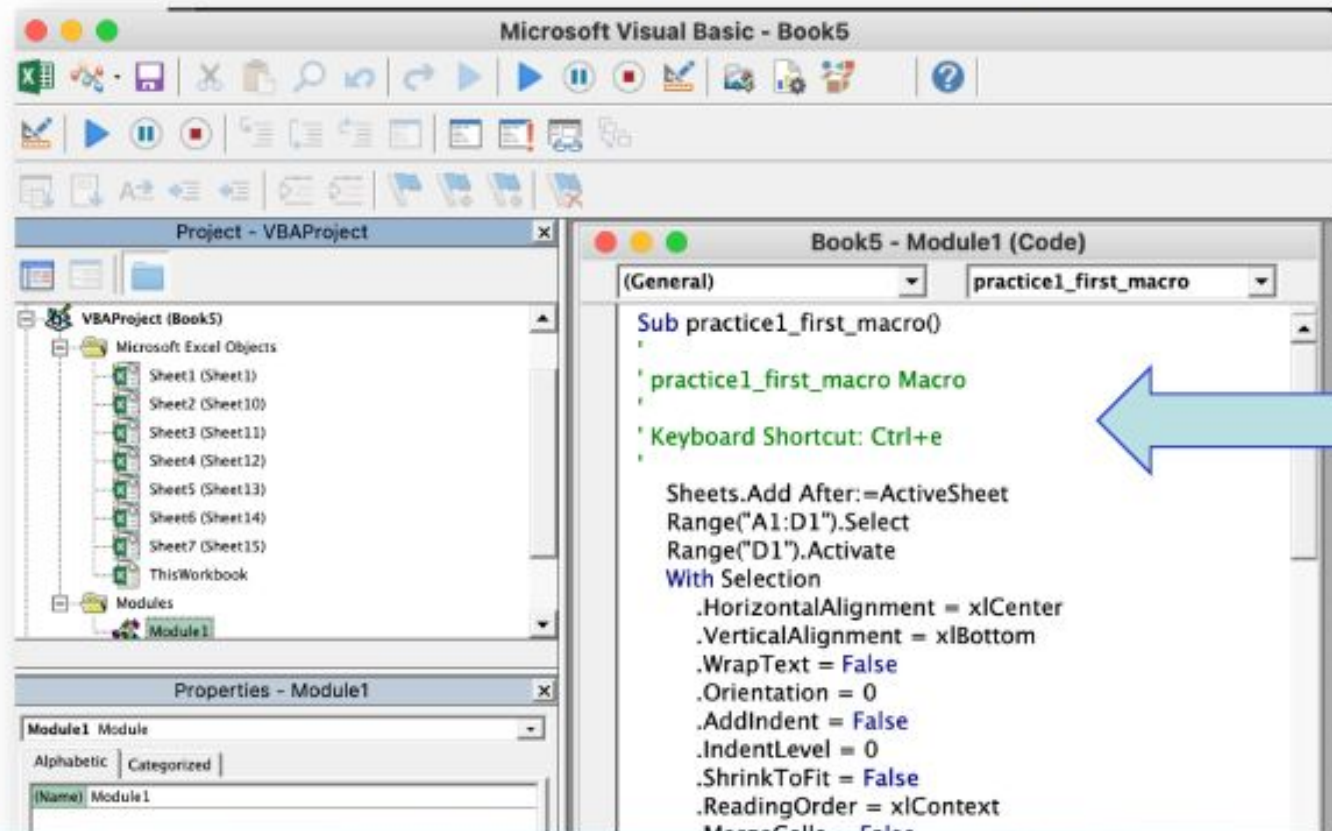
- Integrated Development Environment (IDE)
- Modules = where codes are usually stored



Business Process Automation with VBA

Entering Visual Basic Mode

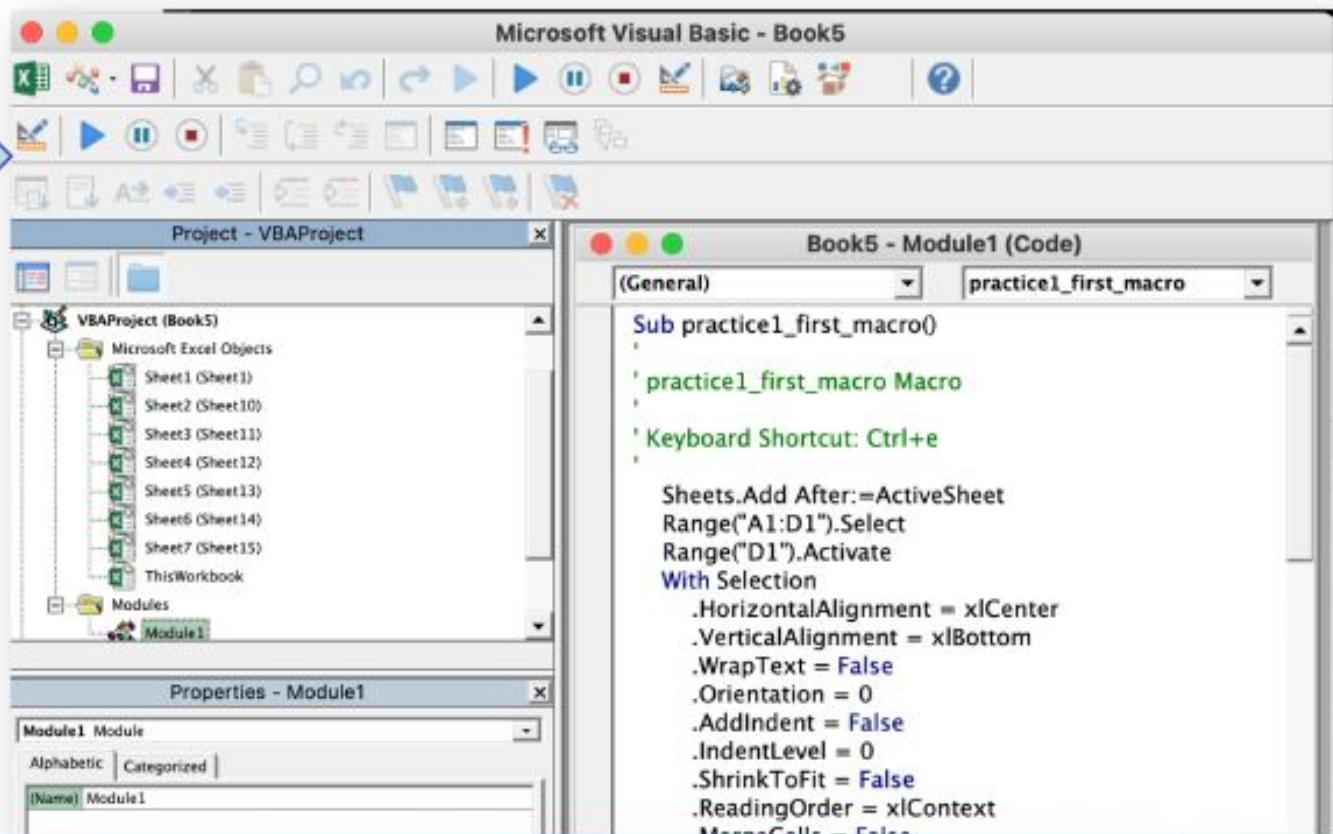
- Main editor on the right
- Modules → 1 or more “Sub” (sub-routines) or “Function”



Business Process Automation with VBA

Entering Visual Basic Mode

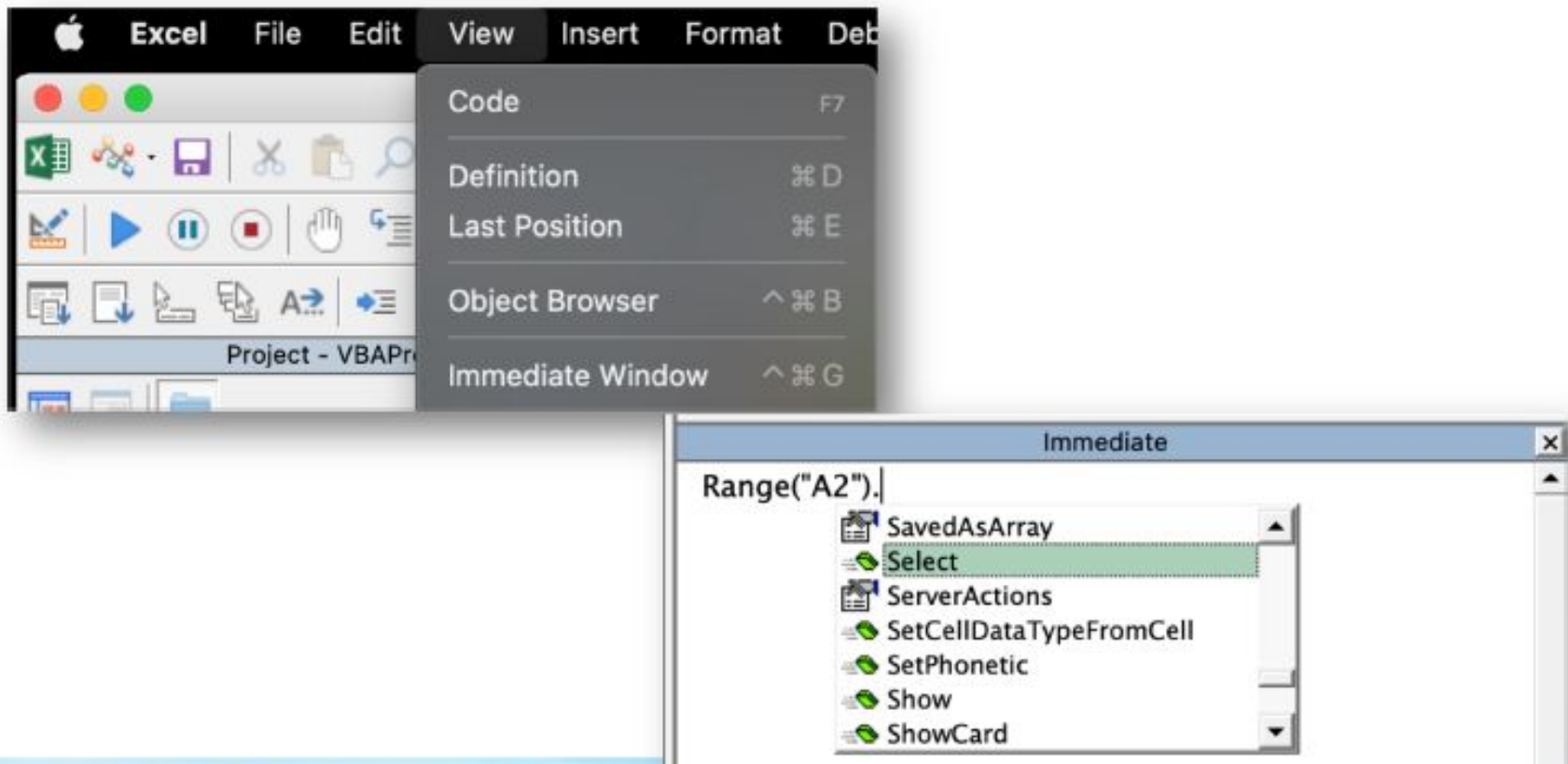
- Debugging toolbar at the top



Business Process Automation with VBA

Immediate Window

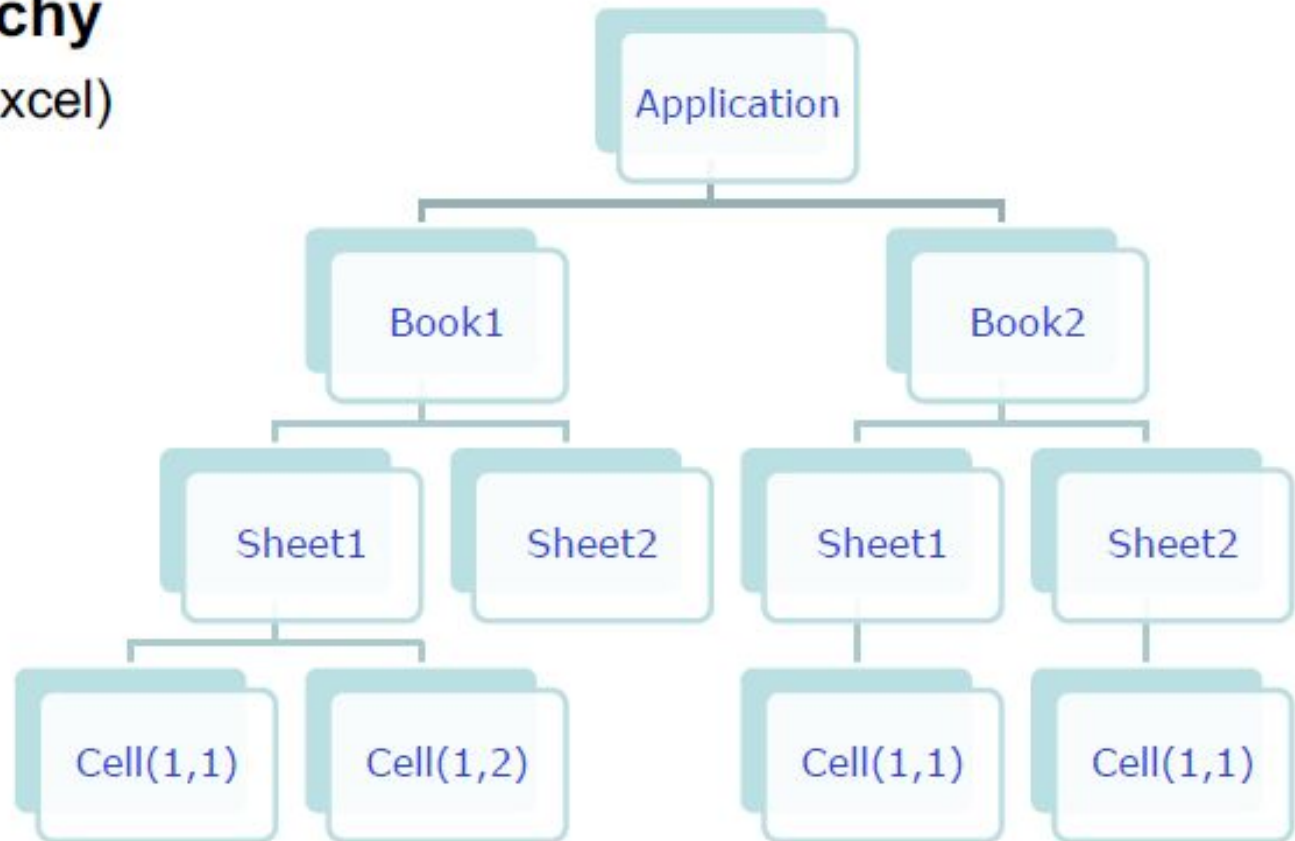
- View → Immediate Window
- Useful for trying out codes (Thanks to Auto-complete)



Business Process Automation with VBA

Object hierarchy

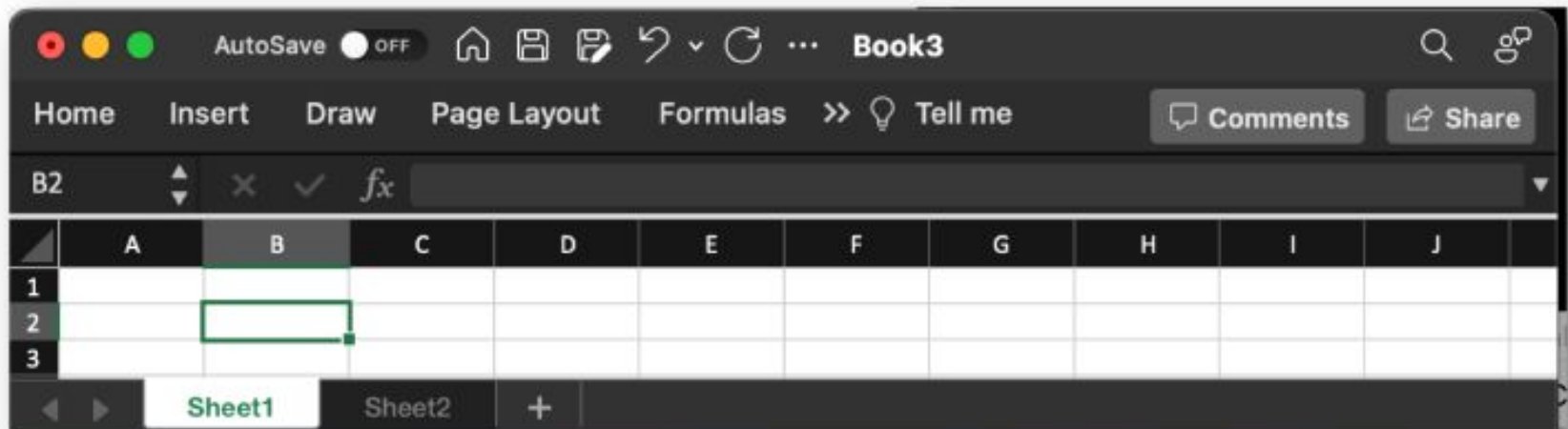
- Application (Excel)
- Workbooks
- Worksheets
- Cells



Business Process Automation with VBA

Object reference – Call something by name

- English: This man / that girl / “Mr. Chan”
- VBA
 - Application → Excel.Application / Outlook.Application / etc
 - Workbooks("Book1")
 - Worksheets("Sheet1") / Worksheets(1)
 - Cells(2,3)
 - Range("B2")



Business Process Automation with VBA

Object reference - Implied naming

- English: Here / Today
- VBA
 - `ActiveWorkbook`
 - `ActiveSheet`
 - `Selection`
 - If you write:
 - `Range("A1:D1").Select`
 - From that moment onwards:
 - `Selection` → Same as `Range("A1:D1")`
- * `ThisWorkbook`
 - This = Where the macro is stored
 - Active = Changeable by user

Business Process Automation with VBA

Object reference - Offset

- English: Is it that building? No, the one on the right
- VBA
 - `Range("B2").Offset(1, 2)`
 - `Worksheets(ActiveSheet.Index + 1).Select`

B2				
	A	B	C	D
1				
2		B2		
3				
4				

Diagram illustrating the VBA formula `Range("B2").Offset(1, 2)`. A blue arrow labeled '1' points down from cell B2 to cell C2. A blue arrow labeled '2' points right from cell C2 to cell D2. The formula bar shows `Range("B2").Offset(1, 2)`.



Business Process Automation with VBA

Object reference – Offset

- Region
 - `Selection.CurrentRegion.Select`
- Boundary
 - `Range("A1").End(xlDown).Select`

B2				Item
	A	B	C	
1				
2		Item	Price	
3		A	1	
4		B	2	
5		C	3	
6		D	4	
7				

	A	
1		1
2		2
3		3
4		4
5		5
6		6

?Range("A1").End(xlDown).Row
6

Business Process Automation with VBA

Object properties – Actions

- English
 - Jackie runs / That person sings
- VBA (Actions are also known as Methods)
 - ActiveSheet.PrintOut
 - Workbooks("Book5").PrintPreview
 - Worksheets("Sheet15").Select



Business Process Automation with VBA

Object properties – More actions

- VBA – Some methods require parameters



- Copy & Paste
 - `Range("B2").Copy Range("B3")`

	A	B	
1			
2		testing	
3		testing	
4			

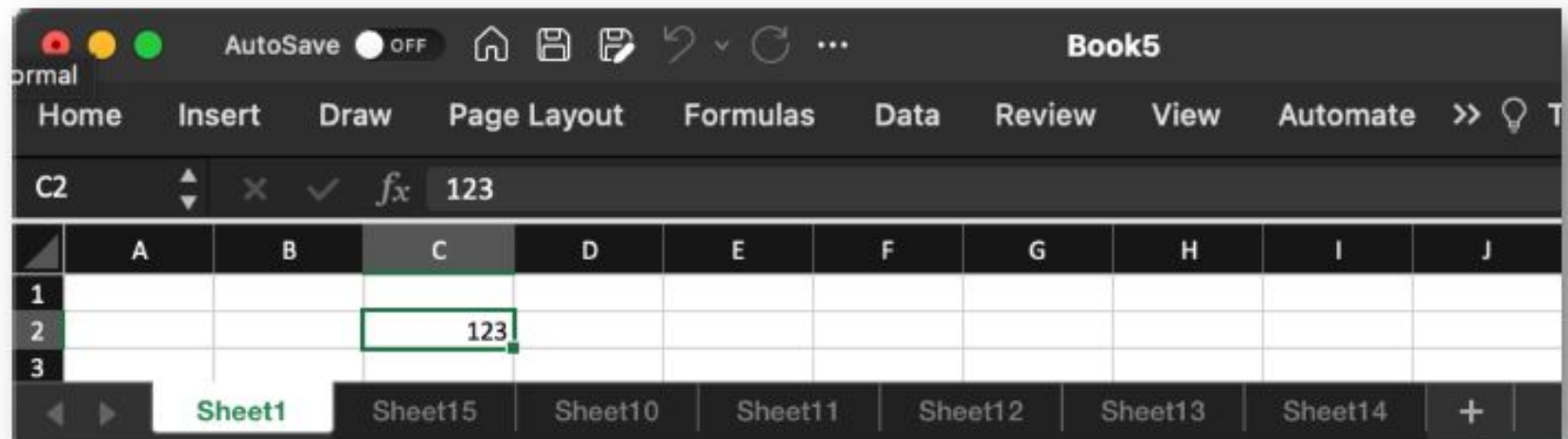
- Export to PDF
 - `ActiveSheet.ExportAsFixedFormat Type:=xlTypePDF, Filename:="demo.pdf"`

Name	Size	Kind
demo.pdf	13 KB	PDF Document
Book5.xlsm	28 KB	Micros... (.xlsm)

Business Process Automation with VBA

Object properties – Value

- English
 - HSBC → Account 123-654321-012 → Value → \$1000
- VBA
 - `Workbooks("Book5").Worksheets("Sheet1").Cells(2,3).Value = 123`



Business Process Automation with VBA

Object properties – Other properties

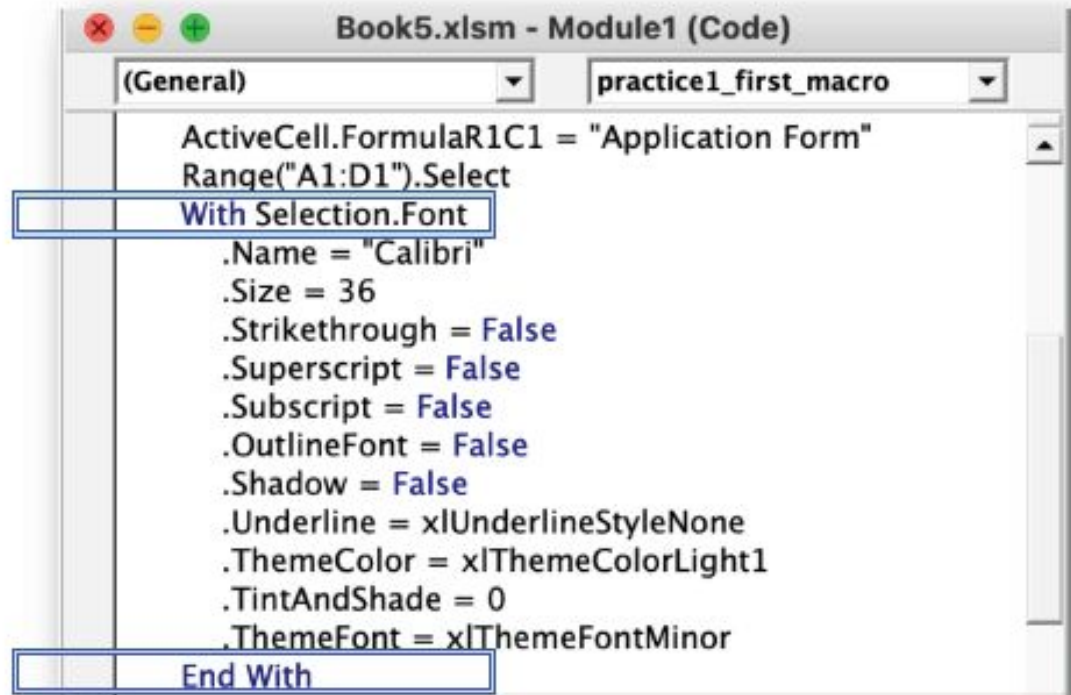
- VBA
 - Worksheets("Sheet1").Range("B2").Font.Size = 36
 - Worksheets("Sheet1").Range("B2").Font.Color = vbRed

	A	B	C
1			
2		testing	
3			

Business Process Automation with VBA

Object properties – Editing many properties

- “With” statement
 - Reduce redundancy
 - Example from Practice 1



The screenshot shows the VBA Editor window titled "Book5.xlsm - Module1 (Code)". The "General" tab is selected, and the macro "practice1_first_macro" is chosen. The code in the editor is as follows:

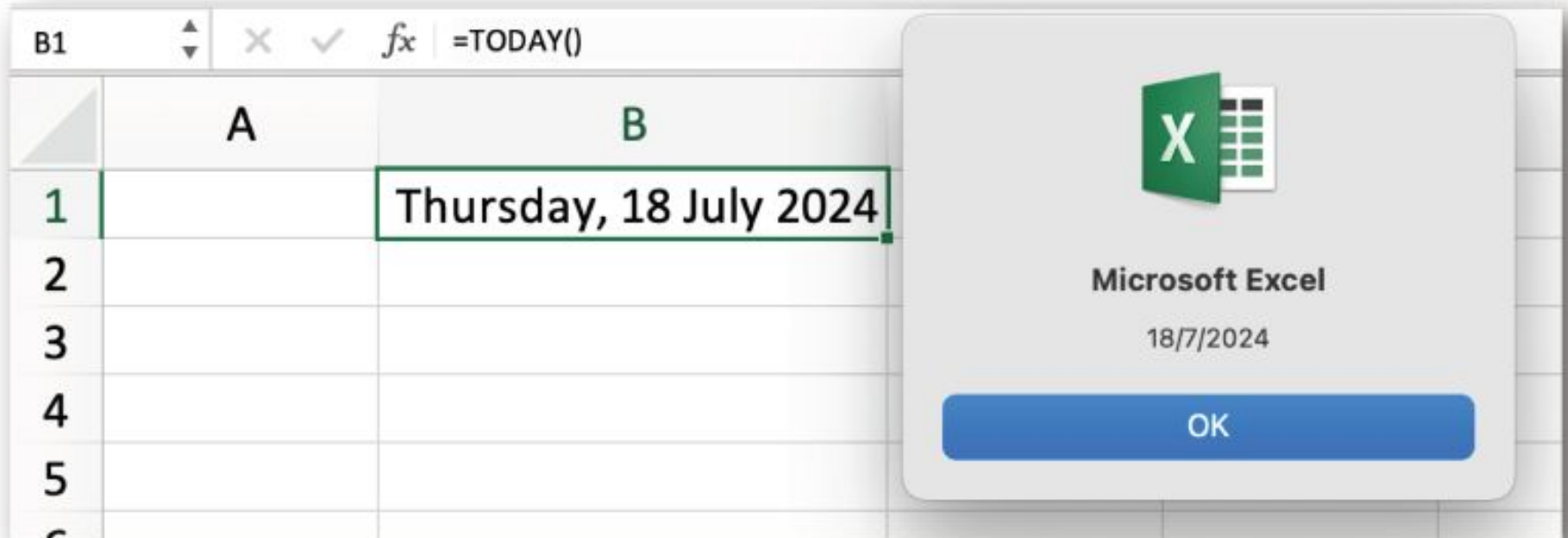
```
ActiveCell.FormulaR1C1 = "Application Form"  
Range("A1:D1").Select  
With Selection.Font  
    .Name = "Calibri"  
    .Size = 36  
    .Strikethrough = False  
    .Superscript = False  
    .Subscript = False  
    .OutlineFont = False  
    .Shadow = False  
    .Underline = xlUnderlineStyleNone  
    .ThemeColor = xlThemeColorLight1  
    .TintAndShade = 0  
    .ThemeFont = xlThemeFontMinor  
End With
```

The "With Selection.Font" line and the "End With" line are highlighted with blue boxes.

Business Process Automation with VBA

Object properties – Value check

- Use “MsgBox”
 - `MsgBox(Worksheets("Sheet1").Range("B1").Value)`

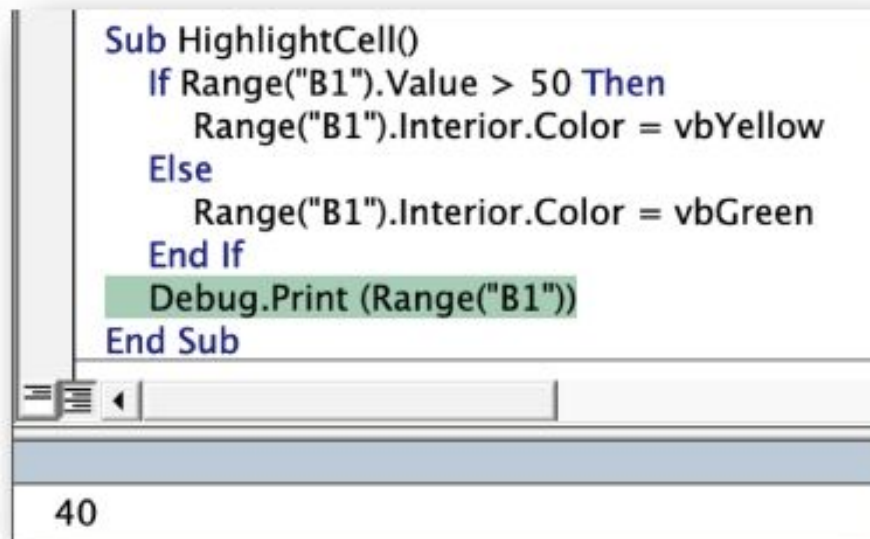


Business Process Automation with VBA

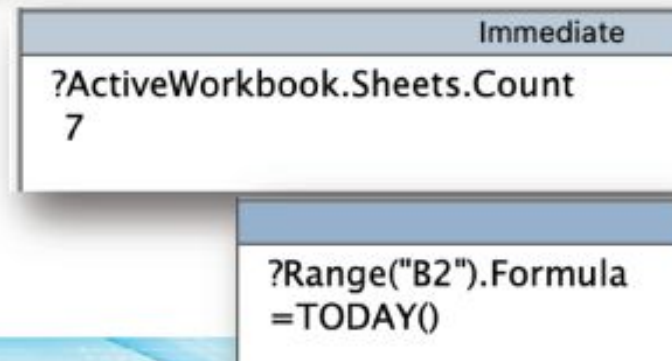
Object properties – Value check

- Use “Debug.Print()”

```
Sub HighlightCell()  
    If Range("B1").Value > 50 Then  
        Range("B1").Interior.Color = vbYellow  
    Else  
        Range("B1").Interior.Color = vbGreen  
    End If  
    Debug.Print (Range("B1"))  
End Sub
```



- Use “?” in the immediate window
 - ?ActiveWorkbook.Sheets.Count
 - ?Range("B2").Formula



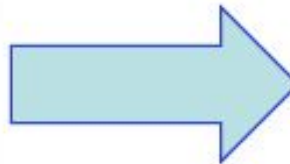
Immediate
?ActiveWorkbook.Sheets.Count 7
?Range("B2").Formula =TODAY()

Business Process Automation with VBA

Checkpoint – Value swapping

- How to swap (switch) values between 2 cells?

	A	B
1		
2		ABC
3		DEF



	A	B
1		
2		DEF
3		ABC

Business Process Automation with VBA

Checkpoint – Value swapping

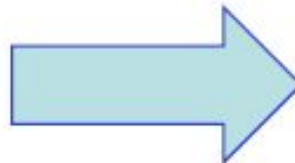
- How to swap (switch) values between 2 cells?
 - 1 solution...

	A	B
1		
2		ABC
3		DEF



	A	B
1		
2	ABC	ABC
3	DEF	DEF

1) Copy



	A	B
1		
2		DEF
3		ABC



3) Clean up

	A	B
1		
2	ABC	DEF
3	DEF	ABC

2) Overwrite

Business Process Automation with VBA

Variables – Motivation

- Storage of values without changing the sheets
- Meaningful references during calculation

	A	B
1		
2		ABC
3		DEF



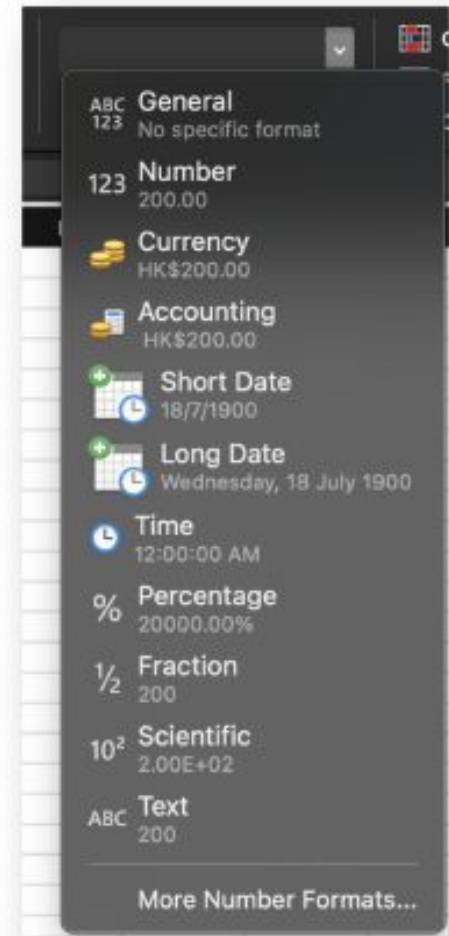
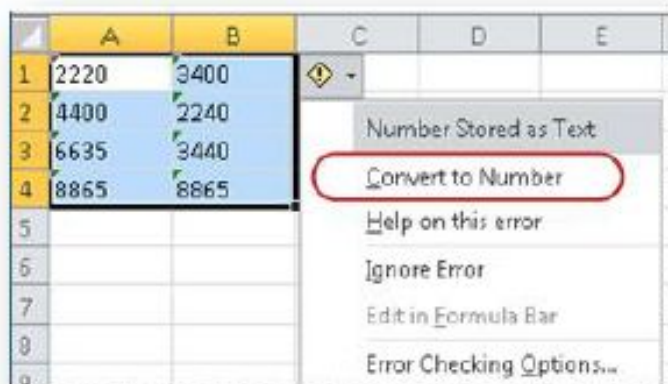
	A	B
1		
2		DEF
3		ABC

```
temp_integer = Range("B2").Value  
Range("B2").Value = Range("B3").Value  
Range("B3").Value = temp_integer
```

Business Process Automation with VBA

Variables – Data Types

- Excel is good at detecting types
 - Numbers / Date / Text
- Though sometimes can still be wrong



1. Fix text-formatted numbers by applying a number format - Microsoft Support. Fix Text-formatted Numbers by Applying a Number Format - Microsoft Support. <https://support.microsoft.com/en-us/office/fix-text-formatted-numbers-by-applying-a-number-format-6599c03a-954d-4d83-b78a-23af2c8845d0>

Business Process Automation with VBA

Variables – Data Types

- Best to define variables with types

Commonly Used VBA Data Types		
Data Type	Bytes Used	Range of Values
Integer	2	-32,768 to 32,767
Long	4	-2.147 Billion to 2.147 Billion
Single	4	45 digits (left or right of decimal point)
Double	8	308 digits (left or right of decimal point)
Currency	8	-922 T to 922 T (4 digits after decimal point)
Date	8	1/1/100 to 12/31/9999
Boolean	2	True or False
String	1 per char	Varies according to number of characters
Variant	Varies	Any data type

Business Process Automation with VBA

Variables – Data Types

- Integer
 - Dim year As Integer
 - year = 2024
- Double
 - Dim pi As Double
 - pi = 3.14125
- Smart conversion during calculation



```
?TypeName( "A" & 3 )  
String
```

```
? "A"&3  
A 3
```

```
?TypeName(2)  
Integer
```

```
?TypeName(3.3)  
Double
```

```
?TypeName(2 * 3.4)  
Double
```

```
?TypeName(5-3)  
Integer
```


Variables – Data Types

- Date
 - Dim today As Date
 - today = DateSerial(2024, 7, 1)
 - today = Date()
- String
 - Dim message As String
 - message = "Welcome!"
- Boolean
 - Range("A1").Font.Bold = True
 - Range("A1").Font.Underline = False



Business Process Automation with VBA

Common functions - Numbers

- IsNumeric() → Check if a text is number
- CInt() / CDbl() → Text to number

```
?IsNumeric("123")
```

```
True
```

```
?IsNumeric("ABCD")
```

```
False
```

```
?CInt("123")
```

```
123
```

```
?CInt("123.45")
```

```
123
```

```
?CDbl("123.45")
```

```
123.45
```

```
?CDbl("123")
```

```
123
```

Business Process Automation with VBA

Common functions - Numbers

- Round() → Rounding to a certain decimal place

```
?Round(5.1234)
```

```
5
```

```
?Round(5.1234, 2)
```

```
5.12
```

- Log() / Exp() → For financial / engineering calculation

```
?exp(2)
```

```
7.38905609893065
```

```
?log(10)
```

```
2.30258509299405
```

```
?log(exp(1))
```

```
1
```

Business Process Automation with VBA

Common functions - Date

- DateSerial → Create a date literal

```
?DateSerial(2024, 12, 25)  
25/12/2024
```

- Date() → Today's date

```
?Date()  
18/7/2024
```

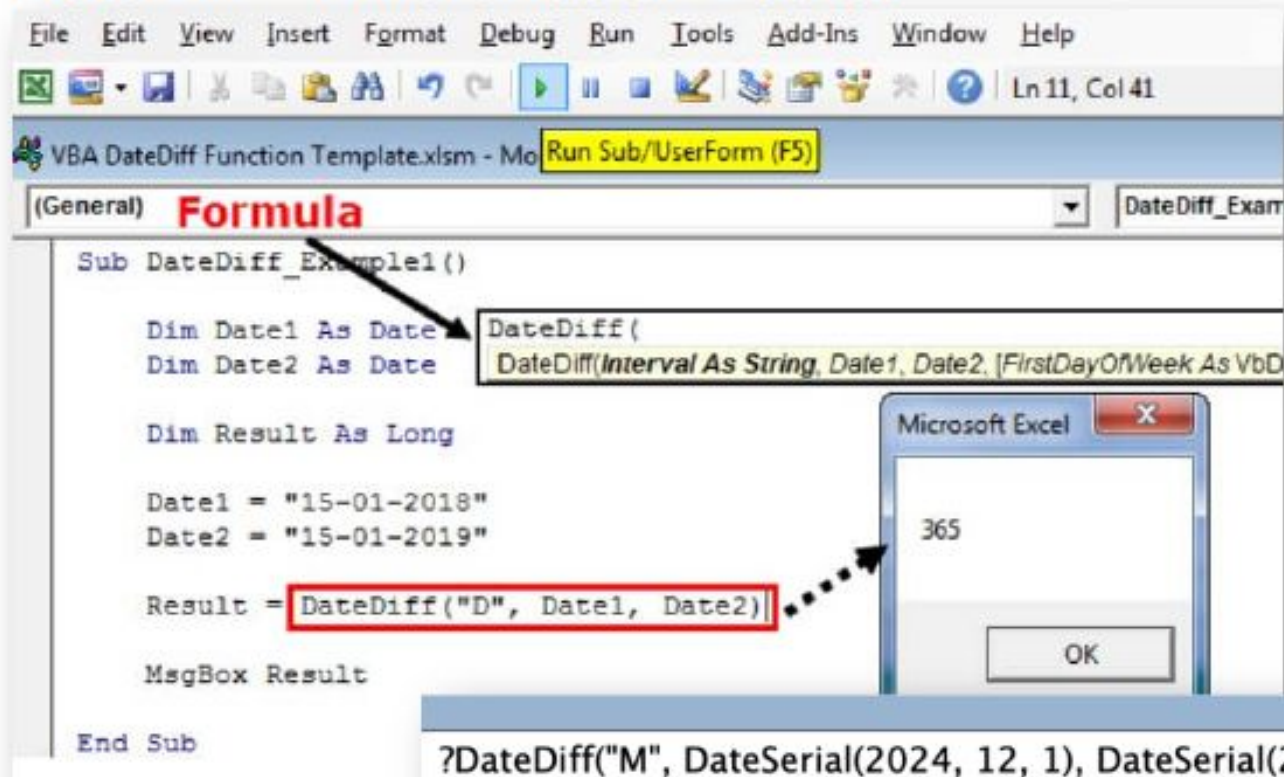
- Year() / Month() / Day() → Extract

```
?Year(DateSerial(2024, 12, 25))  
2024  
  
?Month(DateSerial(2024, 12, 25))  
12  
  
?Day(DateSerial(2024, 12, 25))  
25
```


Business Process Automation with VBA

Common functions - Date

- DateDiff – Difference of 2 dates



?DateDiff("M", DateSerial(2024, 12, 1), DateSerial(2024, 12, 10))
0

?DateDiff("M", DateSerial(2024, 11, 30), DateSerial(2024, 12, 1))
1

Business Process Automation with VBA

Common functions - String

- CDate() → Text to Date

```
?CDate("2023-03-25")  
25/3/2023
```

```
?CDate("April 1, 2023")  
1/4/2023
```

```
?CDate("25/2")  
25/2/2023
```

- Format() → Date to Text

```
?Format(DateSerial(2023, 3, 25), "M")  
3
```

```
?Format(DateSerial(2023, 3, 25), "MM")  
03
```

```
?Format(DateSerial(2023, 3, 25), "MMM")  
Mar
```

```
?Format(DateSerial(2023, 3, 25), "MMMM")  
March
```

```
?Format(DateSerial(2023, 3, 25), "DD-MMM-YYYY")  
25-Mar-2023
```

Business Process Automation with VBA

Common functions - String

- Left() / Right() → Substring

```
?Left("0005.HK", 4)  
0005
```

```
?Right("0005.HK", 2)  
HK
```

- Split() → Tokenizing into a string array

```
?Split("A,B,C", ",")(0)  
A
```

```
?Split("A,B,C", ",")(2)  
C
```

```
?UBound(Split("A,B,C", ","))  
2
```

Business Process Automation with VBA

Common functions - String

- Trim() → Remove empty spaces
 - Note: "" gives 1 quote
- InStr(x, y) → Find y in x
 - Case sensitive

```
?"" & Trim(" Hello! ") & ""  
"Hello!"
```

```
?InStr("Search here", "Search")  
1
```

```
?InStr("Search here", "here")  
8
```

```
?InStr("Search here", "nothing")  
0
```

```
?InStr("Search here", "search")  
0
```

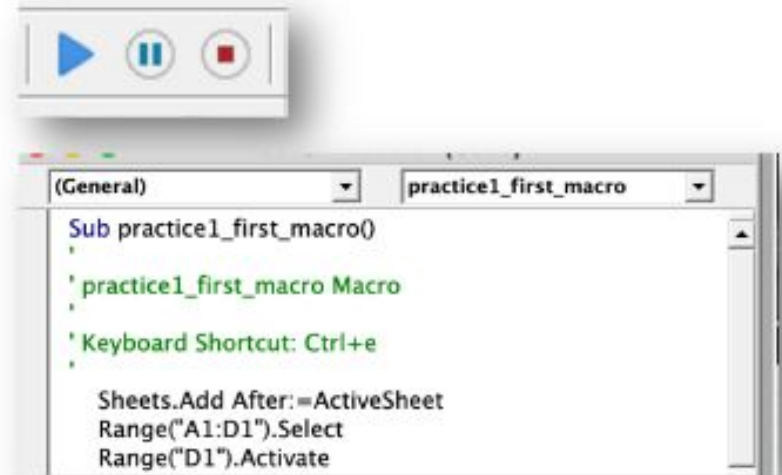
```
?Lcase("HeLlO")  
hello
```

```
?InStr(Lcase("Search here"), "search")  
1
```


Business Process Automation with VBA

Sub-routines

- Macros recorded are wrapped with “Sub ... End Sub”
- Sub-routines are just reusable codes
- Within the scope of a sub-routine
 - Click the “Play” button to execute



Business Process Automation with VBA

Control flow – If, Then, Else

- Example: speed check

	A	B
1	Speed	40
2		

	A	B
1	Speed	50
2		

	A	B
1	Speed	70
2		

```
Sub HighlightCell()
```

```
    If Range("B1").Value > 50 Then  
        Range("B1").Interior.Color = vbYellow  
    Else  
        Range("B1").Interior.Color = vbGreen  
    End If
```

```
End Sub
```

Business Process Automation with VBA

Control flow – If, Then, Else

- Mark → Grade

<i>Mark</i>	<i>Letter Grade</i>
90+	A+
85-89	A
80-84	A-
77-79	B+
73-76	B
70-72	B-
67-69	C+
63-66	C
60-62	C-
57-59	D+
50-56	D
0-49	F

Business Process Automation with VBA

Control flow – Select, Case

- Range check

?GetGrade(90)
A

?GetGrade(80)
B

```
Function GetGrade(score As Integer) As String
    Select Case score
        Case Is >= 90
            GetGrade = "A"
        Case Is >= 80
            GetGrade = "B"
        Case Is >= 70
            GetGrade = "C"
        Case Is >= 60
            GetGrade = "D"
        Case Else
            GetGrade = "F"
    End Select
End Function
```


Business Process Automation with VBA

Control flow – Select, Case

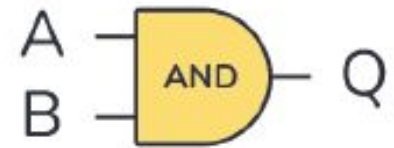
- Grade → Marks

```
Dim letter_grade As String
Dim grade_point As Double

Select Case letter_grade
Case "A", "A+"
    grade_point = 4.0
Case "A-"
    grade_point = 3.7
Case "B+"
    grade_point = 3.3
...
Case Else
    grade_point = 0.0
End Select
```

Control flow – Conditions

- AND
 - (True AND True) → True
 - e.g., Month validity: $1 \leq \text{month}$ AND $\text{month} \leq 12$
- OR
 - (True OR False) → True
 - e.g., Game Area: $\text{gold} > 1000$ OR $\text{level} > 10$
- NOT
 - NOT $x=0$ is same as $x \neq 0$



A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

Business Process Automation with VBA

Loops – Using “For”

- Iterating through the rows
- What is changing?
 - A1 → A2 → A3 → A4...



	A	B	C
1	1		
2	2		
3	3		
4	4		
5	5		
6	6		
7	7		
8	8		
9	9		
10	10		

```
Sub LoopThroughRows()  
    For i = 1 To 10  
        Range("A" & i).Value = i  
    Next i  
End Sub
```

Business Process Automation with VBA

Loops – Using “For”

- Iterating through the columns
- Again, track the change: A1 → B1 → C1 → ...
- Option 1: Make use of offset()



	A	B	C	D	E	F	G	H	I	J
1	1	2	3	4	5	6	7	8	9	10
2										

```
Sub LoopThroughColumns()  
    For i = 1 To 10  
        Range("A1").Offset(0, i - 1).Value = i  
    Next i  
End Sub
```


Business Process Automation with VBA

Loops – What is happening?

- Temporary variable + “Array”

```
Sub LoopThroughRows()  
    For i = 1 To 10  
        Range("A" & i).Value = i  
    Next i  
End Sub
```

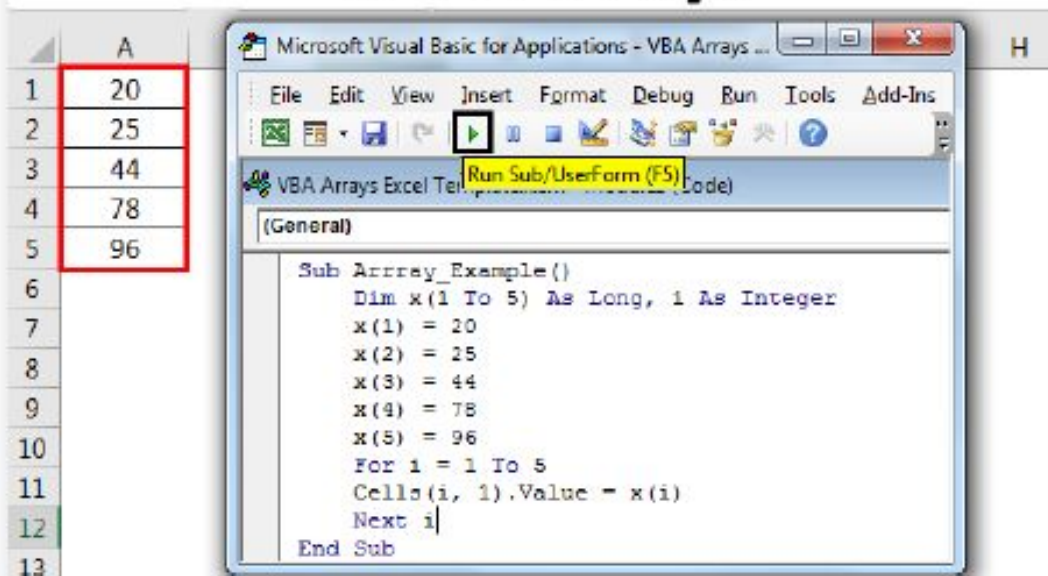
```
Sub LoopThroughColumnsWithCells()  
    Dim i As Integer: i = 1  
    For Each cell In Range("A1:J1").Cells  
        cell.Value = i  
        i = i + 1  
    Next cell  
End Sub
```

Business Process Automation with VBA

Loops – What is happening?

- Array = many variables

VBA Arrays



The screenshot displays an Excel spreadsheet with the following data in column A:

	A
1	20
2	25
3	44
4	78
5	96

The VBA Editor window shows the following code:

```
Sub Array_Example()  
    Dim x(1 To 5) As Long, i As Integer  
    x(1) = 20  
    x(2) = 25  
    x(3) = 44  
    x(4) = 78  
    x(5) = 96  
    For i = 1 To 5  
        Cells(i, 1).Value = x(i)  
    Next i  
End Sub
```





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THANK YOU

